Installation guide

LIQREP solution 2025.02 Installation and Configuration Guide Kubernetes/OpenShift

Wolters Kluwer

Copyright statement

Copyright © 2025 Wolters Kluwer Financial Services Belgium NV (“Wolters Kluwer Financial Services”). All rights reserved.

All other registered or unregistered trademarks and service marks are property of their respective companies and should be treated as such.

No part of this publication may be reproduced, transcribed, transmitted, stored in a retrieval system,computer or otherwise, in any form or by any means, magnetic, mechanical, electronic, optical, manual or otherwise, and may not be translated into any language without the express written permission of Wolters Kluwer Financial Services.

Limit of Liability - Disclaimer of Warranty

Wolters Kluwer Financial Services makes no representations or warranties of any kind with respect to this manual and the contents hereof and specifically disclaim any warranties, either expressed or implied, or merchantability or fitness for any particular purpose. In no event shall Wolters Kluwer Financial Services or its subsidiaries be held liable for errors contained herein or direct, indirect, special, incidental, or consequential damages in connection with or arising from the use of the product, the accompanying manual, or any related materials. Further, Wolters Kluwer Financial Services reserves the right to change both this publication and the software programs to which it relates and to make changes from time to time to the content hereof with no obligation to notify any person or organization of such revision or changes.

All Wolters Kluwer Financial Services publications and computer programs contain proprietary confidential information of Wolters Kluwer Financial Services and their possession and use are subject to restrictions set forth in the license agreement entered into between Wolters Kluwer Financial Services and its licensees. No title or ownership of Wolters Kluwer Financial Services software is transferred, and any use of the product and its related materials beyond the terms on this license, without the written authorization of Wolters Kluwer Financial Services, is prohibited.

Export Restrictions

The recipient agrees to comply in all respects with any governmental laws, orders, other restrictions (“Export Restrictions”) on the export or re-export of the software or related documentation imposed by any applicable public authority (including the government of the United States, the governments of the member states of the European Union and other countries in which the recipient is located). The recipient shall not commit any act or omission that will result in a breach of any such export restrictions.

Statement of Confidentiality

The recipient of this document agrees to keep this document confidential and not to use its contents in any form or by any means without prior written permission from Wolters Kluwer Financial Services.

If such permission is granted, it will be subject to the condition that the recipient ensures that any third party recipient of this document, or of information contained therein, is held responsible to Wolters Kluwer Financial Services for the confidentiality of that document or information.

Gender

Throughout this document the user/reader is referred to as he/him/his. We mean this to include she/her/hers equally.

Wolters Kluwer

Table of contents

1	**About this document**	**5**
1.1	Audience	5
1.2	Related documentation	5
2	**LIQREP solution Installation requirements**	**6**
2.1	Supported components	6
2.2	Delivered components	7
2.3	Additional components	7
2.4	LIQREP prerequisites	7
3	**About the Platform installation guide**	**8**
3.1	Audience	8
4	**The Platform landscape**	**9**
5	**Distribution and deployment overview**	**11**
5.1	Container orchestration	11
5.2	Authentication and authorization	12
6	**Database requirements overview**	**13**
7	**Prerequisites**	**14**
7.1	Access to Product Download Center	15
7.2	Access to Docker Repository	15
7.2.1	Generating tokens	16
7.2.2	Logging in to Wolters Kluwer Docker Registry	17
7.2.3	Alternative for repository setup without direct internet connection	17
7.2.3.1	Downloading and publishing images	18
7.2.4	Alternative for systems without any network interconnection	20
7.2.4.1	Transferring images	20
7.3	Database credentials	22
7.4	Certificates	23
7.5	Configure identity provider	23
7.5.1	Configuring OAuth2 authentication with Keycloak	23
7.5.2	Configuring OAuth2 authentication with Azure AD	27
7.6	OpenID Connect	28
7.6.1	OpenID Connect settings	29
8	**Installing Cert Manager**	**32**
9	**Create custom storage class**	**33**
10	**Installing Platform on Kubernetes/OpenShift with Helm**	**34**
10.1	Downloading Helm chart	34
10.1.1	Helm chart distribution structure	34
10.2	Installing Helm chart	35
10.3	Monitoring deployment status of Helm chart	36
10.4	Parameters	37
10.4.1	Global parameters	37
10.4.2	Azure related parameters	38
10.4.3	Configuration parameters reg and risk	39
10.5	Persistence	39
10.6	Secrets	40
10.7	Managing certificates	41
10.8	Openshift routes	42
10.9	Uninstalling chart	46
11	**Troubleshooting**	**47**
11.1	Time synchronization	47
11.2	Certificates	47
11.3	Access rights	48
11.4	Activate full debug logs	49
11.5	Not found error after login	49
11.6	Cannot connect to Admin Center after successful login	49
11.7	Delivery file does not get generated	49
11.8	WAL errors on Processing Engine restart	50
12	**LIQREP solution Installation procedure**	**51**

13	**Managing workspaces**	**53**
13.1	Creating workspaces	53
13.1.1	Creating workspaces for Oracle databases	55
13.1.2	Creating workspaces for SQL databases	59
13.1.3	Creating workspaces using existing business database	62
13.2	Configuring workspaces	64
13.3	Viewing workspace details	67
13.4	Modifying workspace details	68
13.5	Activating workspaces	69
13.6	Deactivating workspaces	70
13.7	Discarding workspaces	72
13.8	Enabling dual control	74
13.9	Enabling sign-off	75
13.10	Upgrading reporting modules installed in workspace	76
13.10.1	Upgrading reporting module installed in workspace for Oracle databases	77
13.10.2	Upgrading reporting module installed in workspace for SQL databases	80
13.11	Optimizing workspaces	84
13.11.1	Diagnosing workspaces	84
13.11.2	Optimizing workspace for Oracle installation	85
13.11.3	Optimizing workspace for SQL Server installation	89
13.12	Disabling audit comments	91
14	**Managing resource groups**	**93**
14.1	Creating resource groups	93
14.1.1	Granting users access to resource groups	96
14.2	Removing users from resource groups	97
14.3	Viewing resource groups and their details	98
14.4	Adding workspaces to resource groups	99
14.5	Deleting resource groups	101
15	**Risk Engine Model**	**103**
15.1	Setting up environment	103
15.2	Importing Risk Engine model	105
15.3	Updating database	106
15.4	Generating JAR	107
15.5	Creating legal entity	107
15.6	Creating consolidation hierarchy	109
15.7	Updating base currency	111
15.8	Configuring contract selection	112
16	**Contact**	**114**
17	**Glossary**	**115**

1 About this document

This guide describes the installation and configuration of the OneSumX for Basel: Liquidity Reporting (LIQREP) solution.

Note:
This document provides information to set up Platform and the following components:

- Reporting Hub
- Admin Center
- Processing Engine
- Lineage
- Delivery Engine
- Risk Engine

Note:
The FINREP images used in this document are only for illustration purposes.

1.1 Audience

This document is intended for IT administrators responsible for the reporting application installations.

1.2 Related documentation

This section lists all the documents relevant to this release:

- LIQREP solution 2025.02 Installation and Configuration Guide
- LIQREP solution 2025.02 Workflow and User Guide
- LIQREP solution 2025.02 Hardware and Software Requirements
- LIQREP solution 2025.02 Release Notes
- LIQREP solution 2025.02 Standard Records Data Dictionary
- LIQREP solution 2025.02 DataFoundation Interfacing Guide
- Financial Data Model 3.24.0 Data Dictionary
- OneSumX Regulatory Engine User Manual 7.1.0
- OneSumX Regulatory Engine Administrator Manual 7.1.0

2 LIQREP solution Installation requirements

2.1 Supported components

You need to download the components from the Product Download Center according to the instructions in the release email you receive.

Table 1. Engine components are a prerequisite to installing the OneSumX Regulatory Reporting Liquidity 21.2.0 content libraries

Component	Version	Details
Platform	**25.01.02**	**Docker compose script (DXG approved credentials required)**
Reporting Hub	**7.1.2**	
Admin Center	**7.1.2**	
Processing Engine	**11.1.4**	
Lineage	**6.1.2**	
Delivery Engine	**3.1.1**	
Risk Engine	**13.14.1.0**	

Note:

- OneSumX Regulatory Engine is a reporting component of the Platform. It transforms data into regulatory reports and covers the entire reporting process, from data input to delivery file generation.
- When using the Docker version, you only need to download the Platform DockerCompose configuration zip, which will download the:
  - Reporting Hub
  - Admin Center
  - Processing Engine
  - Lineage
  - Delivery Engine
  - Risk Engine

2.2 Delivered components

The LIQREP solution contains the following components.

Table 1. LIQREP solution components

Component	Version
OneSumX Regulatory Reporting Liquidity	***liqrep-distribution-dist-21.2.0.zip (included in the LIQREP 2025.02 package)***
Financial Data Model	**3.24.0 (included in the LIQREP 2025.02 package)**
Data Foundation	**3.53.0 (included in the LIQREP 2025.02 package)**
Regulatory Analytics LQR	**4.1.0 *(included in the LIQREP 2025.02 package)***

2.3 Additional components

You need any one of the following additional components.

Component	Version
Oracle Server	**Oracle 19**
SQL Server	**2019**

2.4 LIQREP prerequisites

The LIQREP solution Hardware and Software Requirement Guide is available on the FRR Knowledge Portal.

Software prerequisites for application server:

- Java runtime environment JRE 21 64-bit version installed and the JAVA_HOME system variable set for this installation.
- Any previous versions of ReportGenerator Desktop or TNG applications have to be removed prior to the OneSumX Regulatory Engine setup.
- Download the latest version of the Reporting Module of LIQREP solution available on the Product Download Center.

Note:
The FINREP images used in this document are only for illustration purposes.

3 About the Platform installation guide

This guide provides instructions for installing and upgrading the Platform. It covers the installation process, system requirements, and troubleshooting steps. It includes detailed information about the installation process, prerequisites, and post-installation tasks.

3.1 Audience

The Platform installation guide is intended primarily for support personnel, administrators, and system administrators.

Installation personnel should have good knowledge of the following technologies:

- Operating systems - Linux
- Containers - Docker and Kubernetes
- Databases - Oracle or SQL
- Identity providers - Azure AD, ADFS, or other OpenID connect compatible providers
- Certificates and key management with Java

4 The Platform landscape

The Platform is a containerized application that uses Docker for deployment. It consists of several components, including:

- Web application servers
- Database servers
- Authentication services
- Container orchestration tools

Platform components

The Platform consists of the following components:

Processing Engine

This component uses distributed cache technology (the Apache Ignite library) to provide the storage and inter-component communications infrastructure for other components. It also provides the run-time environment in which the regulatory logic is executed.

Reporting Hub

This component allows you to define and execute regulatory reporting workflows, such as preparation and enrichment of input data sets, creating result sets and running reports, and allocating and generating reports.

Admin Center

This component provides management functionality for the Reporting Hub component such as user and role management, installation and updating of reporting modules, and job management.

Risk Engine

This contract-centric calculation engine, with extensive support for scenario analysis, provides economic and regulatory risk analysis.

Delivery Engine

This component supports preparing final report files for submission to regulators. XBRL and XML formats are supported, depending on the regulator's request.

Lineage

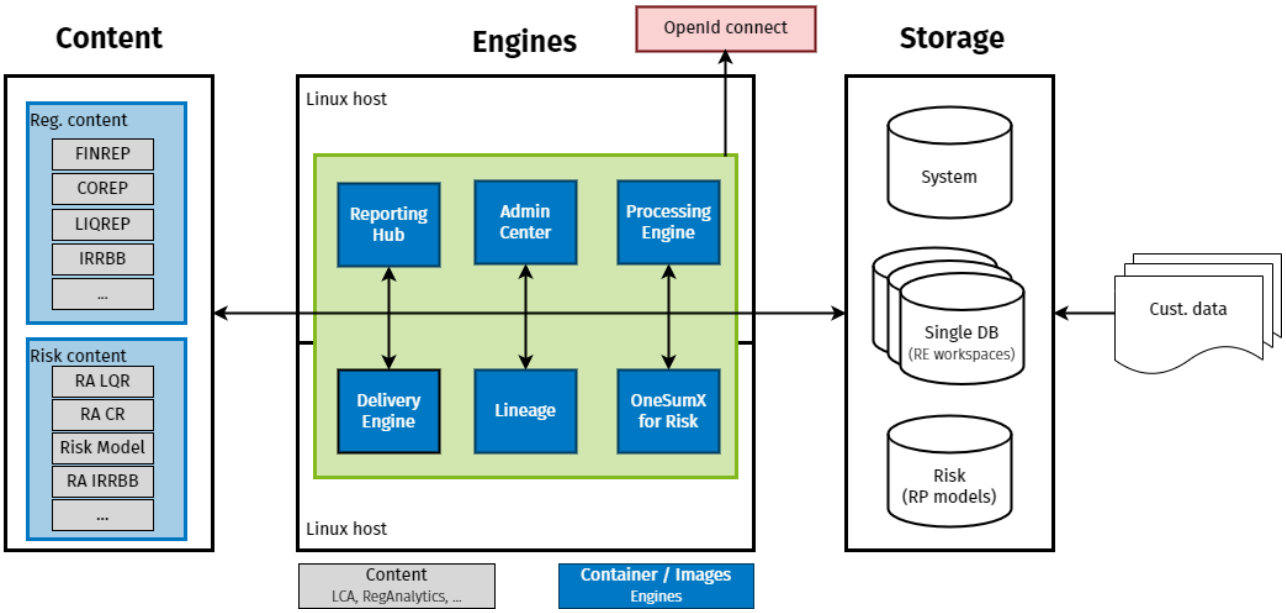
This component allows you to trace the calculation and allocation logic. Starting from any report cell, you can inspect the inputs contributing to this cell and examine each of the calculation steps.

Platform overview

The Platform components require databases for persistent storage.

Specific regulatory support is enabled via content that is packaged risk and regulatory logic, which can be dynamically added to extend the functionality of the solution.

Figure 1. Platform overview



5 Distribution and deployment overview

The Platform uses container technology for packaging, distribution, and deployment:

- Components are packaged as OCI *container images* (commonly known as Docker images) which are a lightweight and portable way to package all the programs, libraries, operating system files, and other dependencies that make up a component.
- Container images are distributed from the Wolters Kluwer *container image repository* using standard tools that allow you to easily obtain these images and verify their authenticity.
- A *container engine* such as Docker or Podman is required to turn a container image (a file) into a *container* (a running process). Container engines make use of Linux kernel features, such as namespaces and control groups features, to isolate containers from each other and from other operating system processes.
- The Docker and Podman container engines are supported on all mainstream Linux distributions (either out of the box or after installing the relevant package) and on the Windows OS. Note, however, that on performance grounds using containers on Windows is not supported for production environments and is limited to light workload environments (for example, development, demonstration, or ad-hoc testing) where performance plays no role.

Compared to distributing applications as standalone files or packaged in virtual machine images, using containers for distribution and deployment offers significant benefits:

- Reduced IT support costs. Wolters Kluwer assumes full responsibility for the technology stack required for running different applications; customers do not need to administer, configure, or update the underlying Java SDK and the Java application server software on which applications are deployed.
- Improved run-time security. Wolters Kluwer product teams can configure stricter security by controlling the application servers on which software components are deployed.
- A better trade-off between isolation and resource usage. Distributing applications as virtual machine images has isolation and configuration advantages, but suffers from high-resource usage. Containers offer similar isolation benefits, but are much less resource-intensive.
- Interoperability. Different components share a common technical DNA and are, therefore, configured and administered in a uniform way.
- Cloud-ready. Images can readily be deployed in cloud environments.
- Easy upgrade and rollback. Changing from one version to the next is as easy as stopping a container that was started from an earlier image and starting up a container from a newer image.
- Scalability. With Kubernetes-based orchestration, increasing the number of calculator instances and speed up calculations is straightforward.

5.1 Container orchestration

When a solution consists of multiple containerized components, it quickly becomes necessary to centrally manage the configuration, deployment, and, in general, the life cycle of the constituent components. This task is known as *container orchestration.*

There are several tools and platforms that support container orchestration, and two of the most commonly used are as follows:

- Docker Compose: a relatively straightforward tool that supports container orchestration using a simple text file that specifies the containers that make up an application or a solution, and various configuration

options required to run these containers. Docker Compose is limited to running containers on a single host and is, therefore, not a suitable orchestration solution for customers with large processing needs that require deploying the application on multiple hosts.

- Kubernetes/OpenShift: this technology, often referred to as K8s, supports large-scale deployments, load balancing, and scaling of containerized applications that need to be deployed on multiple hosts. These advantages come at a complex cost and with the learning curve required to master the relevant tool set.

  Kubernetes/OpenShift applications are specified in terms of *pods*, *services,* and *nodes* which abstract the notions of containers, their interdependencies, and the hosts on which they are deployed. These elements are specified in a *helm chart* which is managed (created, modified, obtained from a repository, and so on) using the helm tool.

5.2 Authentication and authorization

All the Platform components use the OpenID Connect federated authentication protocol to authenticate users and processes. In this protocol, which is an identity layer on top of the OAuth 2.0 protocol, user authentication is delegated to an external OAuth 2.0 server.

The OAuth 2.0 server which performs user authentication must be provided and managed by the customers' IT administrators. At their discretion, this can be an on-premise OAuth 2.0 server solution such as Active Directory Federation Services (ADFS) or KeyCloak. Alternatively, administrators may choose to rely on an external provider such as Microsoft Entra ID Platform.

Two OAuth 2.0 grant types are used:

- Authorization Code grant or auth code flow to obtain authorized access to protected resources such as web APIs, used to protect the endpoints behind user interface requests (for example, Microsoft Identity Platform auth code grant flow).

- Client Credentials grant used for server-to-server interactions to allow platform components to communicate (for example, Microsoft Identity platform client credential grant flow).

Using the OpenID Connect authentication protocol offers important benefits to users and administrators:

- Single Sign On (SSO): Once a user is logged on to one of the Platform components, they are automatically logged on to all of them.

- Since authentication is delegated to the OAuth 2.0 server, it is solely under the control of administrators. In particular:

  - User credentials are never exposed to any of the Platform components.

  - Administrators can set authentication mechanisms (password, authenticator apps, bio-metric authentication, or a combination thereof) that are consistent with their organization's existing policies.

- User administration is centrally performed in the directory server used by the OAuth 2.0 server and not by any of the Platform components.

- Authorization decisions are made based on group membership which is also performed in the directory server.

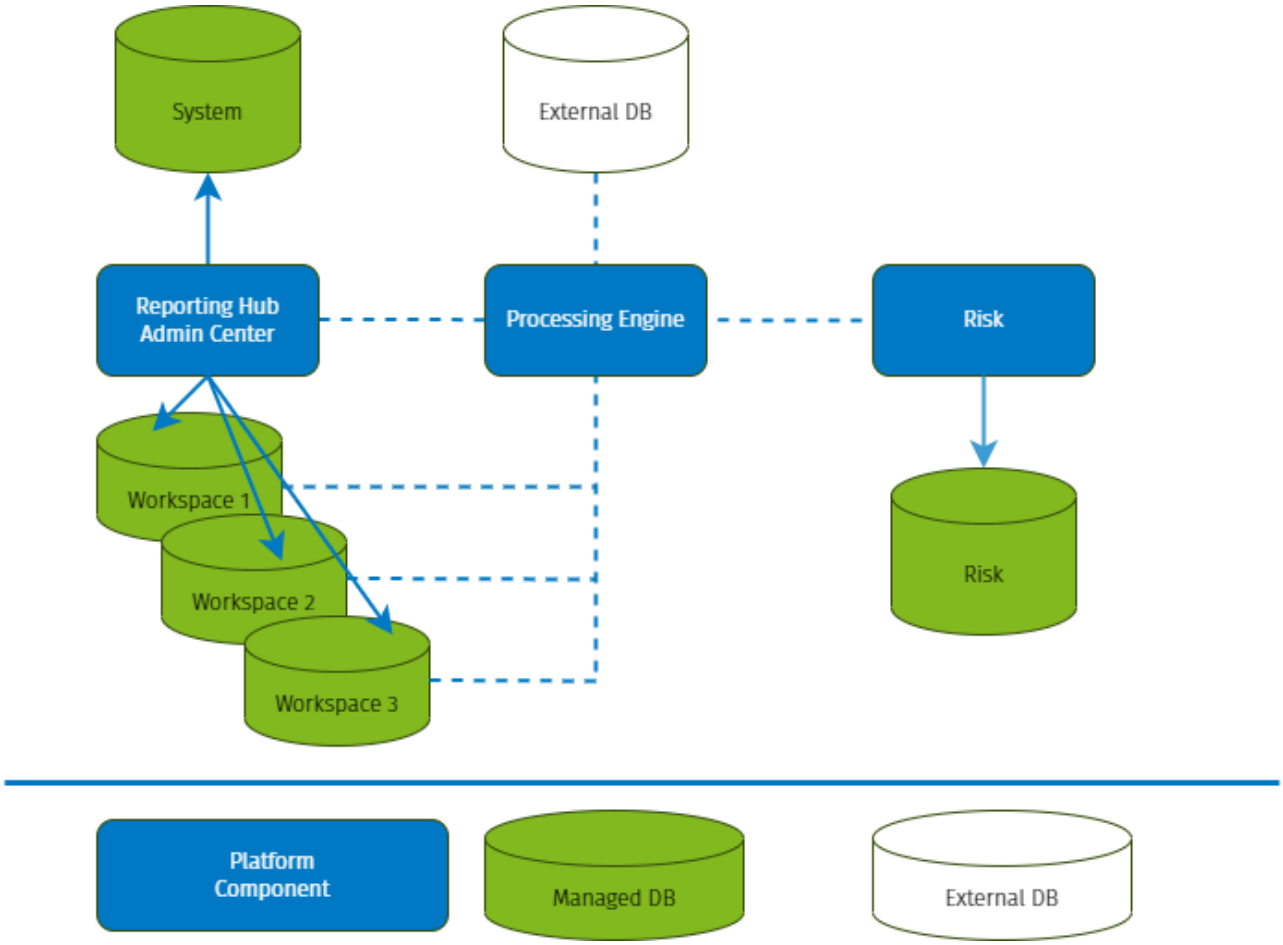
6 Database requirements overview

The Platform consists of multiple components with varying requirements and usage of persistence mechanisms, particularly concerning database handling. Consequently, multiple database schemas need to be created and used concurrently by one or more components.

- Lineage and Delivery Engine have no specific database interaction at all.
- Processing Engine does not need a database on its own but will refer to any database required during processing. While this mostly consists of the physical database of Reporting Hub, it can also link to an external database (in more specific setups), such as OneSumX Report Generator Desktop or OneSumX ReportGenerator TNG GDM or LDM.
- Reporting Hub and Admin Center both share multiple database schemas:
  - A dedicated system database is required to store the internal state, user groups, and user access control definition.
  - Each workspace created in the component requires its own database schema.
- The Risk Engine component also runs on another database schema.

Considering a nominal operation of the platform, with one Reporting Hub workspace, three distinct schemas are required.

Figure 1. Platform Database Overview



7 Prerequisites

The hardware and software recommendations are described in the latest version of the Hardware and Software Requirements Guide.

Before installing Platform, do the following:

- Make sure that you have Wolters Kluwer Container Registry credentials, to be able to download the needed Platform components.
- Make sure that you have database credentials. For details, see Database credentials.
- Make sure that you have access to any Identity Provider; Azure AD, ADFS, or Keycloak. You need the appropriate values to properly set up user authentication and single sign-on. For details, see Configure identity provider.
- Configure applicable secrets (preferably using a vault).

  Note:
  For Azure Vaults, create the secrets with a Helm Azure Key Vault. You can access them with Managed identity and tenant ID.

- Configure the following secrets with the mentioned keys:
  - Installer configuration: App DB and authentication-related properties

admin-auth-token-client-id	**Azure AD app registration client ID**
admin-auth-token-secret	**Azure AD app registration client secret**
hermes-auth-token-client-id	**Azure AD app registration client ID**
hermes-auth-token-secret	**Azure AD app registration client secret**
hub-auth-token-client-id	**Azure AD app registration client ID**
hub-auth-token-secret	**Azure AD app registration client secret**
lineage-auth-token-client-id	**Azure AD app registration client ID**
lineage-auth-token-secret	**Azure AD app registration client secret**
processing-auth-token-client-id	**Azure AD app registration client ID**
processing-auth-token-secret	**Azure AD app registration client secret**

risk-auth-token-client-id	**Azure AD app registration client ID**
risk-auth-token-secret	**Azure AD app registration client secret**
wolterskluwer-application-db-password	**DB admin user name**
wolterskluwer-application-db-username	**DB admin user name**
wolterskluwer-db-connection-url	**SYSTEM_DB URL id Reg application enabled**
wolterskluwer-risk-db-connection-url	**Risk DB URL if risk application is enabled**
wolterskluwer-risk-db-server	**Risk DB host name if risk application is enabled**
wolterskluwer-system-db-password	**DB admin user password**
wolterskluwer-system-db-username	**DB admin user name**

- Docker registry connection-related properties. Docker registry credentials secret: a secret of the kubernetes.io/dockerconfigjson type, used for authentication to the Docker registry.

7.1 Access to Product Download Center

You'll need access to the Product Download Center to obtain the installation files. Several elements of the Platform are manually downloaded from the Product Download Center. If you don't have to access the Product Download Center, contact your account manager.

7.2 Access to Docker Repository

All the Platform components are delivered as OCI-compliant containers, also called images. Those images are not directly available for download in the Product Download Center but are distributed through the Wolters Kluwer Container Registry. This registry is not public and you need a valid access token to log on to it. There is no alternative to retrieving Docker images, but customers can operate their own registry in case their system is not granted proper internet connectivity, typically for security reasons.

The token for accessing the Wolters Kluwer Container Registry offers the following important features:

- Generated tokens have a validity period of one year. After the token expires, you must generate a new one.
- Using a generated token, you can access all the container images that Wolters Kluwer has agreed to provide through their contractual agreements.
- Upon termination of your organization's contractual agreement, the token will be automatically invalidated.
- Wolters Kluwer does not store any generated token locally. If the token is lost or invalid, a new token must be generated and registered again on the local container platform.
- All previous tokens will be invalidated once a new token is generated. You can generate as many tokens as you wish.

For details on generating tokens, see Generating tokens.

Warning:
The Wolters Kluwer Docker Repository is hosted on the Amazon AWS platform, on the Ireland tenant. In case your network infrastructure limits connections on the internet, ensure that the following URLs are on the allowlist for your firewalls and can be accessed from the server hosting Docker, or you won't be able to download the necessary images:

- https://wkfrr.wolterskluwer.io
- https://jfrog-prod-euw1-shared-ireland-main.s3.amazonaws.com

7.2.1 Generating tokens

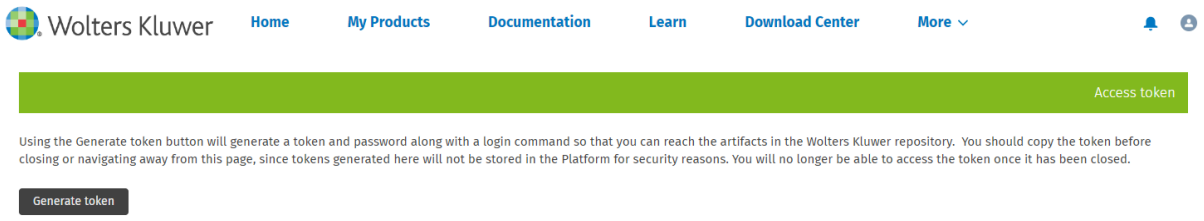
Context

The Wolters Kluwer registry tokens are generated through the FRR Community.

Procedure

1. Log in to FRR Community.
2. Click the user avatar in the upper-right corner, and then click **Access token**.

   The **Access token** page is displayed.



3. Click **Generate token**.

Results

The generated token displays under the **Generate token** button.

Note:
Tokens are not stored on the Platform for security reasons, therefore, it is recommended to copy them before closing the token generation page.

Troubleshooting

In case the self-service generation functionality does not work, customer support can generate a token on your behalf and share it in a secure manner. Should you need help on generating tokens, create a case through FRR Community.

7.2.2 Logging in to Wolters Kluwer Docker Registry

Prerequisites

- You have a valid access token. For details, see Generating tokens.

Procedure

Run the following command.

```
$ docker login wkfrr.wolterskluwer.io -u username
Password: token
```

where items in italic are as follows:

registryUrl

The URL of the registry

username

The username to connect to the FRR Community

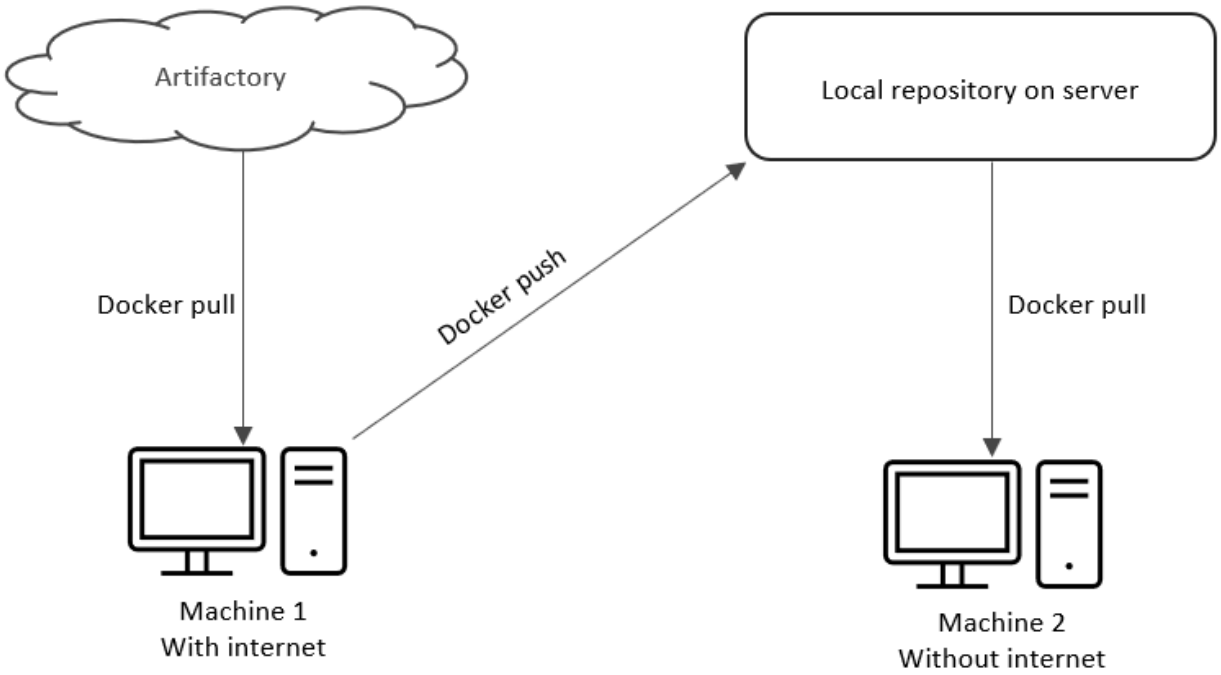
token

The generated token

7.2.3 Alternative for repository setup without direct internet connection

Production servers often don't have direct access to the internet. While it's easy to configure network and firewall settings to allow connection to the Wolters Kluwer Docker Registry, this might not be acceptable from an internal policy perspective. Also, dedicated security scans might be required in addition to internal policies before images are put to production.

In such cases, it is necessary to set up an internal Docker registry that can be reached at the same time from a temporary workstation with internet access and from the production server. This machine will act as a middle tier to bring Docker images to production servers.



7.2.3.1 Downloading and publishing images

Context

Download and publish images using a local Docker registry.

Procedure

1. Set up a local Docker registry.

   It is recommended to deploy the official image published by the Docker team. Use the following command to start the registry on the default port 5000:

   ## Example

   ```
   $ docker run -d -p 5000:5000 --restart=always --name registry registry:2
   ```

2. Pull the required images from the Wolters Kluwer Docker Registry.

   ## Example

   ```
   $ docker pull wkfrr.wolterskluwer.io/wkfrr/wolterskluwer/onesumx/processing-
   engine:9.0.0
   ```

Results

```
gvalvona@CH54GFRRL1422HG:~$ docker pull frr-docker.wolterskluwer.io/wolterskluwer/onesumx/processing-engine:9.0.0-build10
9.0.0-build10: Pulling from wolterskluwer/onesumx/processing-engine
d261077062b2: Already exists
327c893a9020: Pull complete
76a5a10b68a4: Pull complete
42dad66329b4: Pull complete
e29ec07a075a: Pull complete
194d82efec72: Pull complete
66146bd054ac: Pull complete
ccdb890a4a81: Pull complete
098dd5f08caa: Pull complete
aa25bce32b87: Pull complete
Digest: sha256:39c17fe906fec56a93bff3e2fa71842a46162028bdf622d135068ce9167d79fb
Status: Downloaded newer image for frr-docker.wolterskluwer.io/wolterskluwer/onesumx/processing-engine:9.0.0-build10
frr-docker.wolterskluwer.io/wolterskluwer/onesumx/processing-engine:9.0.0-build10
```

3. Perform the necessary security scans required by the internal policies.
4. Tag the pulled images on the local Docker Repository using the tag command.

Example

```
$ docker tag wkfrr.wolterskluwer.io/wkfrr/wolterskluwer/onesumx/processing-engine:9.0.0-build10 mydomain.com:5000/processing-engine
```

5. Push the image to the local Docker Repository.

Example

```
$ docker push mydomain.com:5000/processing-engine
```

Results

```
PS C:\SourceCode\performances> docker push localhost:5000/processing-engine
Using default tag: latest
The push refers to repository [localhost:5000/processing-engine]
bce9c1bd2b57: Pushing [===============>                                   ]  39.52MB/130.9MB
7efb69d56cb4: Pushed
f0f40eafd5d9: Pushing [======================>                            ]  39.52MB/87.76MB
c8e31847ae66: Pushing [=============================================>     ]     39MB/43.11MB
37ae69d4fd0c: Pushed
9d58875f3e77: Pushed
e5e1f0bca777: Pushed
6cc23bc1fc23: Pushing [==============>                                    ]  36.49MB/129.4MB
7b46040b26e8: Pushed
625523cb9cce: Pushed
```

6. Replace the image name in the Docker Compose YAML file for all images, using the appropriate server name and image tag.

Example

```
services:
  installer:
    image: wkfrr.wolterskluwer.io/wkfrr/wolterskluwer/onesumx/reg/installer:${ONESUMX_RE_INSTALLER_RELEASE:-5.0.0-build132}
```

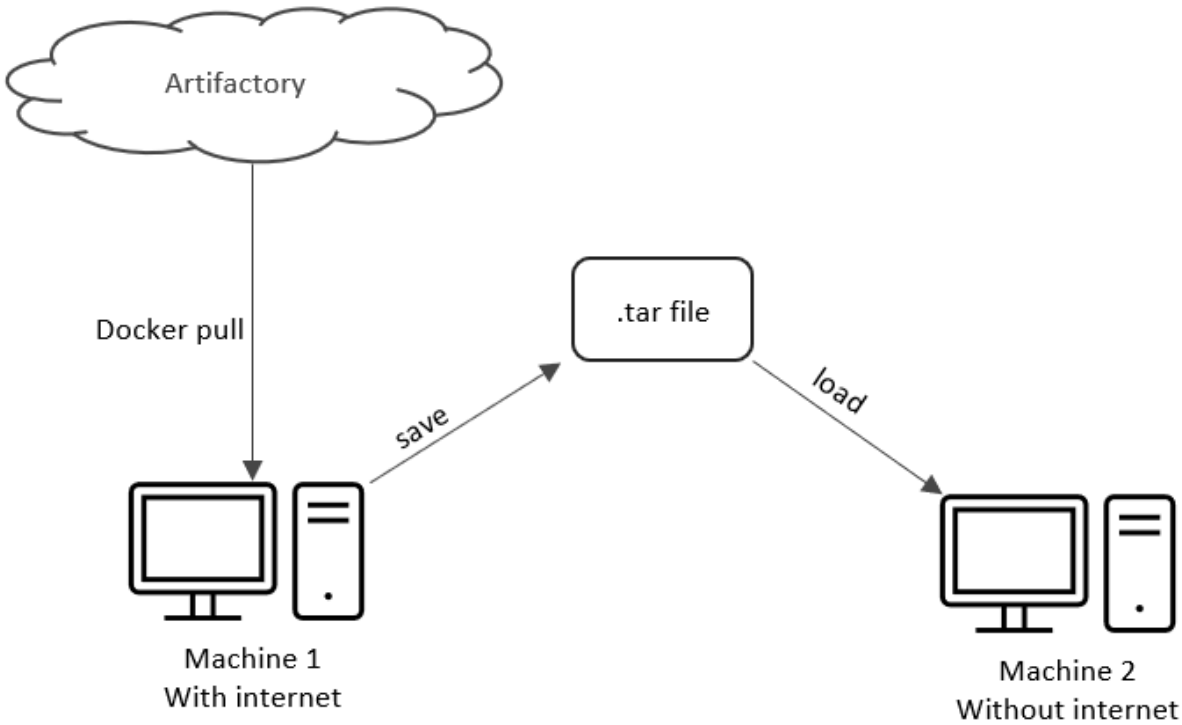
What to do next

To run the docker commands, see Starting.

7.2.4 Alternative for systems without any network interconnection

When institutions have strict policies regarding software downloads from the internet, it can happen that the intermediate machine that has access to the internet is physically connected to a totally separate network from the rest of the infrastructure, to guarantee that security scans are performed in a safe and isolated environment.

In such cases, it is still possible to physically transfer the image from one network to another after the proper scans have run, using Docker save and load features.



7.2.4.1 Transferring images

Context

Transfer images using the save and load method.

Procedure

1. Pull the required images from the Wolters Kluwer Docker Registry on the machine with public internet access to manually download them.

   # Example

   ```
   $ docker pull wkfrr.wolterskluwer.io/wkfrr/wolterskluwer/onesumx/processing-
   ```

```
engine:9.0.0
```

Results

Pulled images can be visualized using the `docker images` command.

```
gvalvona@CH54GFRRL1422HG:~$ docker images
REPOSITORY                                                          TAG                 IMAGE ID       CREATED        SIZE
localhost:5000/processing-engine                                    latest              26a2c61181d1   8 days ago     417MB
frr-docker.wolterskluwer.io/wolterskluwer/onesumx/processing-engine 9.0.0-build10       26a2c61181d1   8 days ago     417MB
processing/refapp-delivery-sqlserver-250                            latest              45cfc7557443   2 weeks ago    2.04GB
frr-docker.wolterskluwer.io/processing/refapp-delivery-sqlserver-250 latest             45cfc7557443   2 weeks ago    2.04GB
wolterskluwer/onesumx/processing-engine                             9.0.0-SNAPSHOT      ef7e623e64fc   2 weeks ago    417MB
frr-docker.wolterskluwer.io/wolterskluwer/onesumx/processing-engine 9.0.0-SNAPSHOT      ef7e623e64fc   2 weeks ago    417MB
wolterskluwer/processing-engine                                     alpine-9.0.0-SNAPSHOT cb7406c0e9fe 2 weeks ago    417MB
frr-docker.wolterskluwer.io/wolterskluwer/processing-engine         alpine-9.0.0-SNAPSHOT cb7406c0e9fe 2 weeks ago    417MB
processing/refapp-delivery-sqlserver-1                              latest              e1efdc52c4a9   3 weeks ago    1.91GB
frr-docker.wolterskluwer.io/processing/refapp-delivery-sqlserver-1  latest              e1efdc52c4a9   3 weeks ago    1.91GB
registry                                                            2                   0d153fadf70b   6 weeks ago    24.2MB
processing/refapp2-oracle19-1                                       latest              fcf8b3f04208   6 weeks ago    15.8GB
frr-docker.wolterskluwer.io/processing/refapp2-oracle19-1           latest              fcf8b3f04208   6 weeks ago    15.8GB
localhost:5000/refapp2-oracle-1                                     latest              fcf8b3f04208   6 weeks ago    15.8GB
```

2. Save the pulled images.

Example

```
$ docker save wkfrr.wolterskluwer.io/wkfrr/wolterskluwer/onesumx/processing-
engine:9.0.0 > processing-engine.tar
```

3. Perform the necessary security scans required by internal policies and transfer the .tar archive to the production server.
4. On the production server, load the image.

Example

```
$ docker load < processing-engine.tar
```

Results

```
gvalvona@CH54GFRRL1422HG:~$ docker load < processing-engine.tar
625523cb9cce: Loading layer [==================================================>]  5.895MB/5.895MB
7b46040b26e8: Loading layer [==================================================>]   14.6MB/14.6MB
6cc23bc1fc23: Loading layer [==================================================>]  129.7MB/129.7MB
e5e1f0bca777: Loading layer [==================================================>]  6.614MB/6.614MB
9d58875f3e77: Loading layer [==================================================>]  12.29kB/12.29kB
37ae69d4fd0c: Loading layer [==================================================>]   2.56kB/2.56kB
c8e31847ae66: Loading layer [==================================================>]  43.12MB/43.12MB
f0f40eafd5d9: Loading layer [==================================================>]  87.77MB/87.77MB
7efb69d56cb4: Loading layer [==================================================>]  3.584kB/3.584kB
bce9c1bd2b57: Loading layer [==================================================>]  130.9MB/130.9MB
Loaded image: frr-docker.wolterskluwer.io/wolterskluwer/onesumx/processing-engine:9.0.0-build10
```

What to do next

To run the docker commands, see Starting.

7.3 Database credentials

The Platform requires the creation of several database schema to properly operate. Before starting the deployment of the platform, the following connection settings and user credentials need to be available.

Setting	Description
Database Server name/port	**Physical server location (FQDN) and port on which the instance is running**
Database Instance name	**Instance name, where applicable**
DBA user name	**Database administrator user, typically 'sa' on Microsoft SQL Server or "system" on Oracle**
DBA user password	**Database administrator credentials**

Tip:
The above table assumes that all schema are created on the same database instance. In practice, each schema can be set up on any set of database instances, but the corresponding parameters and credentials need to be collected for each instance.

At least three database schema need to be created to properly deploy the Platform, and the following parameters need to be prepared before starting the deployment process.

Parameter	Description
RegEngine System DB	**Determine user name, schema name, and password**
RegEngine Workspace DB	**Determine user name, schema name, and password**
Risk DB	**Determine user name, schema name, and password**

Warning:
The database administrator might decide, for security reasons, to spread the schema access rights over two distinct users. A first privileged user actually owns the schema and manages the table and other object creation. A second, less privileged user can only access the data inside the schema. For such cases, both sets of credentials need to be determined upfront. See the corresponding database setup sections for more details.

7.4 Certificates

Note:
Your instance is hosted in *AWS Ireland* and the only S3 bucket you need to allowlist is *jfrog-prod-euw1-shared-ireland-main.s3.amazonaws.com.Whitelisting*. This URL should fix the forbidden errors.

A valid certificate is required to operate the platform so that inter-component communication as well as communication between the client browser and the server are secured through an HTTPS channel. This certificate is usually provided by the internal IT services, but it can also be manually generated (self-signed certificates). In all cases, the signing authority needs to be added to the trust store. This does not require any specific action if the certificate is signed by an official authority. If the certificate is generated using an internal authority, that authority must be added manually.

For security reasons, the distribution does not come with any pre-bundled certificates; they need to be created or registered manually at setup time. Using self-signed certificates triggers a warning message in the browser and that should be avoided.

For details on generating the HTTPS and TLS certificate, see Managing certificates.

7.5 Configure identity provider

OneSumX Regulatory Engine supports single sign-on (SSO) authentication through an identity provider (IdP). There are various identity providers available, two of which can be applied for the OneSumX Regulatory Engine authentication:

- Keycloak
- Active Directory Federation Services (AD FS)

For the inter-component communication, the IdP must support the OAuth2 protocol (see Configuring OAuth2 authentication with Keycloak or Configuring OAuth2 authentication with Azure AD, depending on your identity provider).

The OIDC access token should contain the claims that are conforming to the OpenID Connect standard claim specification, and granting access to the following scopes:

- openid
- profile
- email

7.5.1 Configuring OAuth2 authentication with Keycloak

Context

This procedure is mandatory if you have Keycloak as an identity provider.

First, you need to set up the users in Keycloak, then you copy the authentication details in the *.env* file, as described in this procedure. As a result, a realm, OAuth2 client, and user are created.

Procedure

1. Log in to Keycloak with your admin account.

2. In the Keycloak Administration Console, create a new realm. See the Keycloak documentation for details.
3. On the left menu of the admin console, click **Clients** and then click the **Create** button.
4. Enter a meaningful name in the **Client ID** field.
5. Click **Save**.

Results

The client details page opens.

6. In the **Settings** tab, change the access type from **public** to **confidential**.
7. Turn on the **Service Accounts Enabled** toggle.
8. In the **Valid Redirect URIs** box, add `${protocol}://${host}:${port}/${context}/*`

   Where:

   - `${protocol}` - Application protocol, either http or https
   - `${host}` - Application host
   - `${port}` - Application port
   - `${context}` - Application context, deployment-specific property.

   The default values are:

   - For Admin Center - `admin-webapp`
   - For Reporting Hub - `reporting-webapp`
   - For Lineage - `lineage-webapp`

   There is no context property for Processing Engine, thus the URI should not contain any. See the following examples.

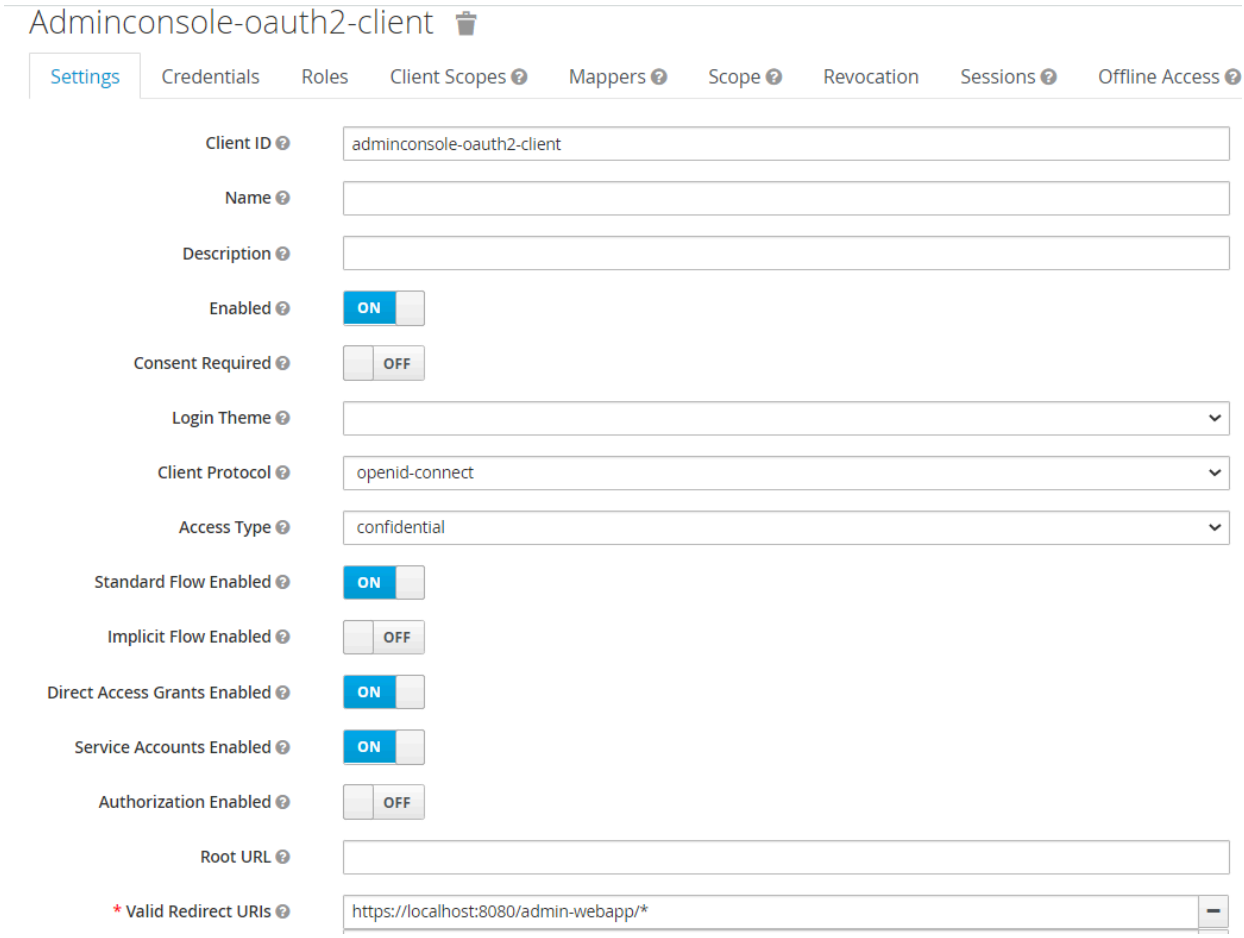
Example

- Redirect URI example for Admin Center - https://mydomain.com:8092/admin-webapp/*
- Redirect URI example for Reporting Hub - https://mydomain.com:8091/reporting-webapp/*
- Redirect URI example for Lineage - https://mydomain.com:8093/lineage-webapp/*
- Redirect URI example for Processing Engine - https://mydomain.com:8094/processing-engine/*
- Redirect URI example for Risk Engine - https://mydomain.com:17443/onesumx-risk/*

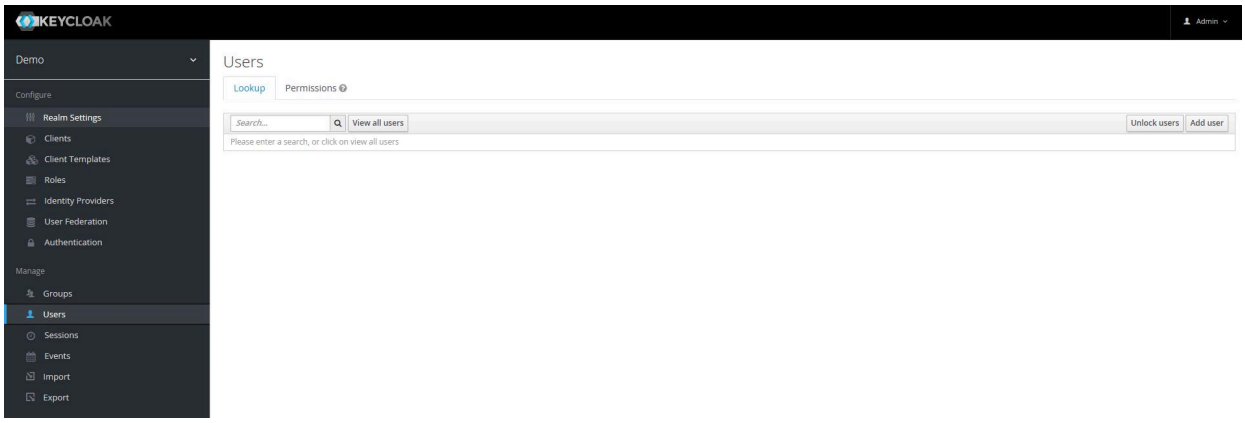
9. Click **Save**.

Example

The following image shows the final client configuration.



10. Add the admin account to Keycloak with the admin console.
11. On the left menu of the Keycloak admin console, click **Users**, and then, on the **Users** page, click **Add user**.



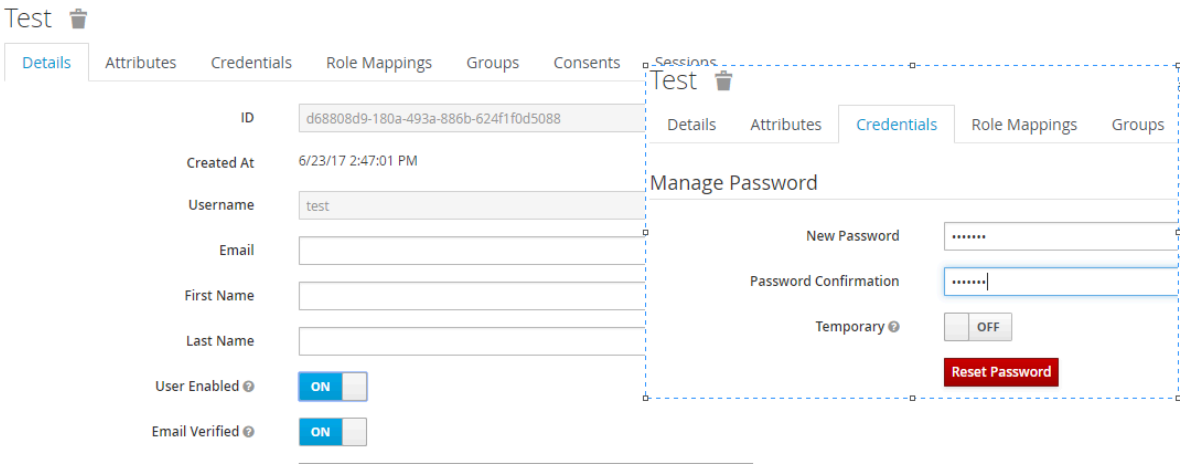
12. On the user's details page, in the **Credentials** tab, set a password.

    Make sure that the **Temporary** toggle is turned off.

13. In the **Details** tab, set the user name to have the same value as the one used for the ONESUMX_RE_DEFAULT_ADMIN and/or ONESUMX_RISK_DEFAULT_ADMIN properties configured in the *.env* file.

Note:

In case you have ONESUMX_RE_DEFAULT_ADMIN= admin and/or ONESUMX_RISK_DEFAULT_ADMIN= admin you need to create user admin in step 11.



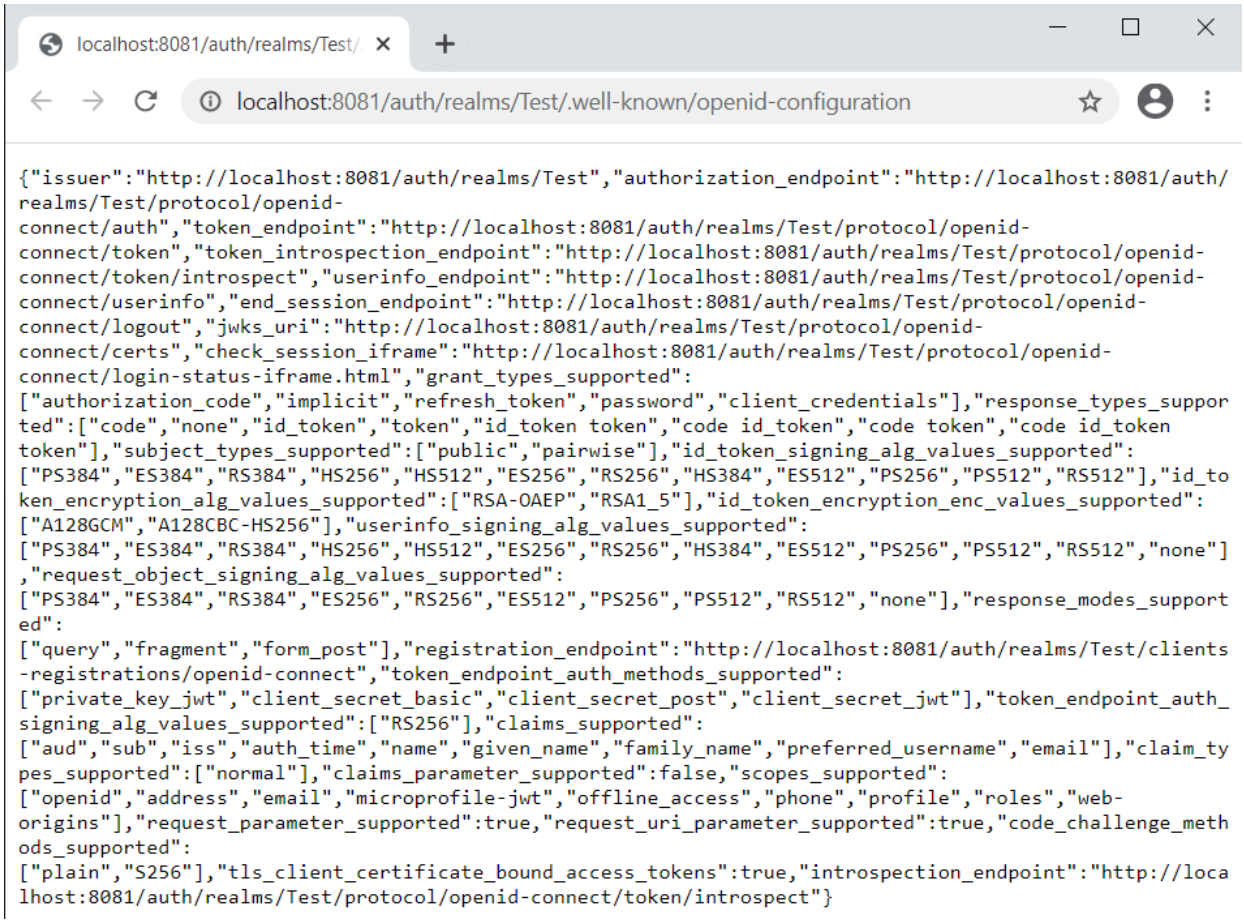
14. Go to http://<KEYCLOAK_HOST>:<KEYCLOAK_PORT>/auth/realms/<REALM>/.well-known/openid-configuration.

    All the OpenID Connect providers have a well-known endpoint to gather the required URI.

Results

The JSON configuration output opens in a separate tab.

Figure 1. Example of an OpenID Endpoint Configuration JSON output




```
{"issuer":"http://localhost:8081/auth/realms/Test","authorization_endpoint":"http://localhost:8081/auth/
realms/Test/protocol/openid-
connect/auth","token_endpoint":"http://localhost:8081/auth/realms/Test/protocol/openid-
connect/token","token_introspection_endpoint":"http://localhost:8081/auth/realms/Test/protocol/openid-
connect/token/introspect","userinfo_endpoint":"http://localhost:8081/auth/realms/Test/protocol/openid-
connect/userinfo","end_session_endpoint":"http://localhost:8081/auth/realms/Test/protocol/openid-
connect/logout","jwks_uri":"http://localhost:8081/auth/realms/Test/protocol/openid-
connect/certs","check_session_iframe":"http://localhost:8081/auth/realms/Test/protocol/openid-
connect/login-status-iframe.html","grant_types_supported":
["authorization_code","implicit","refresh_token","password","client_credentials"],"response_types_suppor
ted":["code","none","id_token","token","id_token token","code id_token","code token","code id_token
token"],"subject_types_supported":["public","pairwise"],"id_token_signing_alg_values_supported":
["PS384","ES384","RS384","HS256","HS512","ES256","RS256","HS384","ES512","PS256","PS512","RS512"],"id_to
ken_encryption_alg_values_supported":["RSA-OAEP","RSA1_5"],"id_token_encryption_enc_values_supported":
["A128GCM","A128CBC-HS256"],"userinfo_signing_alg_values_supported":
["PS384","ES384","RS384","HS256","HS512","ES256","RS256","HS384","ES512","PS256","PS512","RS512","none"]
,"request_object_signing_alg_values_supported":
["PS384","ES384","RS384","ES256","RS256","ES512","PS256","PS512","RS512","none"],"response_modes_support
ed":
["query","fragment","form_post"],"registration_endpoint":"http://localhost:8081/auth/realms/Test/clients
-registrations/openid-connect","token_endpoint_auth_methods_supported":
["private_key_jwt","client_secret_basic","client_secret_post","client_secret_jwt"],"token_endpoint_auth_
signing_alg_values_supported":["RS256"],"claims_supported":
["aud","sub","iss","auth_time","name","given_name","family_name","preferred_username","email"],"claim_ty
pes_supported":["normal"],"claims_parameter_supported":false,"scopes_supported":
["openid","address","email","microprofile-jwt","offline_access","phone","profile","roles","web-
origins"],"request_parameter_supported":true,"request_uri_parameter_supported":true,"code_challenge_meth
ods_supported":
["plain","S256"],"tls_client_certificate_bound_access_tokens":true,"introspection_endpoint":"http://loca
lhost:8081/auth/realms/Test/protocol/openid-connect/token/introspect"}
```


15. To set the correct values for all the properties listed in topic OpenID Connect settings, copy the values of the properties from the JSON output into the *.env* file.

7.5.2 Configuring OAuth2 authentication with Azure AD

Context

This procedure is mandatory if you have Azure ID as an identity provider. As a result, a new managed application will be registered in the Azure AD. You need to copy the values for all the required properties in the *.env* file.

Procedure

1. Log in to the Azure Portal and do the following.
   a. Check the endpoint details.
   b. Ask IT support/customer for details regarding the needed configuration described below.
2. Go to https://login.microsoftonline.com/<TENANT_ID>/v2.0/.well-known/openid-configuration.

   All the OpenID Connect providers have a well-known endpoint to gather the required URI.

Results

The URI list is displayed in the browser.

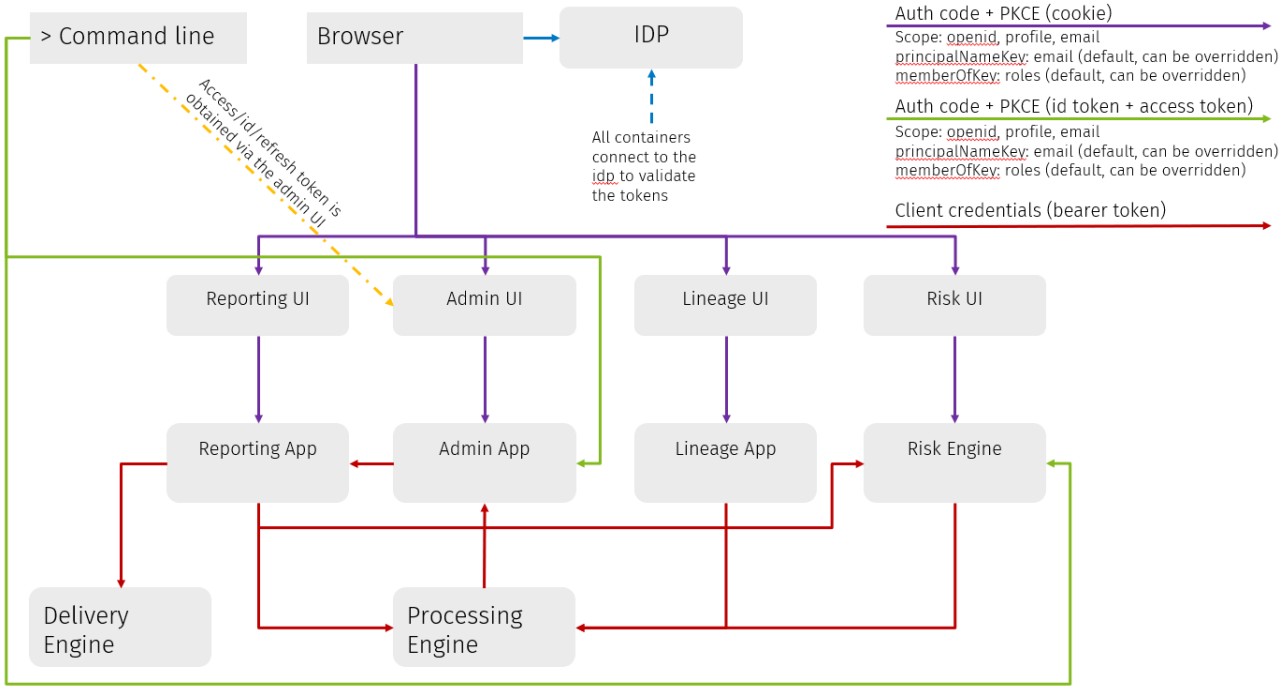
3. Set the correct values for all the properties listed in topic OpenID Connect settings.

7.6 OpenID Connect

The following schema illustrates the grant flows used by the Platform components.

- Client credentials are used for intercomponent communication.
- The auth code flow is used when a user interacts with the user interface of a component.

Figure 1. Platform grant flows



To set up a user authentication and single sign-on, a set of parameters must be provided to the Platform configuration files. The following parameters need to be collected from the OpenID Connect administrator.

Parameter	Description
tenant/realm	**The tenant identifier (a different terminology is used across the various identity providers; for example, realm is used in Keycloak)**
issuer	**The full issuer URL**
keyset uri	**The keyset URI for a client credentials type of connection**
token uri	**The token URI for a client credentials type of connection**
credentials scope	**The scope for a client credentials type of connection**

Parameter	Description
client id	**The client ID**
delivery api id	**The API ID (client scope) for the Delivery Engine authentication**
client secret	**The client secret**
client name	**The given client name**

The **UserInfo** endpoint of the Identity Provider server must be accessible for the OneSumX client application. The UserInfo endpoint is available for specific audiences depending on the chosen Identity Provider. For AD FS, the default audience entitled to access the **UserInfo** endpoint is *urn:microsoft:userinfo*. It must be set as an audience in the issued access token.

7.6.1 OpenID Connect settings

The following table summarizes platform configuration parameters for the identity providers. For more information and examples, see the *values.yaml* file.

Name	Description	Value
auth.oauth2LoginIssuerUri	**Identity provider Entity ID URL- set during installation**	• **For example, for Keycloak: https://<KEYCLOAK_HOST>:<KEYCLOAK_PORT>/auth/realms/<KEYCLOAK_REALM>** • **For example, for MS Entra ID: https://login.microsoftonline.com/<TENANT_ID>/v2.0** • **For example, for AD FS: https://<ADFS_HOST>/adfs**
auth.oauth2LoginIssuerUrl	**Identity provider authority URL - set during installation**	• **For example, for Keycloak: https://<KEYCLOAK_HOST>:<KEYCLOAK_PORT>/auth/realms/<KEYCLOAK_REALM>** • **For example, for MS Entra ID: https://login.microsoftonline.com/<TENANT_ID>** • **For example, for AD FS: https://<ADFS_HOST>/adfs**
auth.oauth2ClientCredentialsTokenUri	**Identity provider token Service URI - set during installation**	• **For example, for Keycloak: https://<KEYCLOAK_HOST>:<KE**

Name	Description	Value
		YCLOAK_PORT>/auth/realms/<KEYCLOAK_REALM>/protocol/openid-connect/token • For example, for MS Entra ID: https://login.microsoftonline.com/<TENANT_ID>/oauth2/v2.0/token • For example, for AD FS: https://<ADFS_HOST>/adfs/oauth2/token
auth.oauth2ClientCredentialsKeysetUri	Identity provider keys URI - set during installation	• For example, for Keycloak: https://<KEYCLOAK_HOST>:<KEYCLOAK_PORT>/auth/realms/<KEYCLOAK_REALM>/protocol/openid-connect/certs • For example, for MS Entra ID: https://login.microsoftonline.com/<TENANT_ID>/discovery/v2.0/keys • For example, for AD FS: https://<ADFS_HOST>/adfs/discovery/keys
auth.tokenIntrospectUri	Identity provider token Introspect URI .	• For example, for Keycloak: https://<KEYCLOAK_HOST>:<KEYCLOAK_PORT>/auth/realms/<KEYCLOAK_REALM>/protocol/openid-connect/token/introspect • For MS Entra ID leave it empty.
auth.oauth2ClientCredentialsScope	Required only for Azure OpenID Connect. It is based on the application registration name - set during installation	• For Example for MS Entra ID: api://<CLIENT_ID>/.default • For example, for Keycloak, leave it empty. • For example, for AD FS: openid The scope is used by the client to acquire the JWT token.
auth.oauth2DeliveryEngineAudience	Identity Providers App registration App ID - set during installation	• For example, for MS Entra ID: api://client-id-string • For example, for Keycloak:

Name	Description	Value
		client_scope_name • For example, for AD FS: "urn:microsoft:userinfo"
oauth2PrincipalNameKey	The key used to identify the user	• Copy the preferred name key from the claims_supported field in the Json output and paste it to this property. This represents how the user wants to log on. By default, users have to use their email address and password to log on.
oauth2DeliveryEngineAudience	The audience claim for the OneSumX Delivery Engine authentication	• For example, for MS Entra ID: api://client-id-string • For example, for Keycloak: client_scope_name • For example, for AD FS: "urn:microsoft:userinfo"
oauth2DeliveryEngineValidateIssuer	Enable or disable the issuer validation	Set it to false for AD FS.
oauth2DeliveryEngineValidateAudience	Enable or disable the audience validation	

8 Installing Cert Manager

Context

Install Cert-Manager on a Kubernetes cluster. You can use Helm, a package manager for Kubernetes.

Procedure

1. Add the Jetstack Helm repository, use the following command.

```
helm repo add jetstack https://charts.jetstack.io
```

2. Update your local Helm chart repository cache:

```
helm repo update
```

3. Install the CustomResourceDefinition resources:

```
kubectl apply -f https://github.com/jetstack/cert-manager/releases/download/
v1.6.1/cert-manager.crds.yaml
```

4. Create the namespace for cert-manager:

```
kubectl create namespace cert-manager
```

5. Install cert-manager in the cert-manager namespace.

Checkpoint:
Replace v1.6.1 with the version of cert-manager you want to install.

```
helm install cert-managerjetstack/cert-manager --namespace cert-manager --
version v1.6.1
```

9 Create custom storage class

Define a storage class in Kubernetes with ReadWriteMany (RWX) access mode.

Warning:
Not all storage provisioners support the ReadWriteMany access mode.

The default storage class is set to azurefile-csi. If you want to use any other custom storage class, override following parameters during the helm installation:

- admin_app.content_store_pvc.storage_class
- processing_engine.pe_storage_pvc.storage_class
- risk_engine.content_store_pvc.storage_class
- common_pvc.workspaces.storage_class
- common_pvc.logs.storage_class

10 Installing Platform on Kubernetes/OpenShift with Helm

Context

Tip:
For information about common errors related to Wolters Kluwer's Helm charts, see Troubleshooting.

The installation of Platform with Kubernetes/OpenShift consists of the following steps:

Procedure

1. Download the Helm chart.
2. Install the Helm chart.
3. Monitor the deployment status of the Helm chart.

10.1 Downloading Helm chart

Context

Run the following commands to download the Helm chart.

Procedure

1. Helm repo Login:
```
helm repo add wkfrr-helm https://wkfrr.wolterskluwer.io/artifactory/api/helm/wkfrr-helm --username <username> --password <TOKEN_from salesforce>
```
2. Helm repo update: `helm repo update`
3. Helm Repo Search: `helm search repo wkfrr-helm/platform --versions`
4. Display default values of Helm chart :
```
helm show values wkfrr-helm/platform --version xx.xx.xx
```
5. Helm repo download: `helm pull wkfrr-helm/platform --version xx.xx.xx`

10.1.1 Helm chart distribution structure

The Platform is composed of Regulatory and Risk components developed by Wolters Kluwer to serve Financial Risk Management and Regulatory Reporting.

The default configuration is set to assume an Azure Deployment, which utilizes various native components of Azure. These include:

- The Azure Application Gateway (AGIC), which is used for tasks such as proxying, reverse-proxying, TLS-termination, and functioning as a web application firewall. If you don't want to use the AGIC ingress, you can disable it by setting ingress.enabled: false. However, this means you'll need to find a way to expose the

services like reporting-app-service, admin-app-service, lineage-app-service, and optionally risk-engine-service to the outside world.

- Azure SQL Server is used as the default database. However, it's important to note that there's no strict dependency on Azure SQL. You can use any SQL server or even an Oracle database.
- Azure Key Vault is used for storing secrets. If you prefer not to use Azure Key Vault, you can disable it by setting useAzureVault: false. As an alternative, you can use Hashicorp vault by setting useHashicorpVault: true.

In case it doesn't run in Azure, make sure you have the equivalent features (for example, HashiCorp Vault instead of Azure Key Vault).

This chart bootstraps Regulatory Engine and Risk Engine deployment on a Kubernetes cluster using the Helm package manager.

The Platform is written in Java and implements the Regulatory Engine and Risk Engine. By default, certificates for inter-component communication are generated based on a root certificate provided as a secret.

Extracting the content of the Helm chart distribution file creates the following directory structure. Only the most important files and folders are listed here.

onesumx-re

Folder containing the LIQREP solution

templates

Main folder for component manifest YML files that can leverage the values in *Values.yaml*.

Chart.yaml

Contains the configuration settings for a Helm chart, including values such as package names, version, and release type.

Values.yaml

Used to store data specific to each environment. It defines parameters such as secrets or external services credentials needed by other charts installed on top of this Helm chart. The Charts directory provides a list of all available charts inside your repository.

10.2 Installing Helm chart

Procedure

Install the chart with the release name `my-release`.

```
helm -n <Namespace> upgrade <Release_Name> frr-helm/platform --version 24.02.07 --
set AKVConfig.keyvaultName=xxxxxxxxxxxx --set
AKVConfig.userAssignedIdentityID=xxxxxxxxxx --set AKVConfig.tenantId=xxxxxxxxxx --
set frontend_public_hostname=xxxx.com --set
auth.idpEntityId=https://login.microsoftonline.com/xxxxxxxxxxxx/v2.0 --set
auth.authority=https://login.microsoftonline.com/xxxxxxxxxxxxxx --set
auth.tokenServiceUri=https://login.microsoftonline.com/xxxxxxxxxxxxx/oauth2/v2.0/
token --install
```

These commands deploy reg and risk on the Kubernetes cluster in the default configuration. For an overview of

the parameters that can be configured during installation, see Parameters.

By default, the Processing Engine component is accessing the K8S API to retrieve the IP addresses of the Processing Engine pods. If the connection to the K8S API is limited or not desired, you can disable it by setting the components.processing_engine.kubernetesIpFinderEnabled parameter to false.

Checkpoint:

For Openshift, if you wish to use Routes, make ingressEnabled=false during the Helm installation.

Tip:

You can list all releases using the command `helm list`.

10.3 Monitoring deployment status of Helm chart

Procedure

1. Check the status of a Helm deployment using the `helm status` command. In this command, you can see the status of a release, including the deployment status of each resource in the release. The basic syntax for the command is:

   ```
   helm status [RELEASE_NAME] -n <namespace>
   ```

   #### Example

   In the the following command, you can see the status of the release, including the status of each release resource.

   ```
   helm status my-release
   ```

2. For more detailed information about the resources you can use the `kubectl` command to view the resources directly.

   #### Example

   In the following command, you can see the pods in the namespace.

   ```
   kubectl get pods -n <namespace>
   ```

Results

Once this chart is deployed, it is possible to access the applications using default user credentials via SSO.

- Reg application: `https://<wolterskluwer_application_host>/admin-webapp`
- Risk application: `https://<wolterskluwer_application_host>/onesumx-risk`

10.4 Parameters

Specify each parameter using the `--set key=value[,key=value]` argument with `helm install`.

```
helm install my-release \ --set RegEnabled=false,createschemarisk=false \
Platform.1.0.0.tgz
```

10.4.1 Global parameters

Name	Description	Value
wk_registry	Wolter Kluwer image registry	wkfrr.wolterskluwer.io/wkfrr
default_registry	Global Docker image registry	
appName	Global Application Name	platform
riskEnabled	make this flag true if risk app to be deployed	false
regEnabled	make this flag true if Reg app to be deployed	true
ingress.enabled	Flag to enable or disable the ingress	true
frontend_public_hostname	Hostname used to configure redirect between the application	xxx.xxx.com
version.installer	Image tag for installer	6.1.0
version.admin_app	Image tag for adminCenter	6.1.0
version.risk_engine	Image tag for risk	12.14.0.0
version.reporting_app	Image tag for reportingHub	6.1.0
version.lineage_app	Image tag for lineage	5.1.0
version.delivery_engine	Image tag for hermes	1.1.0

Name	Description	Value
version.busyBox	**Image tag for busyBox**	**1.35.0**
version.processingEngine	**Image tag for hermes**	**10.2.0**
securityContext.setReadOnlyFilesystem	**Setting this to true will force the container filesystem to be readonly. Set to true in production.**	**false**
securityContext.setUidAndGid	**Setting to true will add the following use configuration to the security context.** **Setting it to false will ignore the user UID and GID block.**	**false**

10.4.2 Azure related parameters

Note:
Applicable in case of Azure Kubernetes service deployment.

Name	Description	Value
AKVConfig.keyvaultName	**Azure keyvault name - Set during installation**	**XXXX**
AKVConfig.userAssignedIdentityID	**Azure Managed identity ID - Set during installation**	**xxxxxxxxxxxxx**
AKVConfig.tenantId	**Azure tenant ID -Set during installation**	**xxxxxxxxxxx**
AKVConfig.useVMManagedIdentity	**Azure useVMManagedIdentity**	**true**
AKVConfig.useServicePrincipalMode	**User servicePrincipalMode**	**false**
AKVConfig.servicePrincipalSecretName	**Set servicePrincipal if useServicePrincipalMode true**	

10.4.3 Configuration parameters reg and risk

Name	Description	Value
frontend_public_hostname	**Application Host name - Set during installation**	**xyz.com**
auth.defaultUser	**Default Admin user Email - Set during installation**	**customer@wk.com**
auth.defaultDisplayedName	**Default Admin user Name - Set during installation**	**false**
volumeMounts.storage.enabled	**Enable the option of sending certain exported files from Reporting Hub and Admin Center to volume storage.**	**false**
volumeMounts.storage.root	***Note:*** **Property volumeMounts.storage.enabled should be set totrue** **Specifies the main path inside the volume where users will be able to upload certain exported files from Reporting Hub and Admin Center to volume storage.**	**/tmp/usr/storage**
volumeMounts.storage.path	***Note:*** **Property volumeMounts.storage.enabled should be set totrue** **Specifies the sub path inside the volume where users will be able to upload certain exported files from Reporting Hub and Admin Center to volume storage.**	**/onesumx-re-file-store/content**

10.5 Persistence

The Wolters Kluwer reg and risk image stores the reg and risk data and configurations at the */config* path of the container.

Persistent volume mount point permissions

As the image runs as non-root by default, you need to adjust the ownership of the persistent volume so that the container can write data into it.

By default, the chart is configured to use Kubernetes Security Context to automatically change the ownership of the volume. However, this feature does not work in all Kubernetes distributions. As an alternative, this chart supports using an init container to change the ownership of the volume before mounting it to the final destination.

Additional containers

If additional containers are needed in the same pod (such as additional metrics or logging exporters), they can be defined using the `sidecars` config parameter. Similarly, extra init containers can be added using the initContainers parameter.

Pod affinity

This chart allows you to set custom Pod affinity using the `affinity` parameter. For more details about Pod affinity, See Kubernetes documentation.

10.6 Secrets

Context

Provide the docker container registry credentials.

Procedure

1. Deploy the secret in the namespace. Use the following command to deploy the chart.

```
kubectl -n [namespace]  create secret docker-registry docker-configuration--
docker-server=<DOCKER_REGISTRY_SERVER> \  --docker-username=<DOCKER_USER> \  --
docker-password=<DOCKER_PASSWORD> \  --docker-email=<DOCKER_EMAIL
```

2. Provide the secrets for the open ID connect and the database connection. Use the AzureKeyVault or Hashicorp vault. Set the one you need to *true* and leave the other one *false*. If you do not want either of them, set them to *false* and see the next paragraph to create the secrets.

```
# Use Azure Key Vault or Hashi Corp Vault # If no Vault is selected, then
secrets must be loaded directly as kubernetes secrets. # If useAzureVault is
set to true, refer to the AKVConfig section to configure the vault # refer to
section AKVConfig below

useAzureVault: true

# If useHashicorpVault is set to true, please refer to the AKVConfig section to
configure the vault # please refer to section HCVConfig below

useHashicorpVault: false
```

3. If you choose not to use a vault, you need to provide the secrets directly into the cluster. The secret must be loaded in the namespace, the application is deployed in the following command.

```
kubectl -n [namespace]  create secret installer-configuration --from-
file=installer-configuration.yaml
```

4. Run the installer-configuration.yaml as in the following example.

```
apiVersion: v1
kind: Secret
data:
  # base64 encoding of the client id for admin app
  admin-auth-token-client-id: XXXXXXXXX
  # base64 encoding of the secret for admin app
  admin-auth-token-secret: XXXXXXXXX
  # base64 encoding of the client id for delivery engine
  delivery-engine-auth-token-client-id: XXXXXXXXX
  # base64 encoding of the secret for delivery engine
  delivery-engine-auth-token-secret: XXXXXXXXXXXXXXXXX
  # base64 encoding of the client id for delivery engine
  hub-auth-token-client-id: XXXXXXXXX
  # base64 encoding of the secret for delivery engine
  hub-auth-token-secret: XXXXXXXXX
  # base64 encoding of the client id for lineage app
  lineage-auth-token-client-id: XXXXXXXXX
  # base64 encoding of the secret for lineage app
  lineage-auth-token-secret: XXXXXXXXX
  # base64 encoding of the client id for processing engine
  processing-auth-token-client-id: XXXXXXXXX
  # base64 encoding of the secret for processing engine
  processing-auth-token-secret: XXXXXXXXX
  # base64 encoding of the client id for risk engine
  risk-auth-token-client-id: XXXXXXXXX
  # base64 encoding of the secret for risk engine
  risk-auth-token-secret: XXXXXXXXX
  # SQL Server Admin Username
  database-admin-username: XXXXXXXXX
  # SQL Server Admin Password
  database-admin-password: XXXXXXXXX
  # Password for application level credentials
  database-application-username: XXXXXXXXX
```

10.7 Managing certificates

You can decide to either leverage the cert-manager or to provide the certificates directly. If you choose to leverage the cert-manager, install the CRDS:

```
kubectl apply -f https://github.com/cert-manager/cert-manager/releases/download/
v1.13.3/cert-manager.crds.yaml
```

You can either enable or disable the cert-manager by set enabled to either true or false. It is possible to choose between a self signed certificate or to provision a certificate authority keypair as a secret.

```
certManager:# Enabling cert managers. enabled: true# A key pair for the certificate
authority issuer ca_key_pair_secretName: ca_keypair # If selfSigned is set to true
then the ca_key_pair_secretName is ignored selfSigned: true# version: "v1.13.3"#
namespace: cert-manager# installCRDs: true
```

If cert manager is disabled, certificates and keystores need to be loaded in a secret called onesumx-platform-tls and it must contain the following data elements.

```
# truststore.jks: a java trustore in jks format# truststore.p12: a java trustore in
p12 format# ca.crt: a certificate for the certificate authority# keystore.jks: a
java keystore in jks format. This one contains the key for encryption# keystore.p12:
a java keystore in p12 format# tls.crt: a certificate in crt format# tls.key: a
private key in pem format
```

If you're not using the certificate manager, you can utilize the utility script provided with the Helm chart.

First, download and extract the Helm chart. Next, navigate to the tools folder and execute the following script, replacing <keystore-password>and <truststore-password> with your actual passwords:

```
./k8s-onesumx-cert-tool.sh -k <keystore-password> -p <truststore-password>
```

The certificates will be generated in the current directory of the script.

The k8s-onesumx-cert-tool.sh accepts other additional attributes, to handle additional truststore, CA certificate and additional SAN attributes. To view the supported options of the script, execute ./k8s-onesumx-cert-tool.sh -h:

```
-------------------------------------------------------
k8s-onesumx-cert-tool.sh: start
-------------------------------------------------------
Usage: k8s-onesumx-cert-tool.sh [OPTIONS]
  -h | --help                  Display help and exit
  -k | --keystore-password     Password for the server's keystore
  -t | --truststore            (Optional) Path to the truststore in JKS format
containing trusted certificates
  -p | --truststore-password   Password for the truststore
  -c | --ca-cert               (Optional) Path to the CA certificate
  -kca | --ca-key              (Optional) Path to the CA private key
  -r | --root-certificates     (Optional) Path to root certificates, defaults to
JAVA_HOME/lib/security/cacerts
  -s | --san-attributes        (Optional) Add custom SAN attributes, comma
separated, e.g., dns:myhost1,dns:myhost2
```

Once the certificates are generated, a secret named onesumx-platform-tls should be created from the certificated files:

```
kubectl -n <NAMESPACE> create secret generic onesumx-platform-tls --from-
file=tls.crt=./tls.crt --from-file=tls.key=./tls.key --from-file=ca.crt=./ca.crt
--from-file=keystore.jks=./keystore.jks --from-
file=truststore.jks=./truststore.jks --from-file=truststore.p12=./truststore.p12
--from-file=keystore.p12=./keystore.p12
```

10.8 Openshift routes

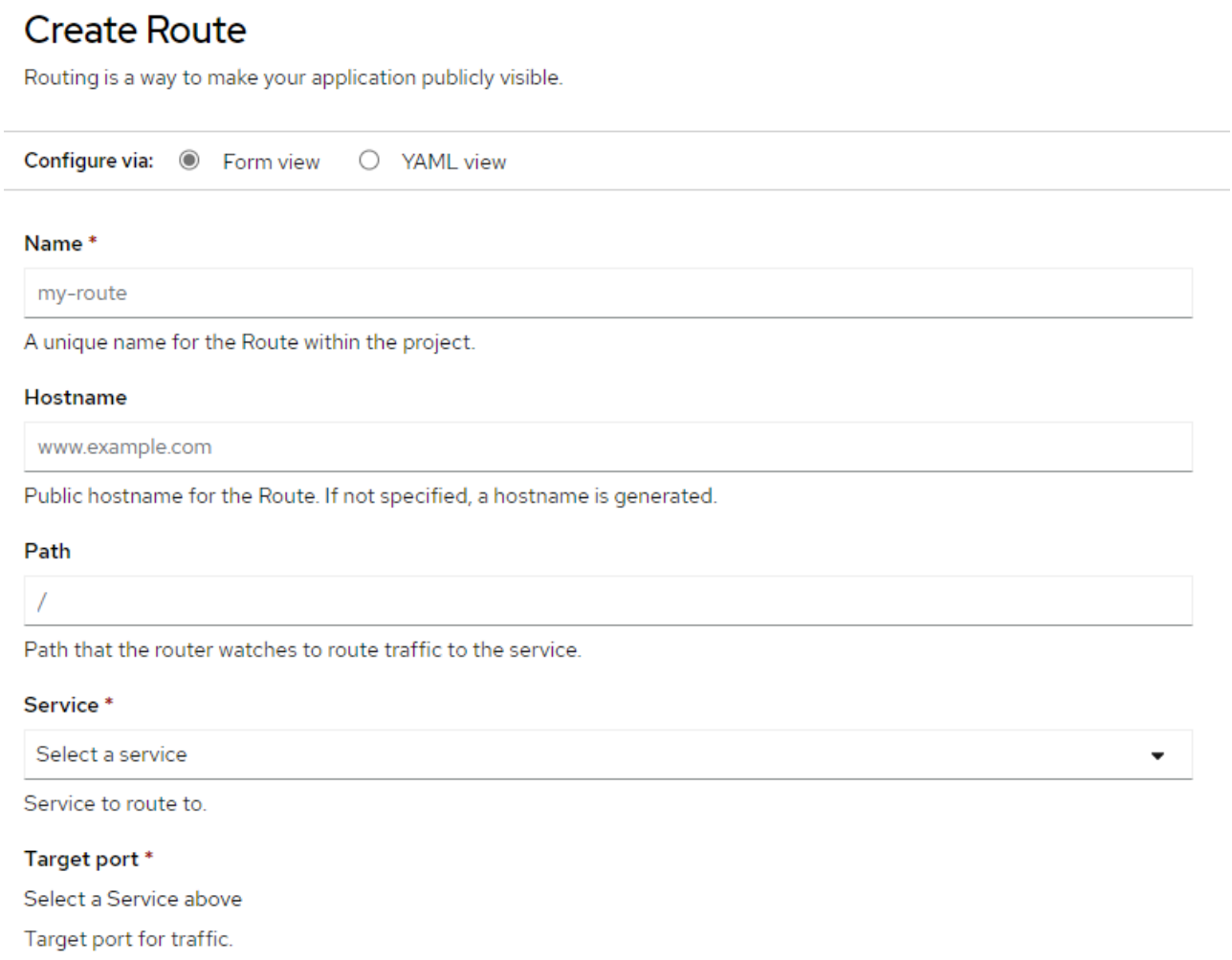
Routes are Openshift-specific implementations of exposing your service to the external world. You need to create 4 routes for your application.

Name	Status	Location	Service
admin-app-route	Accepted	https://os1.demo-eap-mercury.com/admin-webapp	admin-app-service
lineage-route	Accepted	https://os1.demo-eap-mercury.com/lineage	lineage-app-service
onesumx-risk-route	Accepted	https://os1.demo-eap-mercury.com/onesumx-risk	risk-engine-service
reporting-webapp-route	Accepted	https://os1.demo-eap-mercury.com/reporting-webapp	reporting-app-service

For each route, provide the following information:

1. Name
2. Hostname

3. Path -Application context root
4. Select the desired service from the drop-down
5. Target port - use HTTP port mapping
6. Select the check box of secure route
7. TSL termination Edge
8. Insecure traffic - redirect
9. Cert for the hostname
10. Private key for the hostname



Security

☑ Secure Route

Routes can be secured using several TLS termination types for serving certificates.

TLS termination *

Edge

Insecure traffic

Redirect

Policy for traffic on insecure schemes like HTTP.

Certificates

TLS certificates for edge and re-encrypt termination. If not specified, the router's default certificate is used.

Certificate

Browse...

The PEM format certificate. Upload file by dragging & dropping, selecting it, or pasting from the clipboard.

Private key

Browse...

The PEM format key. Upload file by dragging & dropping, selecting it, or pasting from the clipboard.

CA certificate

Browse...

The PEM format CA certificate chain. Upload file by dragging & dropping, selecting it, or pasting from the clipboard.

Create Cancel

Sample Yaml to create the routes

```
kind: Route
apiVersion: route.openshift.io/v1
metadata:
  name: admin-app-route
  namespace: mercury-n
  labels:
    app: platform
    app.kubernetes.io/managed-by: Helm
    name: admin-app-service
spec:
  host: xx.demo-eap-mercury.com
  path: /admin-webapp
  to:
    kind: Service
    name: admin-app-service
    weight: 100
  port:
    targetPort: http
  tls:
    termination: edge
    certificate: "xxxxxxxxxx"
    insecureEdgeTerminationPolicy: Redirect
  wildcardPolicy: Non
```

10.9 Uninstalling chart

Procedure

Uninstall/delete the `my-release` deployment, use the following command.

```
helm delete my-release
```

11 Troubleshooting

This section offers solutions to common issues you might encounter during the installation or upgrade process.

11.1 Time synchronization

Issue:

The authentication process may fail if OneSumX Regulatory Engine and the Identity Provider server are running on different computers and the time setting of the respective computers differs by more than 60 seconds.

This issue could result in the error message `Response issue time is either too old or with date in the future` in the log of OneSumX Regulatory Engine after a failed authentication.

Solution:

To avoid this issue, the time settings of the computers must be synchronized.

11.2 Certificates

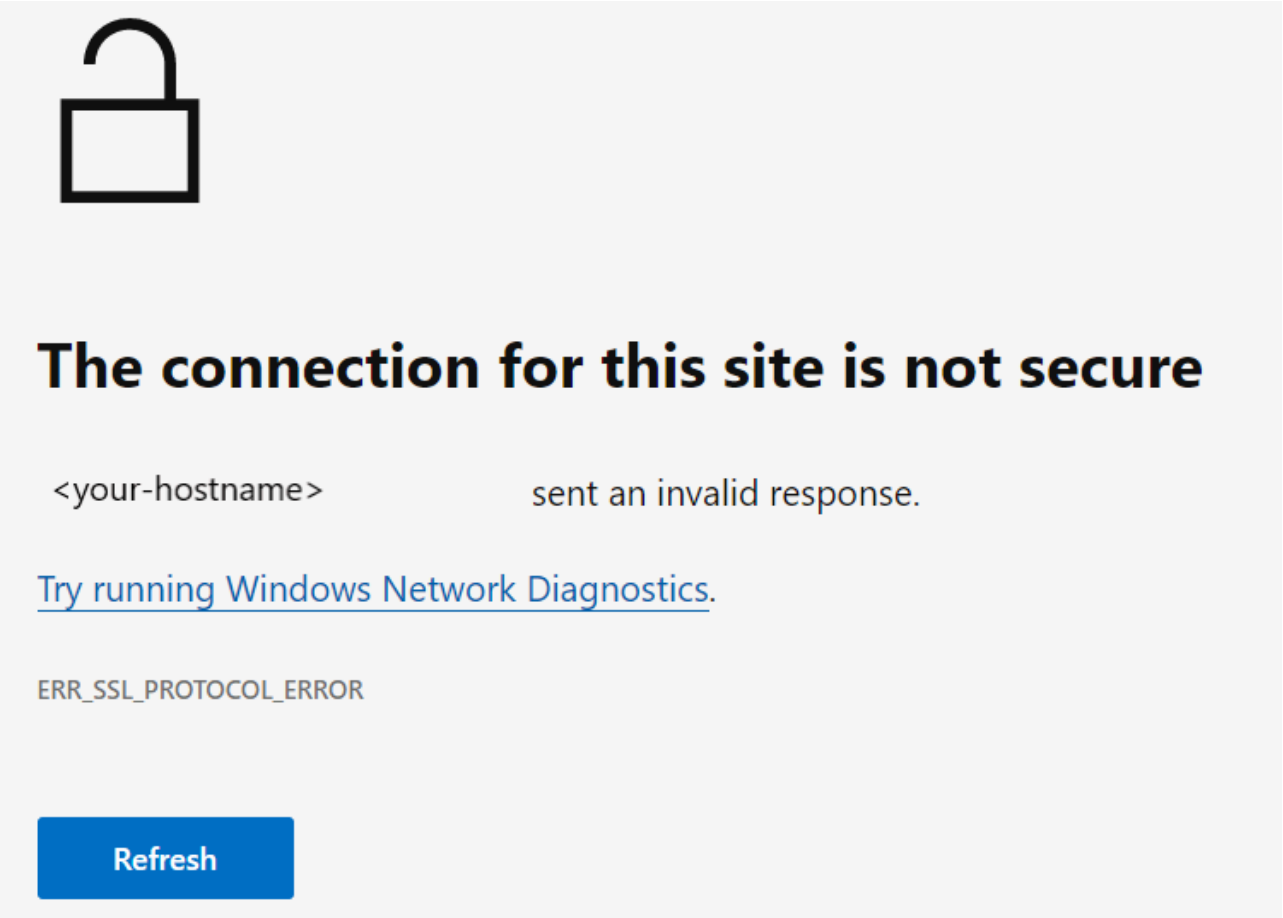
Issue:

The installation fails at the first health checkpoint with Admin Center, and the log output shows the following information:

```
onesumx-re-admincenter | 2023-04-26 07:14:07,188 INFO [o.j.as] (Controller Boot
Thread) WFLYSRV0025: WildFly EE 25.0.0.Final (WildFly Core 17.0.1.Final) started
in 22692ms - Started 375 of 443 services (133 services are lazy, passive or on-
demand) dependency failed to start: container onesumx-re-admincenter is unhealthy
```

The ONESUMX_KEYSTORE_ALIAS setting is incorrect in the *.env* file: it does not match the value set in the certificate.

If you try to access the application via a browser, you will receive the following message:



Solution:

Configure the certificate settings with the correct values. For details, see HTTPS and certificates settings.

11.3 Access rights

Issue:

When you try to perform the Docker Compose up command, the following error is displayed:

```
stat .: permission denied
```

Solution:

Set the correct permissions for the *onesumx-platform* folder and files:

```
chmod -R a+rw onesumx-platform-compose
```

For details, see Setting up access rights using alternate user belonging to onesumx group.

11.4 Activate full debug logs

To activate debug logs, add the following lines into the file located under the main extracted folder, at `config/onesumx-re/sharedConfig/datasources.cli`:

```
/subsystem=logging/root-logger=ROOT:change-root-log-level(level=DEBUG)
/subsystem=logging/root-logger=ROOT:write-attribute(name=level,value=DEBUG)
/subsystem=logging/console-handler=CONSOLE:write-attribute(name=level,value=DEBUG)
```

11.5 Not found error after login

In case the application displays a single-line error message `Not Found` after login, cross-check the OpenID Connect credentials, settings, parameters, and configuration. This typically happens when the authentication layer is not properly configured.

Note:
Ensure that Client ID and Client Secret are properly configured in the environment settings and restart the system.

11.6 Cannot connect to Admin Center after successful login

When you first connect to the Admin Center, despite having been successfully authenticated on the Identity Provider, the `User does not exist` message appears in the application.

The default admin setting (`ONESUMX_RE_DEFAULT_ADMIN`) in the environment file is very important and case-sensitive. It must precisely match the user name and casing that is returned by the Identity Provider, otherwise the user will not be recognized.

In case the setting is wrong, you will get the message that the `user does not exist`. You will need to do the following:

- Shut down the system
- Fix the user in the environment file
- Delete the RegEngine system database schema entirely
- Re-run the `create-re-db` Docker profiles to recreate it with the correct user name.

Once the database is recreated, you should be able to seamlessly log in. Confer to your OpenID Connect administrator in case of doubt on the exact name returned by the Identity Provider.

11.7 Delivery file does not get generated

In case the delivery file is not generated properly, increase the generation timeout to a greater value. For this purpose, open the environment configuration file, locate the `ONESUMX_DE_DATA_RX_TIMEOUT` parameter and set it to a greater value. The timeout value is in seconds - typically, increase the value to 3600 for a one-hour timeout.

11.8 WAL errors on Processing Engine restart

In case the Processing Engine (PE) crashes or is not stopped properly, it may lead to WAL errors preventing the system from restarting. The issue might appear because the PE is not properly stopped. According to this page https://stackoverflow.com/questions/66686985/apache-ignite-reached-logical-end-of-the-segment-for-file. The line of code should be called when PE is stopped: *ignite.cluster().state(INACTIVE)*.

Note:

If you receive these errors you need to start PE without the WAL mode (if no WAL mode is present, the mode **NONE** is the default one) or with the WAL mode set to **NONE**, stop it, and restart it, in any mode.

To restart PE, do the following:

1. Go to the **Ignite Home** (if it is not set the path is similar to this: *C:\Users\<UserName>\AppData\Local\Temp\processing-engine\ignite)* and then go to the *\work\db* folder.

2. **Delete** the WAL + the node folders which are created.

Tip:

All the folders whose names start with node.

12 LIQREP solution Installation procedure

Following are the high-level steps to ensure you have completed the Platform installation process:

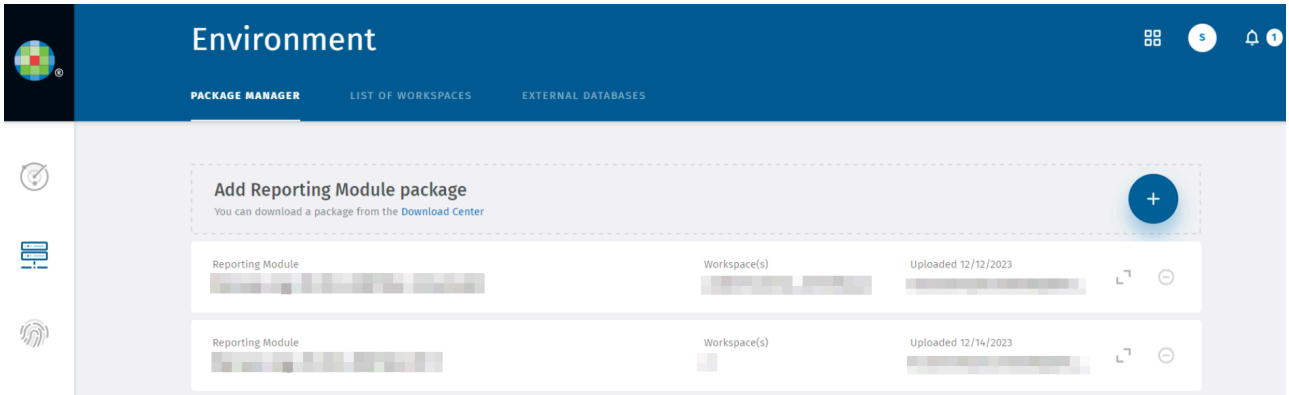
1. Download and extract the distribution ZIP file.

   Note:
   It contains the following folders:

   - onesumx-re- This folder contains the LIQREP RE Reporting Module (*liqrep-distribution-dist-21.2.0.zip*)
   - onesumx-risk- This folder contains the Risk Model (*risk-liquidity-prod-model-4.1.0.zip*)

2. Configure the properties.
3. Install and configure an Identity Provider and applications for SAML authentication.
4. Generate all required security resources for the application.
5. Run the installation process.
   a. Fresh installation or upgrade, as applicable.
6. Launch the application.

After you have completed the Platform installation process, for the LIQREP solution package installation process, you must upload LIQREP RE Reporting Module to the Admin Center of the Platform:



Note:

1. To upload, click .
2. On the **Upload Reporting Module** pop-up window, you can either drag and drop the LIQREP RE Reporting Module package to be installed or browse for the package and then click **Upload** .

Postrequisites

After installing LIQREP solution, you must create the Workspace on Platform.

13 Managing workspaces

A workspace is a OneSumX Regulatory Engine environment resource, a business database where you can create and store business objects.

When you create specific objects in the Reporting Hub application, you must specify the workspace where they will be stored. Therefore, workspaces play a key role in the Reporting Hub application.

You can perform the following actions on workspaces:

- Creating workspaces - to add workspaces to OneSumX Regulatory Engine.
- Viewing workspace details - to view information such as database details, the reporting module installation date, upgrade possibilities, and adding other reporting modules.
- Modifying workspace details - to modify the workspace name and description.
- Activating workspaces - to activate a workspace following its creation soyou can use it.
- Deactivating workspaces - to render a workspace inactive so that it can't be used anymore in Reporting Hub.
- Upgrading reporting modules installed in workspace - to upgrade the reporting modules installed in a workspace.
- Discarding workspaces - to delete a workspace from the application.
- Enabling dual control- to enable dual control on a workspace.
- Enabling sign-off- to enable the possibility to sign off submissions.

13.1 Creating workspaces

Prerequisites

- The appropriate reporting module is added to the OneSumX Regulatory Engine environment. For details, see Adding reporting modules to environment.
- You are logged in to OneSumX Regulatory Engine, to Admin Center.

Context

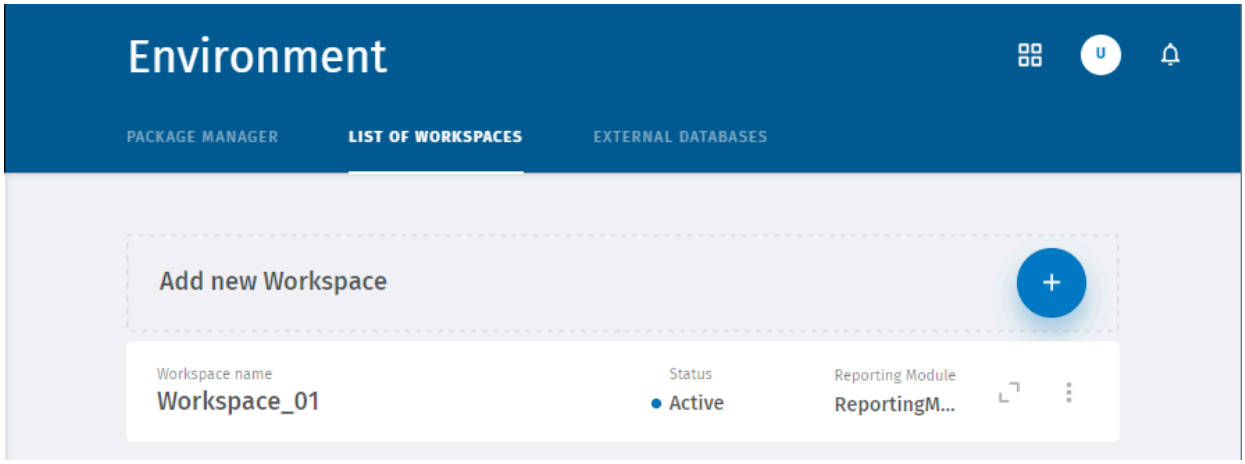
When you create a workspace, you need to take into account the details of your specific implementation: for example, the database options depend on the server type (Oracle/SQL Server). The workspace creation procedure will guide you through all these options.

Procedure

1. In the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.

Results

A list of all the created workspaces is displayed, in the form of cards, sorted by status and then by the workspace name, in ascending order.

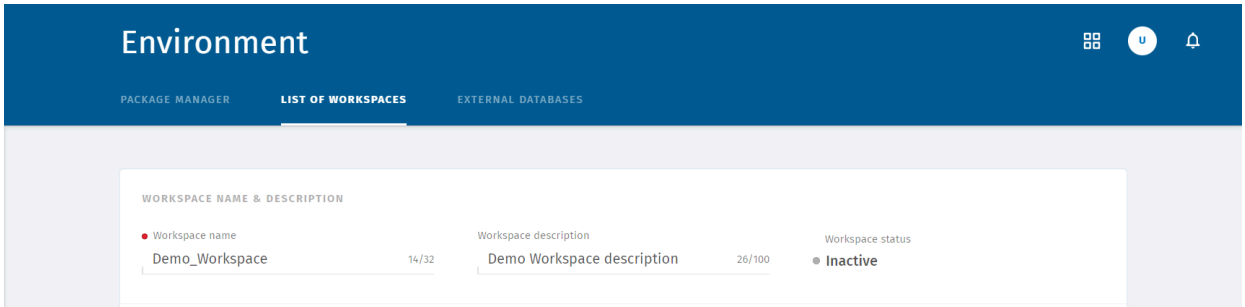


2. In the content pane, in the **Add new Workspace** area, click .

Results

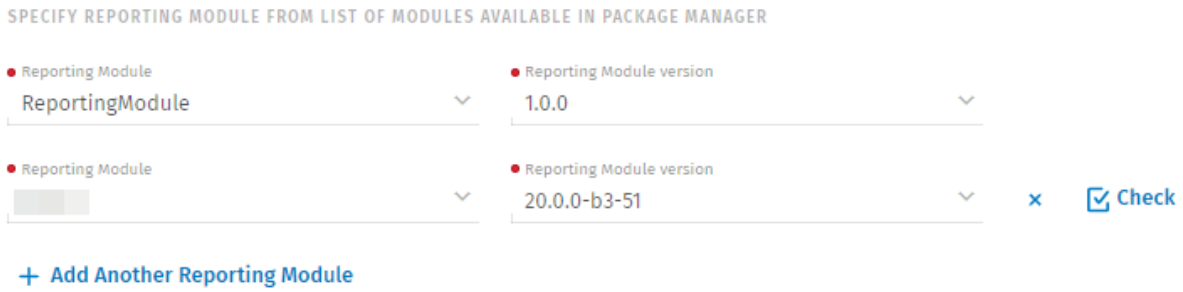
A workspace form is displayed. A red dot precedes the mandatory fields.

3. Under **WORKSPACE NAME & DESCRIPTION**, enter the details of the new workspace in the following fields:
   - **Workspace name** - Enter a name for the new workspace.
   - **Workspace description** - Enter the description of the workspace.



4. For **Database approach**, select **New DB**.
   Figure 1. Database approach.
   - **New DB** - workspace creation with Reporting Module installation
   - **Attach DB** - workspace creation by attaching an existing database with Reporting Module already installed
5. In the **Reporting Module** dropdown list, select the module you want to install in the workspace that you are creating.
6. In the **Reporting Module version** dropdown list, select the module version you want to install.
7. Optionally, add another reporting module and repeat steps 5 and 6.

Results



8. Under **DATABASE SERVER TYPE**, select **Oracle** or **SQL Server**, depending on your implementation.
9. In the **SPECIFY DATABASE DETAILS** area, under **Database Approach**, select one of the following options:
   - **RE will provision new schema/database** - Choose this option if you want to create a new database for this workspace.
   - **RE will use customer provisioned schema/database** - Choose this option if you want to use an existing empty database for this workspace.
10. Continue with one of the following actions:
    - If you selected **Oracle**, perform the steps described in section Creating workspaces for Oracle databases.
    - If you selected **SQL Server**, perform the steps described in section Creating workspaces for SQL databases.

13.1.1 Creating workspaces for Oracle databases

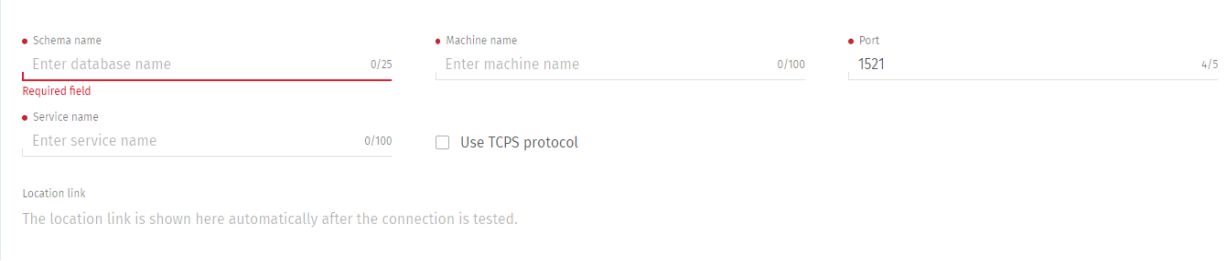
Prerequisites

- You performed the steps described in Creating workspaces.

Procedure

1. Under **SPECIFY DATABASE(S) DETAILS**, provide the following information:
   - **Schema name** - Enter a name for the schema that will be created for this workspace, otherwise enter a name of already existing empty schema.
   - **Machine name** - Enter the machine name on which the database server is stored.
   - **Port** - Enter the port number used to access the Oracle service.
   - **Service name** - Enter the service name that is configured on the database server.
   - **Use TCPS protocol** - Select this option in case an encrypted connection is required.

   Note:
   To benefit from an encrypted connection, the database server must be configured to accept incoming TLS connections. For TLS connection configuration, see the respective database server documentation.



2. Click **Proceed**.

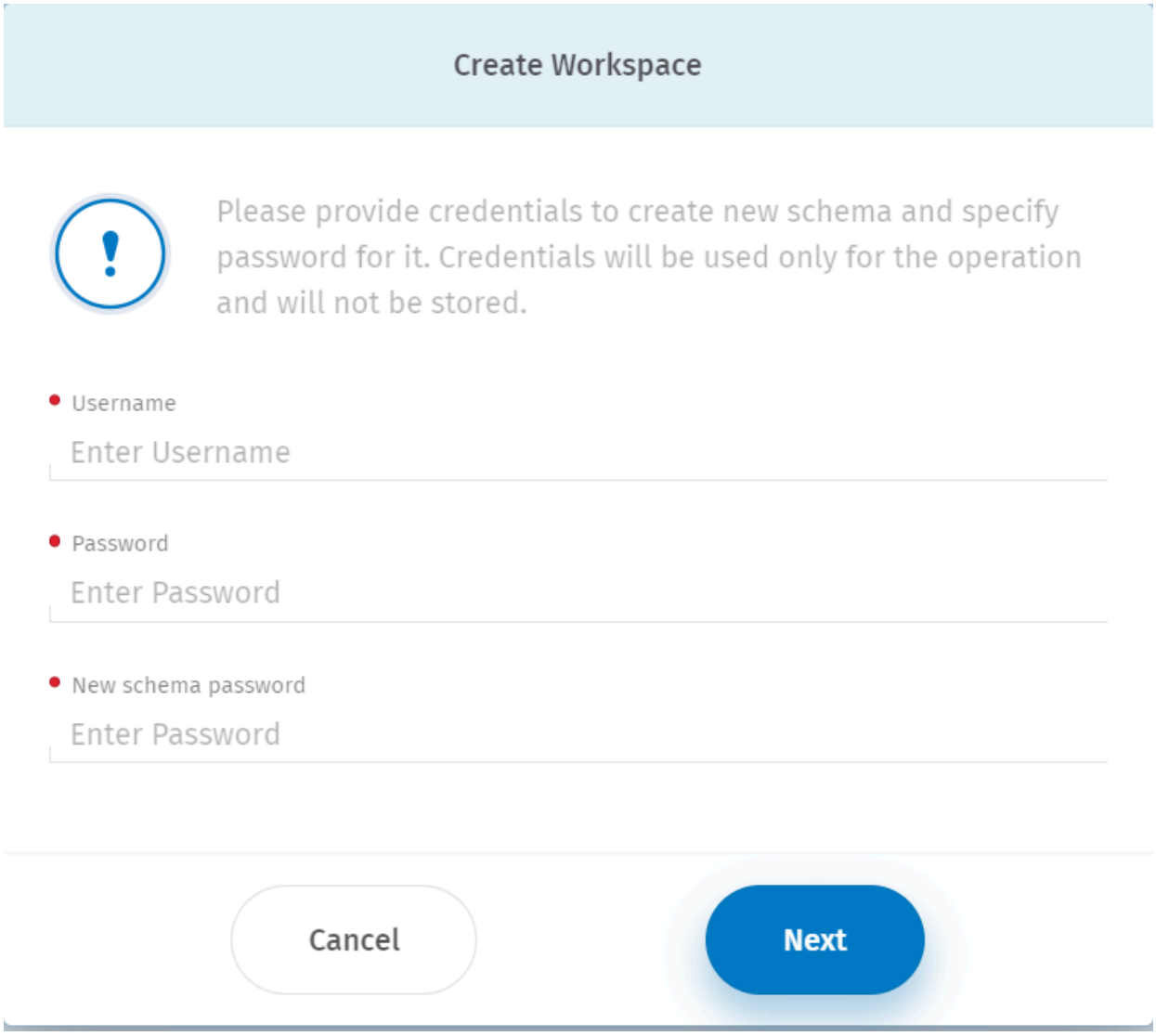
Results

Based on the selection made at step 9, one of the next options will be applicable:

- **RE will provision new schema/database** - A new database will be created by the engine.
- **RE will use customer provisioned schema/database** - An empty database must be created upfront by connecting to the database server with a user that has create database rights. The following example illustrates the instruction that needs to be executed.

```
CREATE USER NEW_SCHEMA IDENTIFIED BY newShemaPassword DEFAULT TABLESPACE users
QUOTA UNLIMITED ON users;
```

3. A dialog will appear, prompting you to enter **Username**, **Password**, and **New schema password**.



Type the user name and password for workspace installation (level 2 user credentials, granting the permission to alter the schema).

The list of required permissions varies depending on the user. You have the option to use the schema owner. Then, the username and password are not editable, but the schema password input will become editable and must be provided.

- Grants required for system user (default admin user who created the schema, for example, system)

Note:
These SQL Statements have to be executed with an external tool, which is not provided by Wolters Kluwer. Ensure you have sufficient user rights to execute these administrative SQL statements and use a compliant tool for your database type.

```
GRANT inherit privileges ON USER system to NEW_SCHEMA;
```

- Grants required for schema owner (for example, NEW_SCHEMA)

```
GRANT CREATE SESSION, ALTER SESSION, CREATE TABLE, CREATE PROCEDURE, CREATE
```

```
SYNONYM, CREATE SEQUENCE, CREATE VIEW, CREATE TYPE, CREATE TRIGGER, CREATE ROLE
TO NEW_SCHEMA;
```

- Grants required for custom level 2 user / alter user (for example, level2user)

```
CREATE USER level2user identified by 123level2_user DEFAULT TABLE SPACE users
QUOTA unlimited ON users;
grant create session to level2user with admin option;
grant alter session to level2user with admin option;
grant create synonym to level2user with admin option;
grant create role to level2user with admin option;
grant select_catalog_role to level2user with admin option;
grant drop user to level2user;
grant create user to  level2user;
grant alter user to level2user;
grant select any table to level2user;
grant create any table to level2user;
grant insert any table to level2user;
grant update any table to level2user;
grant delete any table to level2user;
grant alter any table to level2user;
grant drop any table to level2user;
grant alter any role to level2user;
grant drop any role to level2user;
grant grant any role to level2user;
grant grant any object privilege to level2user;
grant create any index to level2user;
grant create any sequence to level2user;
grant select any sequence to level2user;
grant drop any sequence to level2user;
grant create any trigger to level2user;
grant create any procedure to level2user;
grant execute any procedure to level2user;
grant drop any procedure to level2user;
grant create any type to level2user;
grant create any view to level2user;
```

Results

If all the details that you provided are correct, the connections are established successfully, and a connection successful message is displayed. The workspace creation process starts, as well as the installation of the reporting module, which starts automatically.

Note:

The installation process could last between several minutes and several hours, depending on multiple factors, including the complexity of the reporting module. When the process is finished, you are notified through a message that appears in the **top bar options** > **Notifications**. For details on viewing notifications, see Viewing notifications.

To monitor the status of the workspace creation process, you have the following options:

- In the main menu, click **Environment** and then click the **LIST OF WORKSPACES** submenu. Look for the card corresponding to the workspace you just created and check the status: a progress bar is displayed if the workspace is still being created.

- In the main menu, click **Monitoring** and then click the **RUNNING JOBS** submenu for ongoing processes or the **HISTORY** submenu for finished processes. For details on monitoring processes in OneSumX Regulatory Engine, see Monitoring.

Warning:

- If you encounter an error or failure during the workspace creation, we strongly recommend discarding the workspace and trying again from the start, as there are some known issues. For details, see the Release Notes document.
- We recommend creating workspaces one by one (not to trigger two or more simultaneous reporting module installations), especially in the case of a shared GDM.
- If the workspace creation fails, you will not be able to create another workspace with the same database names, even if you delete the failed workspace. Check the logs for more details. You need to drop the created database, otherwise, the following error will be displayed: `Name should be unique across all database servers`.
- If you are working with Docker and using the workspaces, you must restart OneSumX Regulatory Engine for the changes to be visible in the application. For example, you must restart OneSumX Regulatory Engine after you create a workspace, otherwise, you may encounter errors in the application.

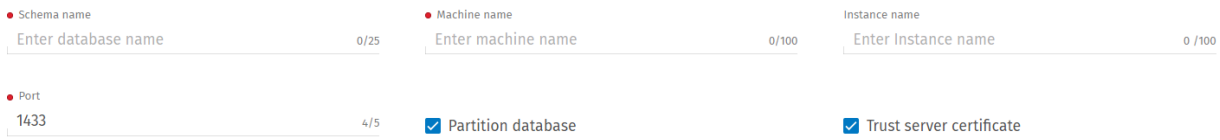
13.1.2 Creating workspaces for SQL databases

Context

- You performed the steps described in Creating workspaces.

Procedure

1. Under **SPECIFY DATABASE(S) DETAILS**, provide the following information:
   - **Schema name** - Enter a name for the database that will be created for this workspace, or enter the name of an already existing empty database.
   - **Machine name** - Enter the database server name on which this workspace will be stored.
   - **Instance name** - Enter the name of the instance on which this workspace database will be stored. If you are using an SQL database without a default instance, leave this field empty.
   - **Port** - Enter the port number used to connect to the workspace database. The default value is `1433`, which is the default port for SQL server.
   - **Partition Database** - Enable this option if the database associated to this workspace should be partitioned.



2. Click **Proceed**.

Results

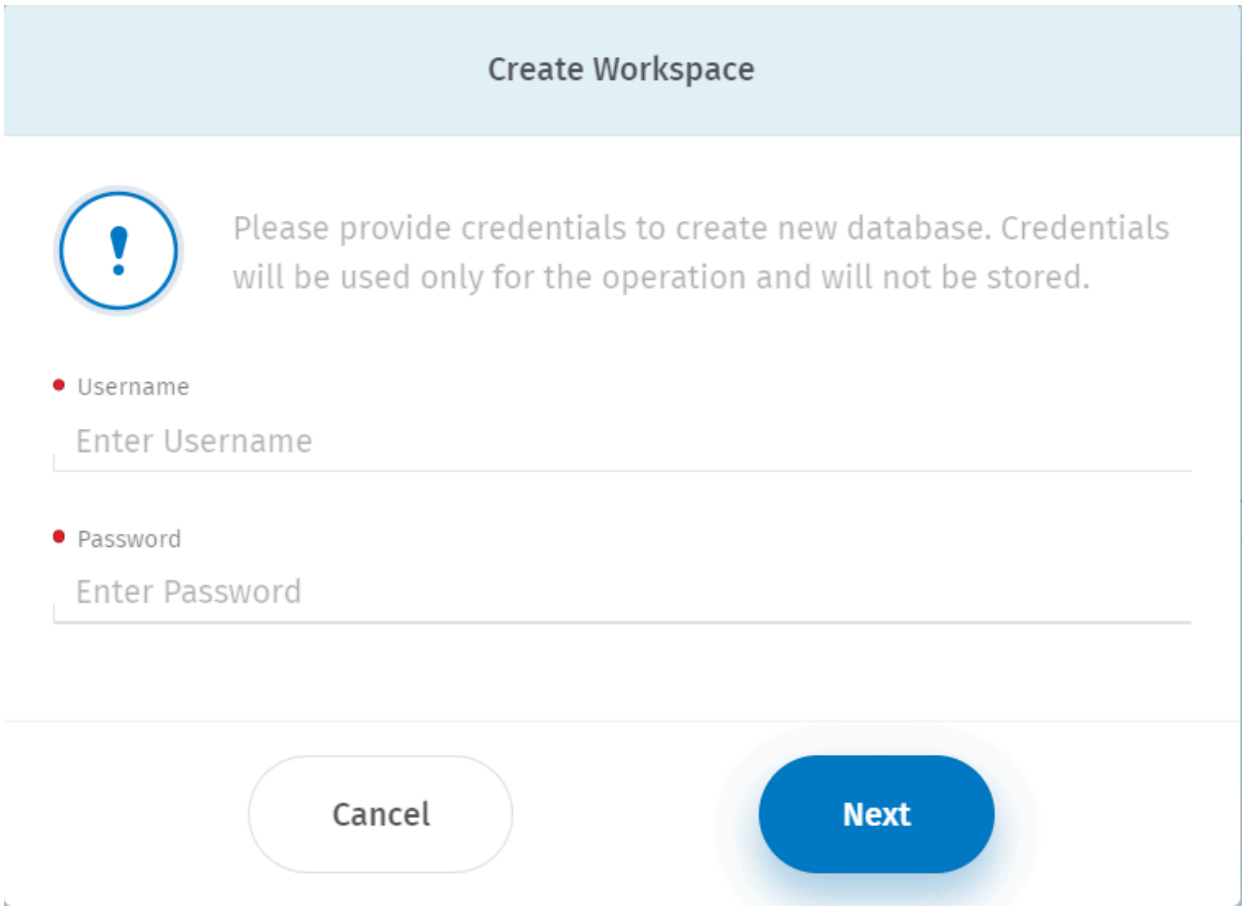
Based on the selection made at step 9, one of the next options will be applicable:

- **RE will provision new schema/database** - A new database will be created by the engine.

- **RE will use customer provisioned schema/database** - An empty database must be created upfront by connecting to the database server with a user who has "create database" rights. The following example illustrates the instruction that needs to be executed.

```
CREATE DATABASE NEW_SCHEMA ;
```

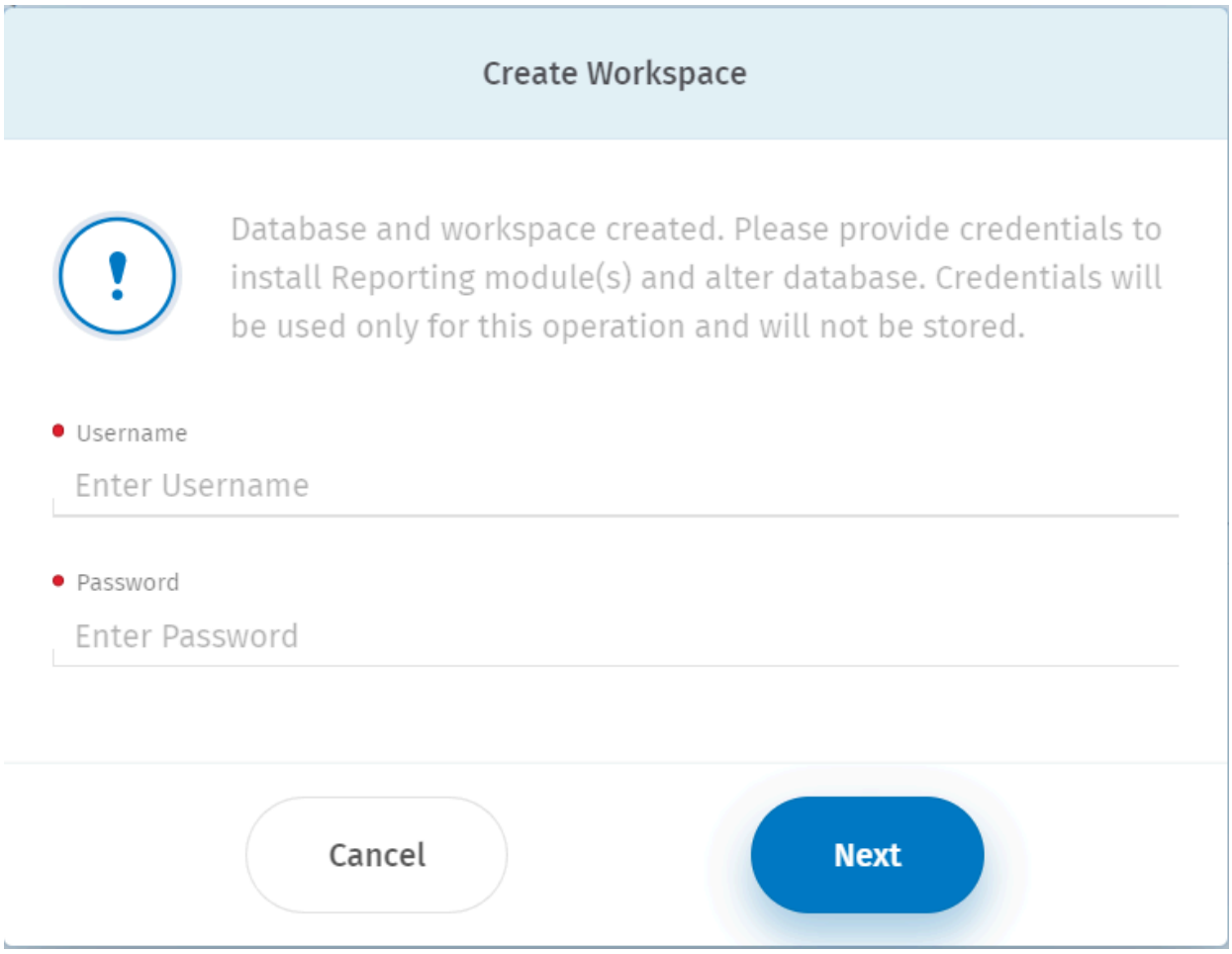
3. A dialog will appear, prompting you to enter **Username** and **Password**.



Type the username and password (level 1 user credentials, granting permission to create a new database, the default admin user from the database server, or a new custom user created with the required grants).

```
USE master;
CREATE LOGIN level1user WITH PASSWORD = '123level1_user';
CREATE USER level1user FOR LOGIN level1user;
GRANT CREATE ANY DATABASE TO level1user;
GO
```

4. A second dialog will appear, prompting you to enter **Username** and **Password**.



The default admin user ("sa") can be used, without any additional grants, or a custom user can be created and have the required grants assigned.

```
use master;
CREATE LOGIN level2user WITH PASSWORD = '123level2_user';
GO;
use NEW_SCHEMA;
CREATE USER level2user FOR LOGIN level2user;
GRANT CREATE TABLE TO level2user;
GRANT ALTER ON SCHEMA::dbo TO level2user;
GRANT REFERENCES TO level2user;
GRANT SELECT TO level2user;
GRANT UPDATE TO level2user;
GRANT INSERT TO level2user;
GRANT DELETE TO level2user;
GRANT CONNECT TO level2user;
GRANT CREATE PROCEDURE TO level2user;
GRANT ALTER TO level2user;
GRANT EXECUTE to level2user;
GO
```

Results

If all the details that you provided are correct, the connections are established successfully, and a connection successful message is displayed. The workspace creation process starts, as well as the installation of the reporting module, which starts automatically.

Note:

The installation process could last between several minutes and several hours, depending on multiple factors, including the complexity of the reporting module. When the process is finished, you are notified through a message that appears in the **top bar options** > **Notifications**. For details on viewing notifications, see Viewing notifications.

To monitor the status of the workspace creation process, you have the following options:

- In the main menu, click **Environment** and then click the **LIST OF WORKSPACES** submenu. Look for the card corresponding to the workspace you just created and check the status: a progress bar is displayed if the workspace is still being created.
- In the main menu, click **Monitoring** and then click the **RUNNING JOBS** submenu for ongoing processes or the **HISTORY** submenu for finished processes. For details on monitoring processes in OneSumX Regulatory Engine, see Monitoring.

Warning:

- If you encounter an error or failure during the workspace creation, we strongly recommend discarding the workspace and trying again from the start, as there are some known issues. For details, see the Release Notes document.
- We recommend creating workspaces one by one (not to trigger two or more simultaneous reporting module installations), especially in the case of a shared GDM.
- If the workspace creation fails, you will not be able to create another workspace with the same database names, even if you delete the failed workspace. Check the logs for more details. You need to drop the created database, otherwise, the following error will be displayed: `Name should be unique across all database servers`.
- If you are working with Docker and using the workspaces, you must restart OneSumX Regulatory Engine for the changes to be visible in the application. For example, you must restart OneSumX Regulatory Engine after you create a workspace, otherwise, you may encounter errors in the application.

13.1.3 Creating workspaces using existing business database

Prerequisites

To be able to attach an existing business database to the system, the Reporting Modules on which the database was created should be uploaded in the **PACKAGE MANAGER**.

Context

A pre-existing, well-formed business database can be integrated into a new Regulatory Engine instance. To proceed, the database must include at least one installed Reporting Module. Upon integration, a new workspace is created with the business database attached. Once added to a resource group, the business objects within the attached database become available for use in Reporting Hub.

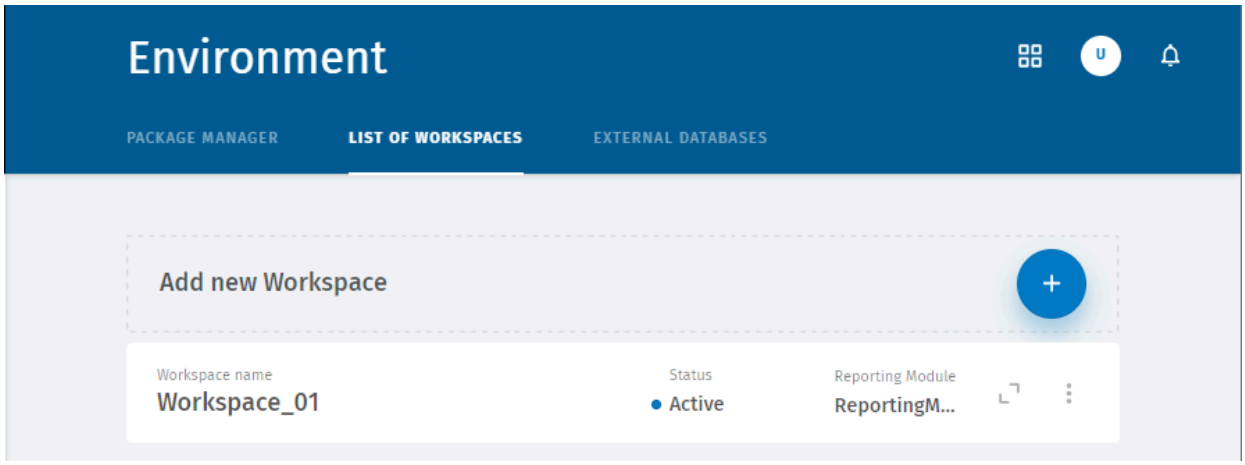
Attach a database as follows:

Procedure

1. In the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.

Results

A list of all the workspaces created is displayed, sorted by status and then by workspace name, in ascending order.



2. In the content pane, in the **Add new Workspace** area, click .

Results

A workspace form is displayed. A red dot precedes the mandatory fields.

3. Under **WORKSPACE NAME & DESCRIPTION**, enter the details of the workspace in the following fields:
   - **Workspace name** - Enter a name for the new workspace.
   - **Workspace description** - Enter the description of the workspace.
4. For **Database approach** , select **Attach DB**.

   The two options are:
   - **New DB** - workspace creation with Reporting Module installation
   - **Attach DB** - workspace creation by attaching an existing database with Reporting Module already installed
5. Choose the type of database server and provide specific connection details of the existing business database.
6. After clicking **Proceed**, a dialog appears, prompting you to enter **Username** and **Password**. Choose the database user for the connection. This can be a less privileged user (a level 3 user), an admin user, or even a level 2 user. The last two options are not recommended but can be used for testing purposes. Credentials for this user will be saved in the data sources and used for regular operations in the workspace.

Results

The result of the attachment will be a configured workspace in an Inactive state.

7. Activate the workspace.
8. Add the workspace to your resource group in Security Center, section Resource Groups.

   Select the resources from the attached database — Data sets and Submissions — that should be available in Reporting Hub.

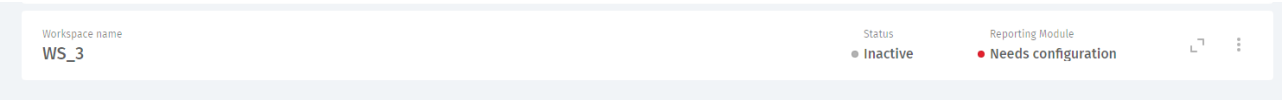
Results

The business objects from the attached database can be used in Reporting Hub.

13.2 Configuring workspaces

Context

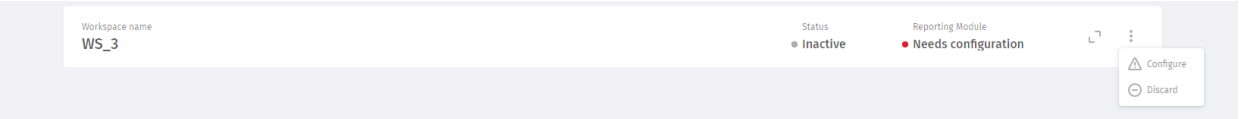
After a successful installation, the workspace will be in a Needs configuration state.



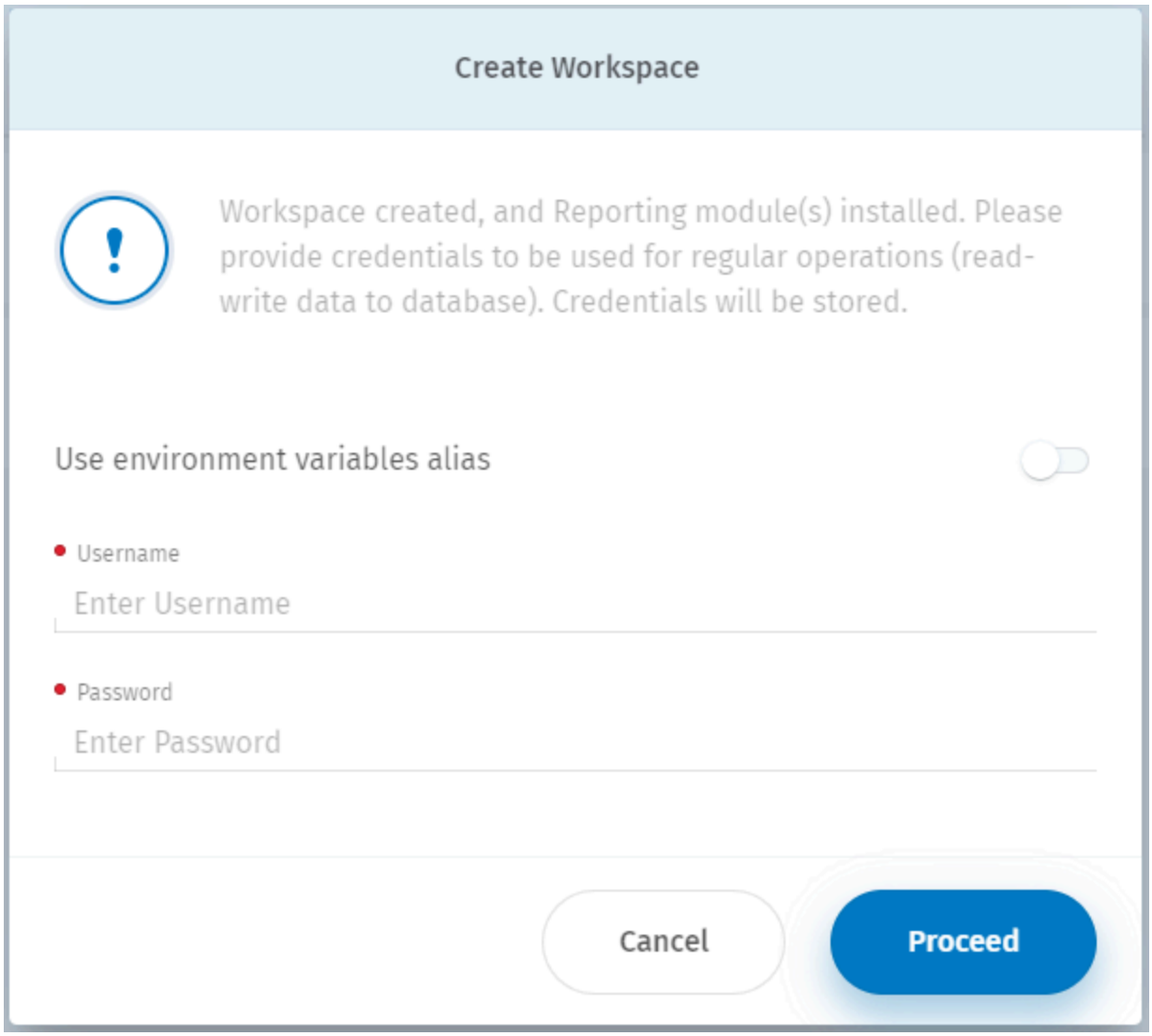
Procedure

1. Click ⋮ and select option **Configure**.

Results



2. A dialog is displayed, prompting you to enter **Username** and **Password**.



Choose the database user for the connection. This can be a less privileged user, an admin user, or even a level 2 user. The last two options are not recommended but can be used for testing purposes. Credentials for this user will be saved in the data sources and used for regular operations in the workspace.

Based on the selection you made at step 8 during the workspace creation, one of the following options is available:

- For Oracle: A script with the instructions is present as follows.

```
-- Create user for connection -level3
create user level3user identified by 123level3_user;
--Privilege setup for connection user, roles may be replaced with some other
role, list of grants in the future
grant connect to level3user;
grant NEW_SCHEMA$DYN to level3user;
grant NEW_SCHEMA$ROL to level3user;
grant NEW_SCHEMA$USR to level3user;
grant NEW_SCHEMA#APP to level3user;
commit;
```

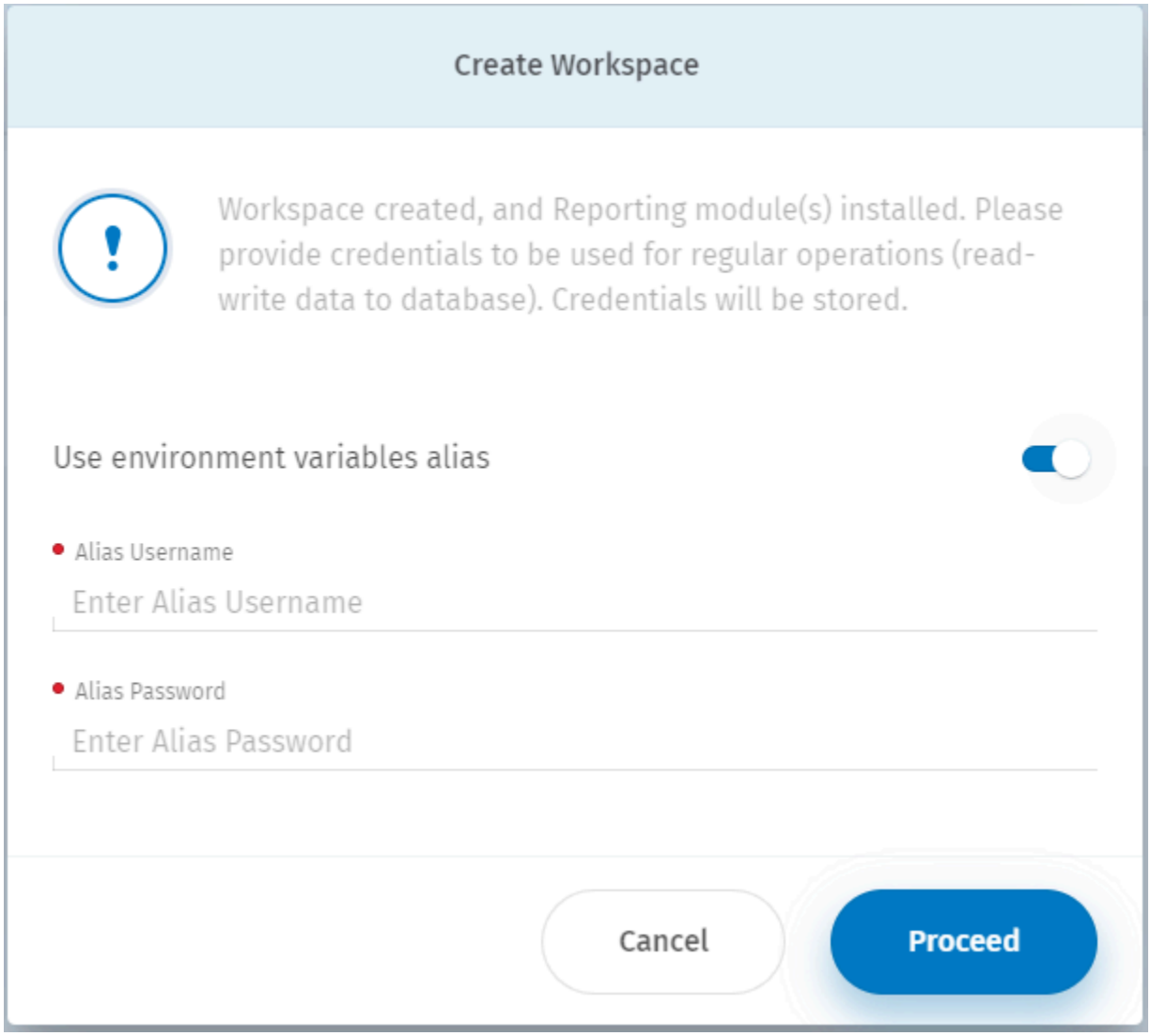
- For MSSQL: A script with the instructions is present as follows.

```
use master;
CREATE LOGIN level3user WITH PASSWORD = '123level3_user'
GO
use NEW_SCHEMA;
CREATE USER level3user FOR LOGIN level3user;
GRANT SELECT TO level3user;
GRANT UPDATE TO level3user;
GRANT INSERT TO level3user;
GRANT DELETE TO level3user;
GRANT CONNECT TO level3user;
GO
```

Note:

If you don't want to provide the database username and password for connection, you can enable the Use environment variables alias option.

If this option is enabled, environment variables aliases can be provided for **Alias Username** and **Alias Password**.

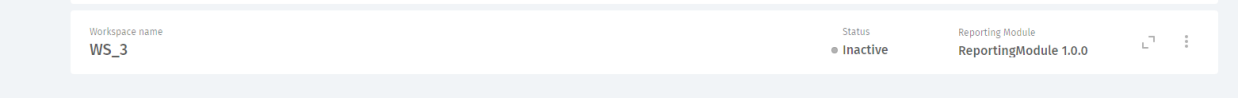


Alias username and alias password must be set before starting the solution, in the *.env.datasources-variables* file. These aliases will be saved in data sources instead of the username and password.

3. Click **Proceed**.

Results

After the process is successfully finished, the workspace will be in an Inactive state. For activation,see [Activating workspaces](#).



13.3 Viewing workspace details

Prerequisites

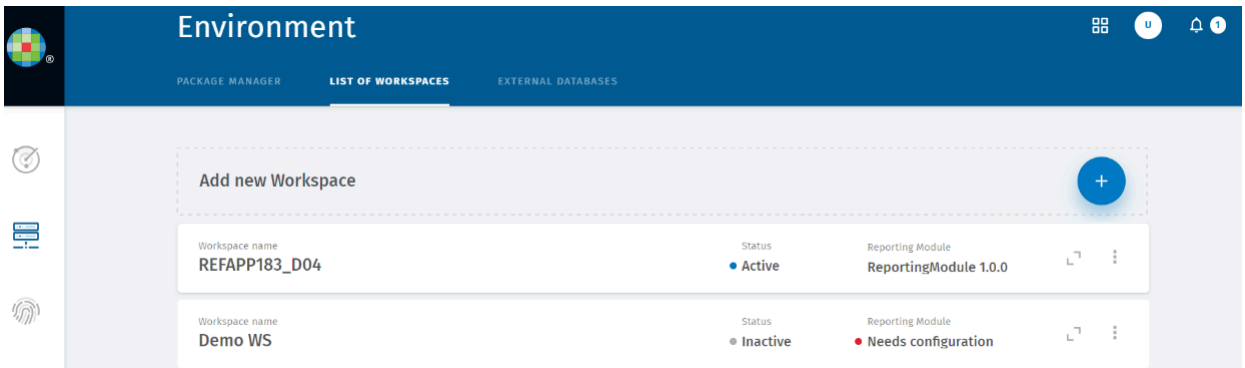
- You are logged in to OneSumX Regulatory Engine, to Admin Center.

Procedure

1. In the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.

Results

A list of all the workspaces created is displayed, in the form of cards, sorted by status and then by workspace name, in ascending order.



The following details are displayed for each workspace:

- **Workspace name**
- **Status** - Displays the status of a workspace (active or inactive).
- **Reporting Module** - Displays the name and version of the installed reporting module.

2. To view more details of a specific workspace, click its corresponding card, or click **Expand** to the right of the card.

Results

The workspace card is expanded in the content pane, displaying all the details. The details displayed

depend on the configuration of the workspace:

- **Workspace name**
- **Workspace description**
- **Workspace status** - Displays the status of a workspace (active or inactive).
- **ABOUT DATABASE** area - Displays database details of the selected workspace, for example, the database type, the machine name, the port, the service name.
- **DATABASE** area - Displays the name of the database.
- **Location link** - Displays a link to the database location.
- Information on the reporting module installed in the selected workspace - for example: the reporting module name and version, the installation date, and the user who installed it - and buttons to launch the upgrade or installation of the reporting module.

3. To close the detailed view, click (collapse), in the upper-right corner of the card.

13.4 Modifying workspace details

Prerequisites

- You are logged in to OneSumX Regulatory Engine, to Admin Center.

Procedure

1. On the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.

   #### Results

   A list of all the workspaces created is displayed, in the form of cards, sorted by status and then by workspace name, in ascending order.

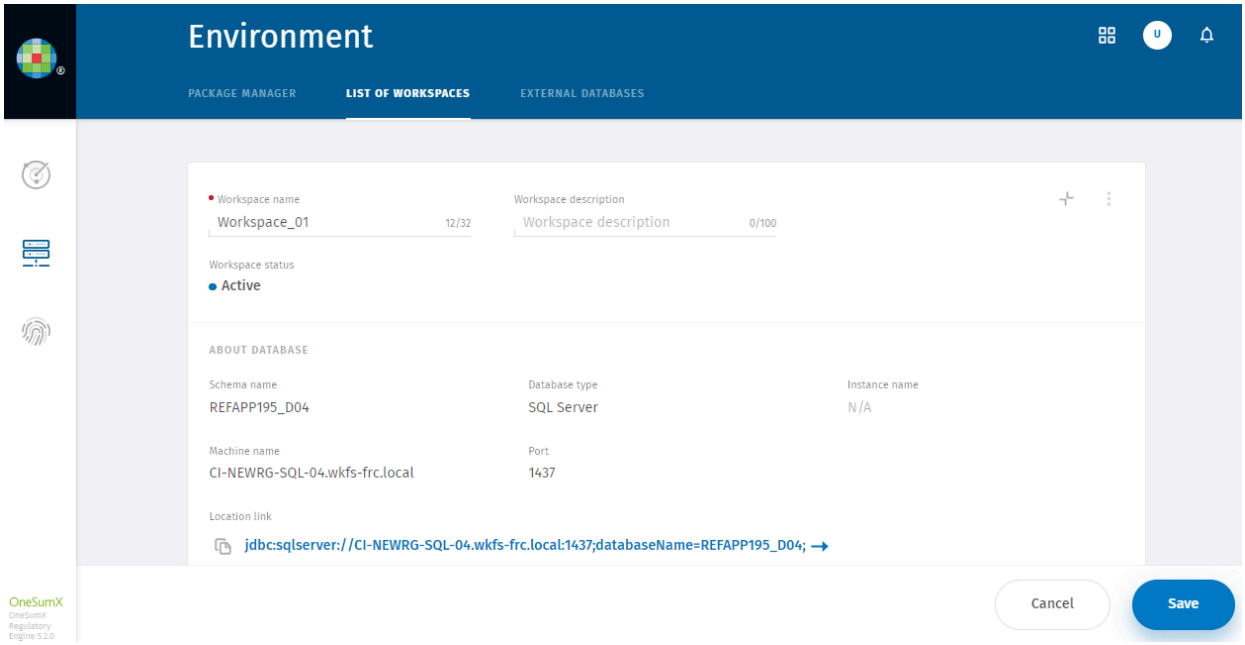
2. In the content pane, look for the workspace whose details you want to modify, then, to the right of its corresponding card, click the menu, and then click **Edit**.

   #### Results

   The workspace card is expanded in the content pane, displaying all the details of the selected workspace in edit mode.

   The only details that you can modify are:

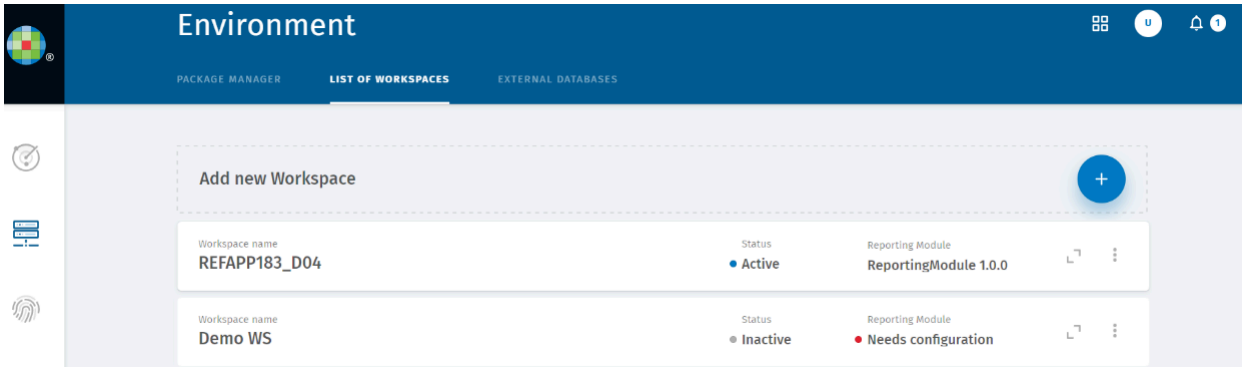
   - **Workspace name**
   - **Workspace description**



3. Make the required changes and click **Save**.

Results

The workspace is displayed in the collapsed view in the content pane, showing the modified details.



13.5 Activating workspaces

Prerequisites

- You are logged in to OneSumX Regulatory Engine, to Admin Center.

Context

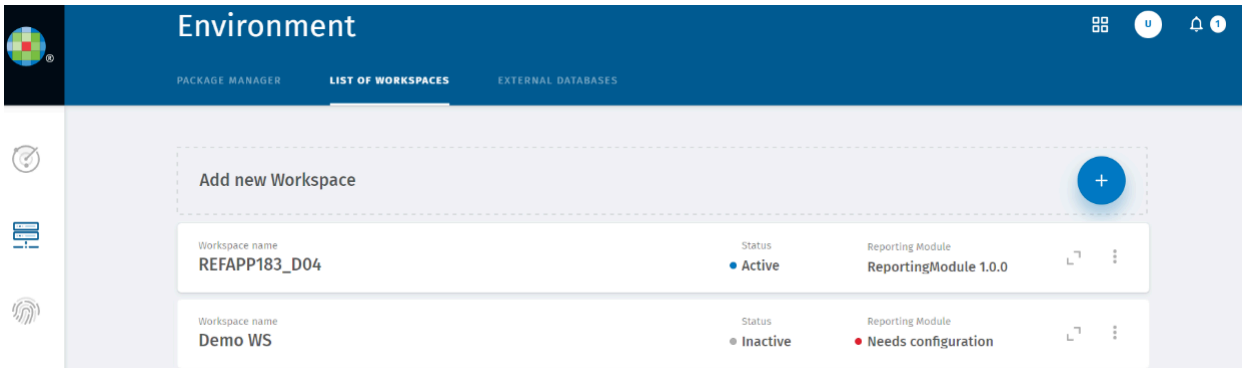
After you create and configure a workspace, you must activate it to be able to use it.

Procedure

1. On the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.

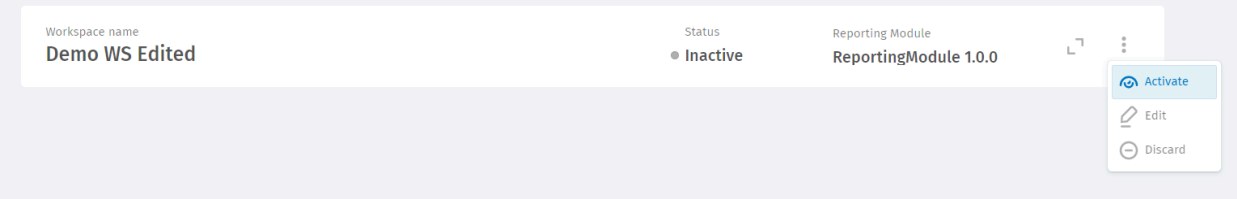
Results

A list of all the workspaces created is displayed, in the form of cards, sorted by status and then by workspace name, in ascending order.



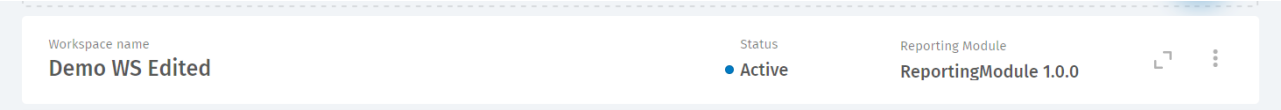
2. In the content pane, look for the workspace you want to activate, then, to the right of its corresponding card, click the menu, and then click **Activate**.

Results



Results

The status of the workspace changes to Active and you can now use it in OneSumX Regulatory Engine.



13.6 Deactivating workspaces

Prerequisites

- You are logged in to OneSumX Regulatory Engine, to Admin Center.

Context

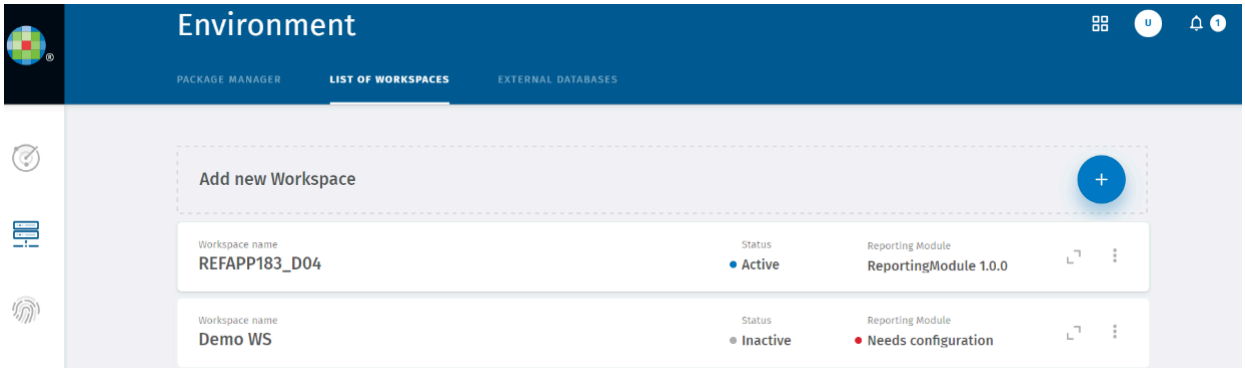
If you don't want a workspace to be used in Reporting Hub anymore, you can deactivate it.

Procedure

1. In the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.

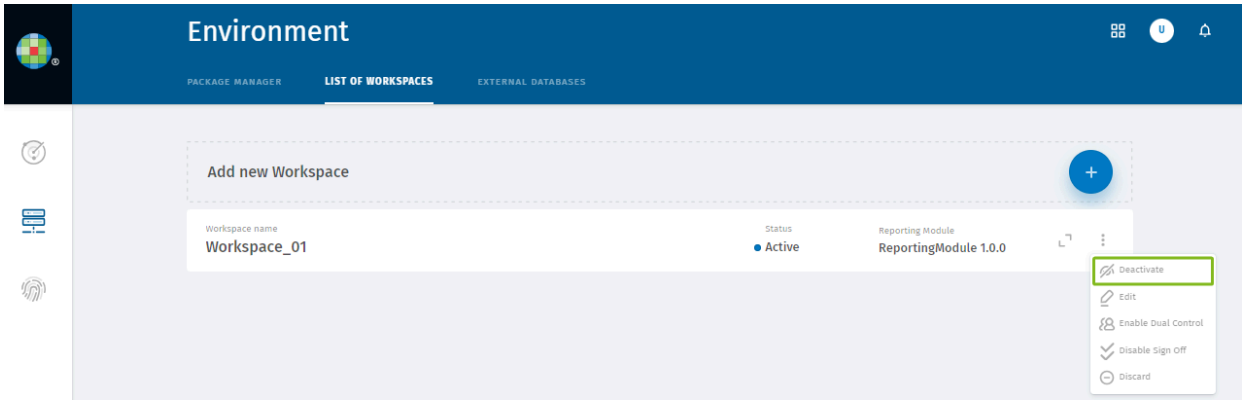
Results

A list of all the workspaces created is displayed, in the form of cards, sorted by status and then by workspace name, in ascending order.

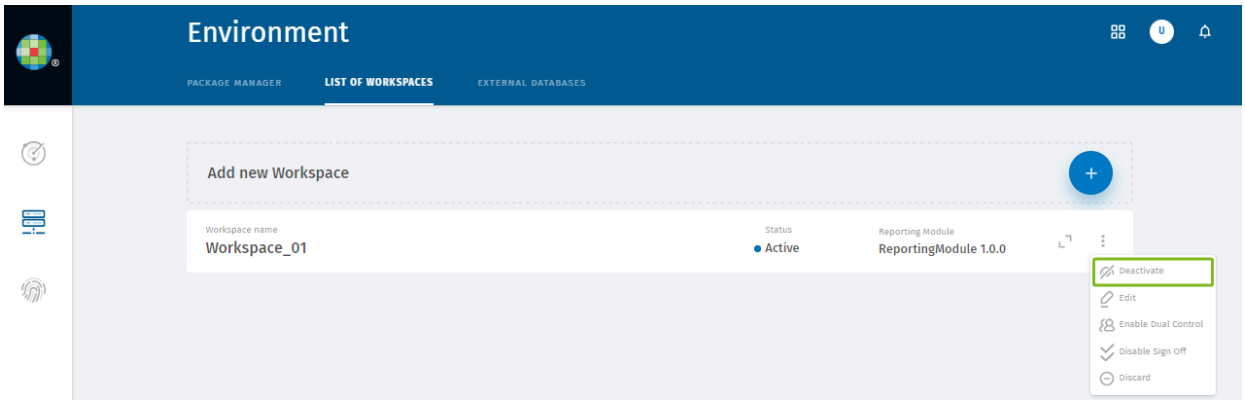


2. In the content pane, look for the workspace that you want to render inactive, then, to the right of its corresponding card, click the menu, and then click **Deactivate**.

Results

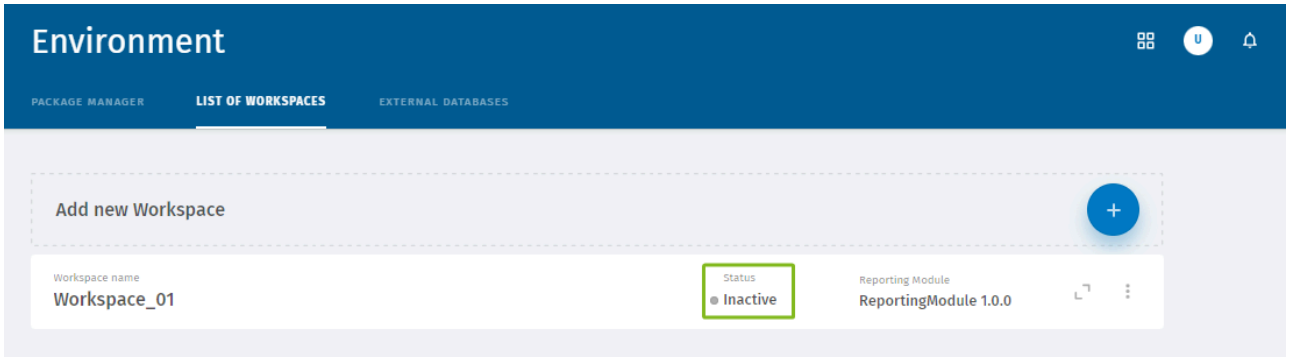


3. In the confirmation dialog, click **Deactivate**.

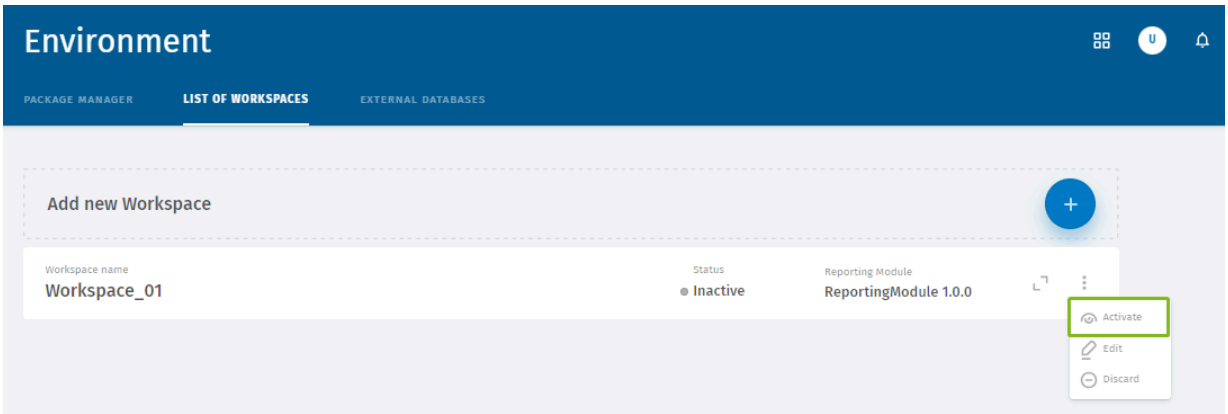


Results

The status of the workspace changes to Inactive.



Note:
Inactive workspaces can still be modified, discarded, or made active.



13.7 Discarding workspaces

Context

When you discard a workspace from OneSumX Regulatory Engine, you remove it from the application. The databases will not be removed.

You can discard active as well as inactive workspaces. You can't discard workspaces that have processes running in Reporting Hub.

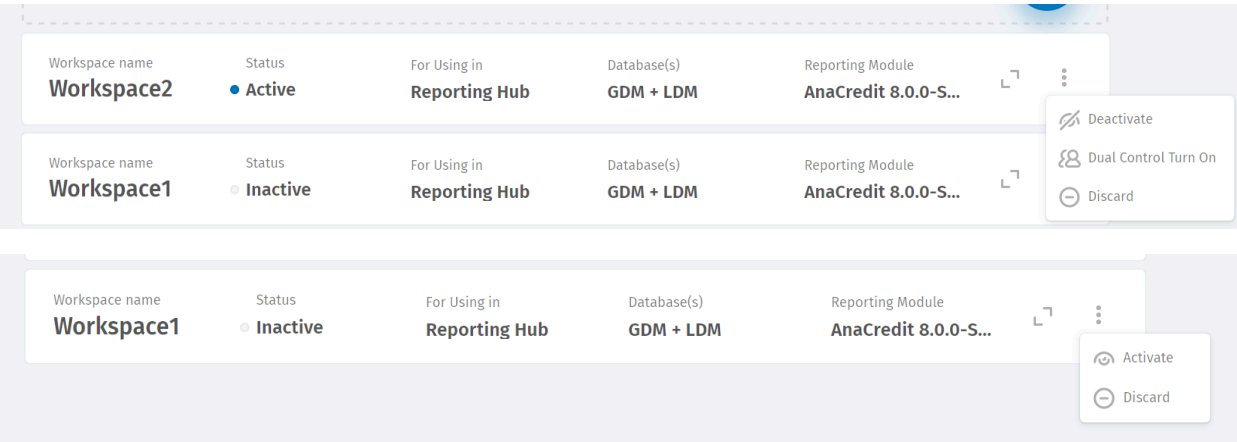
Warning:
Discarded workspaces cannot be recovered.

Procedure

1. On the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.
2. In the content pane, look for the workspace that you want to discard. Then, to the right of its corresponding

card, click the menu, and then click **Discard**.

This option is available for both inactive and active workspaces.

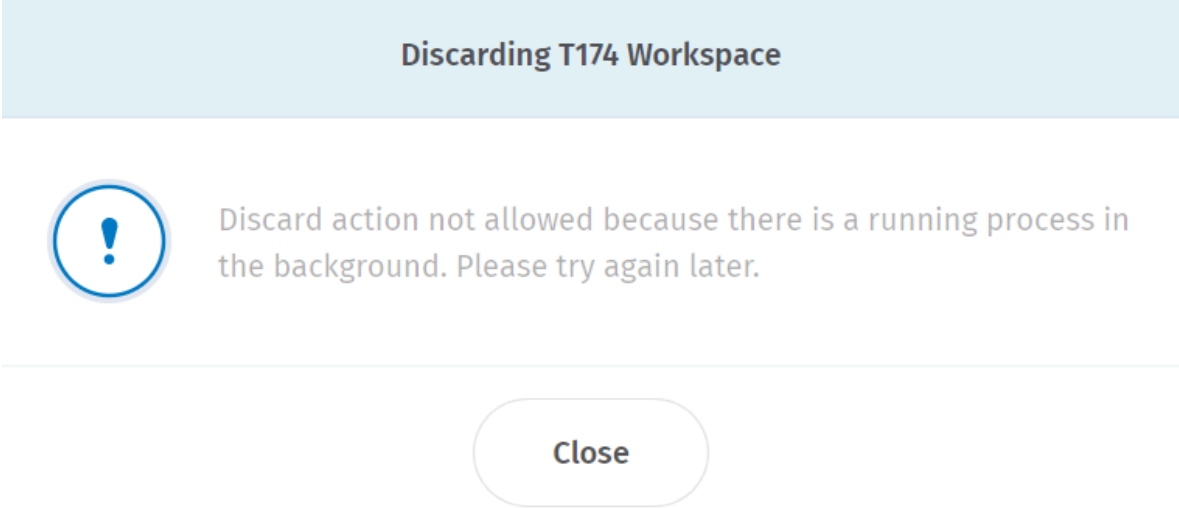


3. Click **Discard** in the confirmation dialog.

Results

The discarding process starts and in the workspace overview, a progress bar is displayed on the card corresponding to the discarded workspace, indicating that the workspace removal is in progress.

Note:
If a process is running in Reporting Hub for the selected workspace, the following warning is displayed.



Results

When the discarding process is finished, the workspace is removed from the application and you are notified through a message that appears in the **top bar options** > **Notifications**. For details on viewing notifications, see Viewing notifications.

Tip:
If discarding the workspace fails, you can download the log.

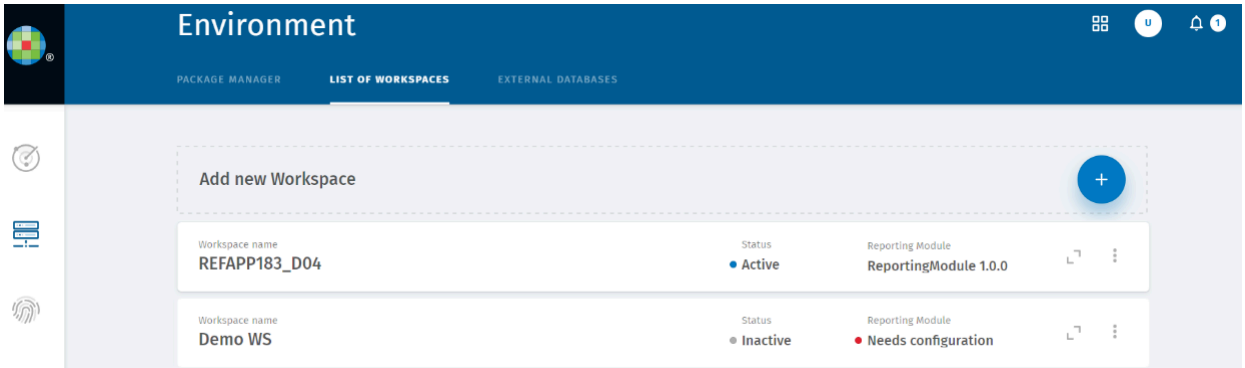
1. Go to **Monitoring** > **HISTORY**.
2. Look for the discarding workspace job.
3. Click the **Download log** icon to the right of the corresponding card.

13.8 Enabling dual control

Procedure

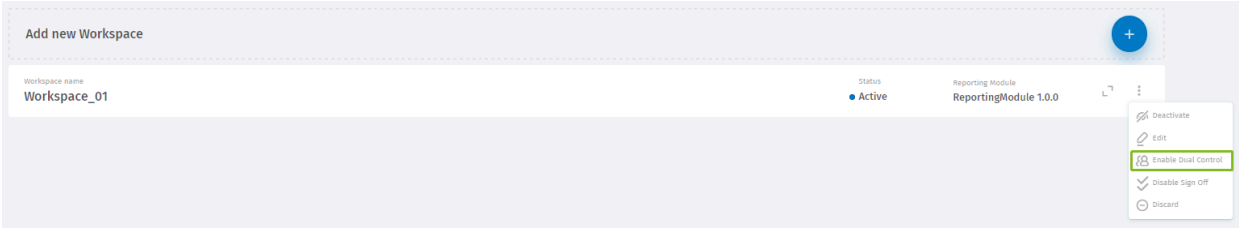
1. In the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.

Results



2. Select the workspace that requires dual control, click and select option **Enable Dual Control**.

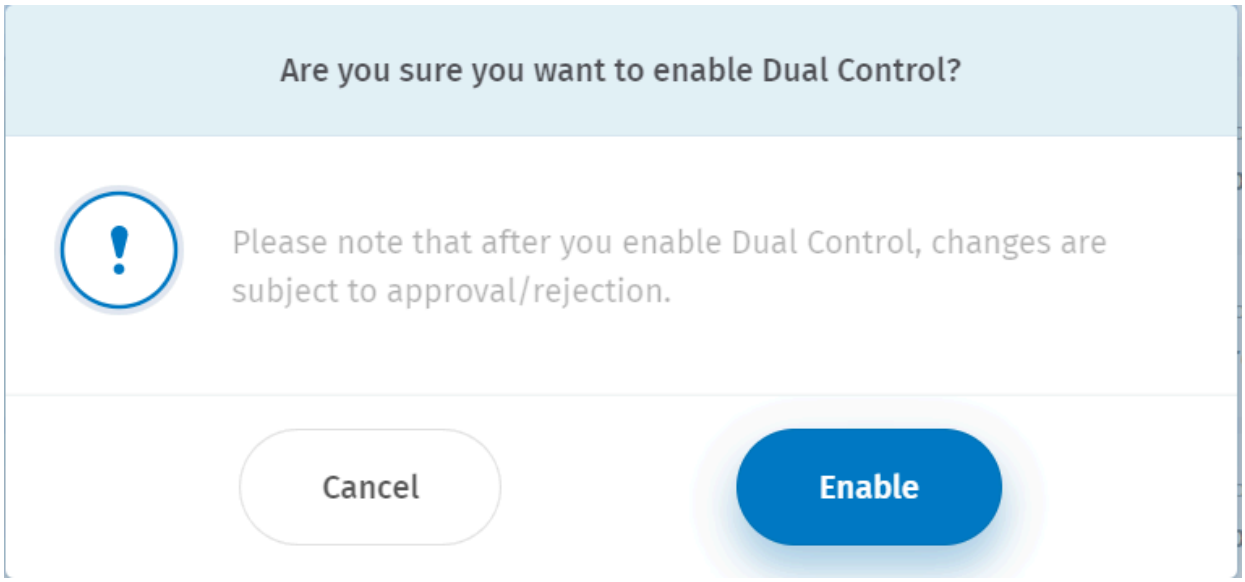
Results



Note:
If you want to disable dual control, select **Disable Dual Control** in the same manner for a workspace where this option was enabled.

3. Click **Enable**.

Results



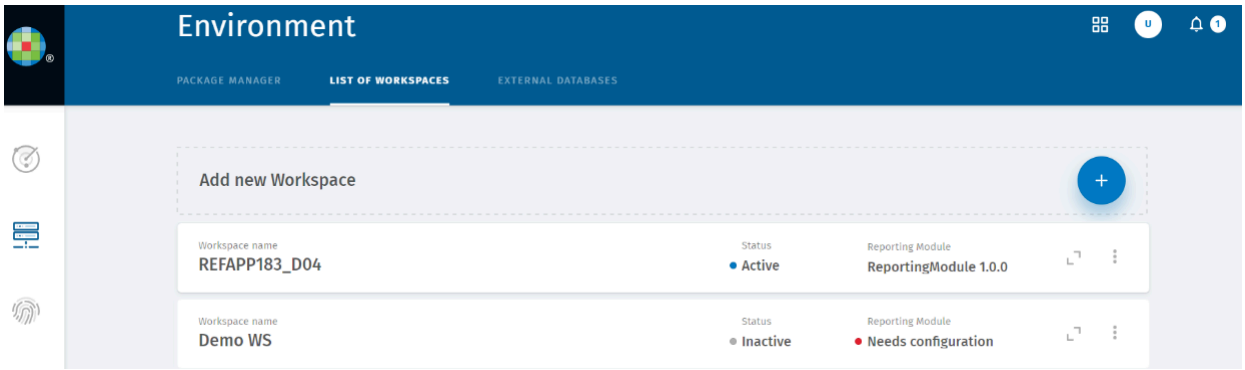
With this option enabled and permission **Approve/Reject** available for the role, you can **Approve/Reject/Cancel** changes.

13.9 Enabling sign-off

Procedure

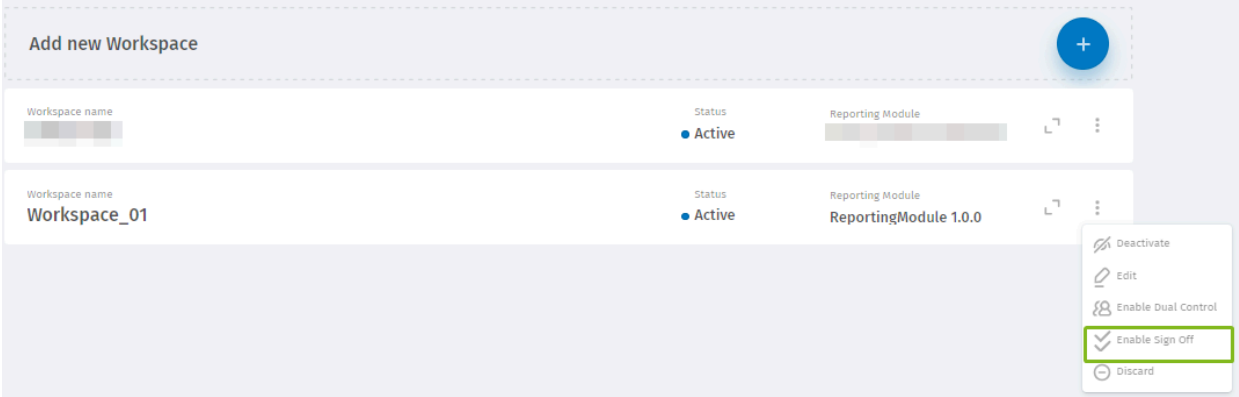
1. In the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.

Results



2. Select the workspace that requires sign-off enabled, click ⋮ and select **Enable Sign Off**.

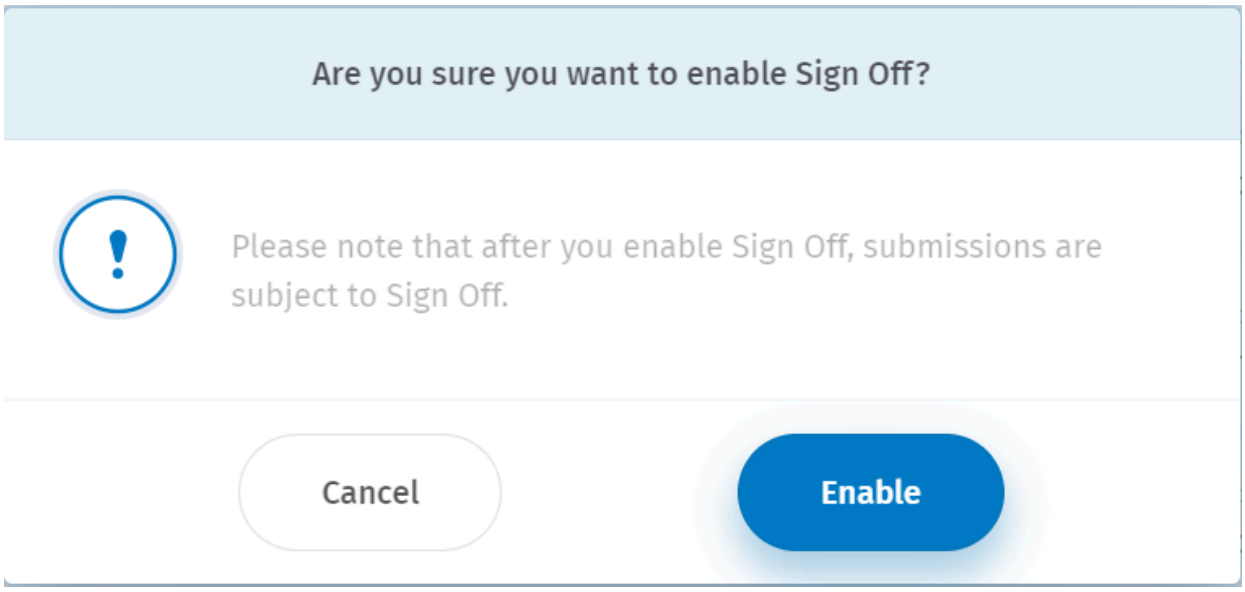
Results



Note:
If you want to disable sign-off, select **Disable Sign Off** in the same manner for a workspace where this option was enabled.

3. Click **Enable**.

Results



With the sign-off option enabled you will see it at the submission level.

13.10 Upgrading reporting modules installed in workspace

When a new version of a reporting module installed in a workspace is available, you can upgrade that specific module. To be able to perform the upgrade, you need to have the new version uploaded to the package manager.

Depending on your installation, refer to the applicable topic for the upgrade procedure:

- Upgrading reporting module installed in workspace for Oracle databases

- Upgrading reporting module installed in workspace for SQL databases

Note:
After the workspace has been upgraded successfully, currently active submissions will be updated in terms of their corresponding report set, reports, and delivery files to the new Reporting Module structure.

When you perform an upgrade to a new reporting module version, do not delete the older one. Deleting the old module that was used may cause issues.

Warning:
During a reporting module upgrade, the overridden process parameter values from the older version reporting module will be migrated to the new version reporting module (if a default process parameter of a previous release is manually updated, all parameters will be migrated to next version). After the upgrade workspace step, check the parameters for all processes starting with **Risk Enrichment Process** to make sure parameter logical-model-version is set to 3.23.2 (the version used for COREP solution 2025.01). If this is not the case, change it to 3.23.2.

Warning:
After the workspace has been upgraded, new resources (for example, new reports, new business processes) have to be added manually to the resource group.

Warning:
When you upgrade the platform, a new resource group will need to be created after the platform upgrade within the existing workspace. This will ensure users have access to the new Large Exposure enrichments added with LIQREP solution 2025.02.

13.10.1 Upgrading reporting module installed in workspace for Oracle databases

Procedure

To check if there is an upgrade available for a specific workspace, perform the following steps.

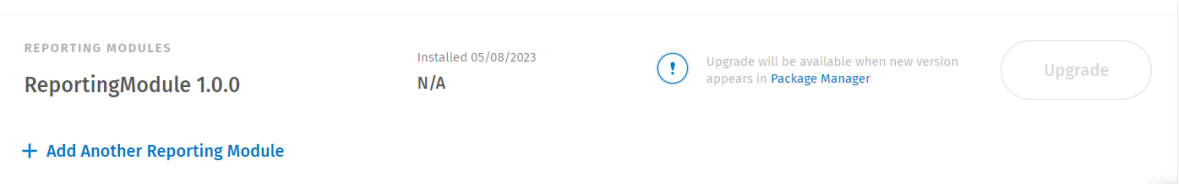
1. On the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.
2. Browse to the workspace whose details you want to view, then click its corresponding card, or click (expand) to the right of the card.

Results

The workspace card is expanded in the content pane, displaying all the details of the selected workspace in view mode.

If new versions are available for the reporting module installed in this workspace, the **Upgrade** button will be available in the bottom-right corner.

Note:
If there are no available new versions, the **Upgrade** button will be disabled.

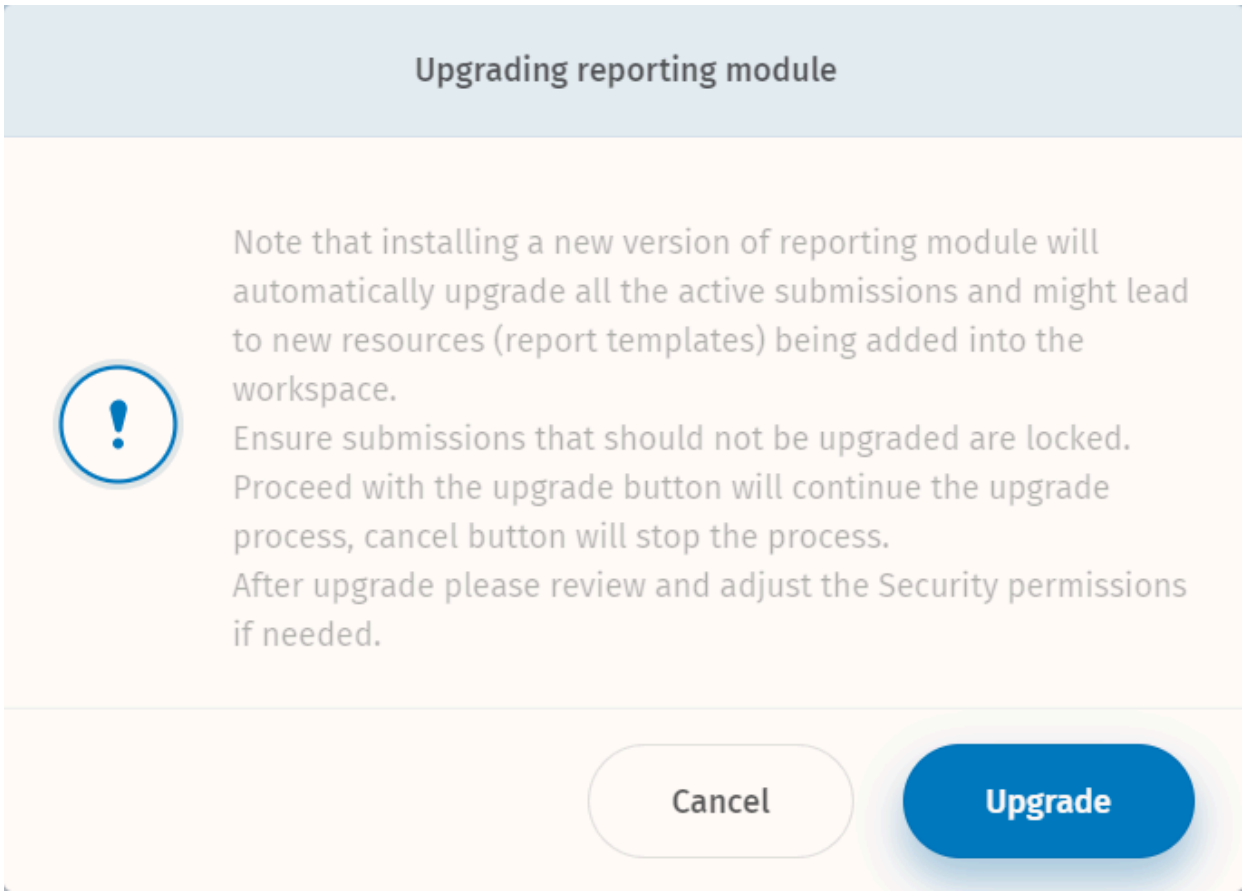


To upgrade the reporting module installed in the selected workspace, perform the following steps.

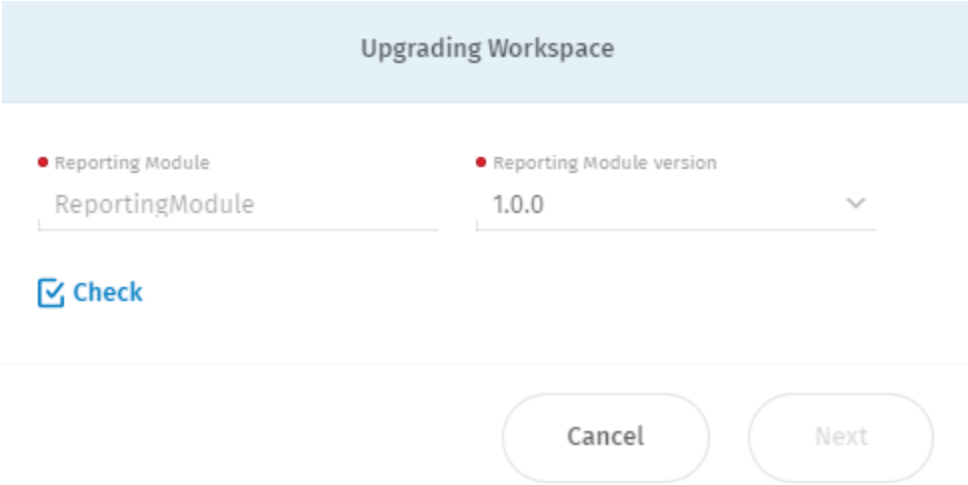
3. Click **Upgrade**.

Results

A new dialog will inform you that currently active submissions will be upgraded automatically to the new reporting module. To proceed with the process of upgrading the workspace, click **Upgrade**.

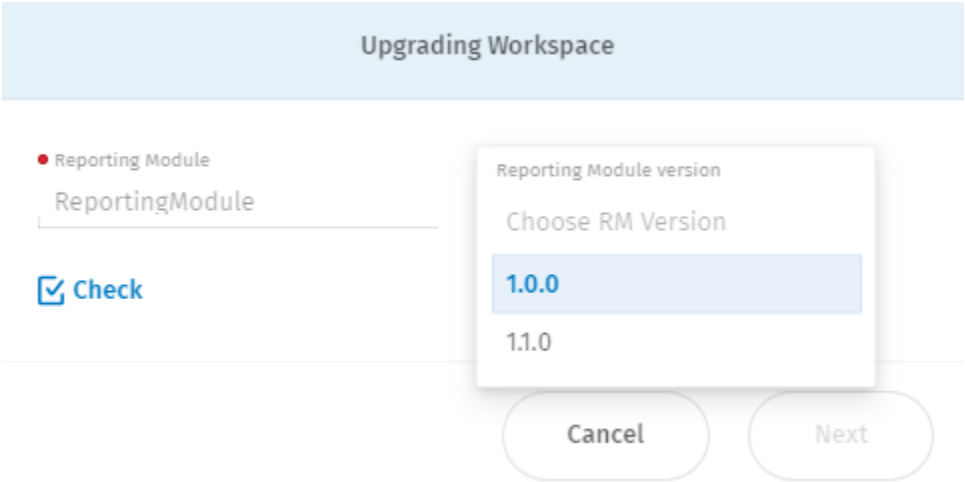


For the selected workspace, the **Upgrading workspace** dialog is displayed.



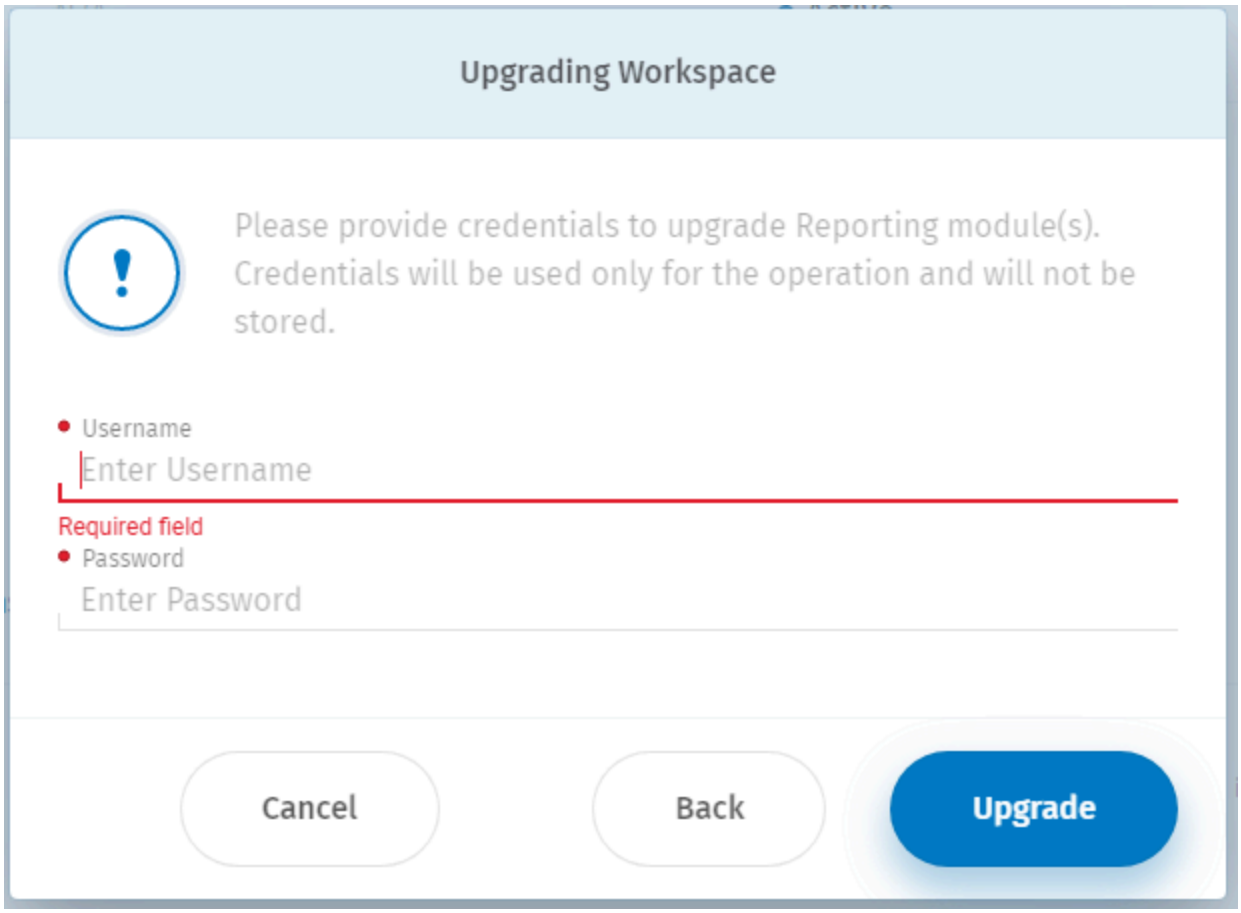
4. In the **Version** list, select the reporting module version you want to upgrade to and click **Next**.

Results



5. In the dialog, provide the applicable **Username** and **Password**.

Note:
To upgrade, the system user or schema owner can be used.



6. Click **Upgrade**.

Results

If all the details you provided are correct, the connections are established successfully. The upgrade process starts and you are prompted to the expanded view of the workspace that you are upgrading. A progress bar indicates that the upgrade is in progress.

Results

The upgrade process may take several minutes up to several hours, depending on multiple factors, including the complexity of the reporting module. When the process is finished, you are notified through a message that appears in the **top bar options** > **Notifications**. For details on viewing notifications, see Viewing notifications.

13.10.2 Upgrading reporting module installed in workspace for SQL databases

Procedure

To check if there is an upgrade available for a specific workspace, perform the following steps.

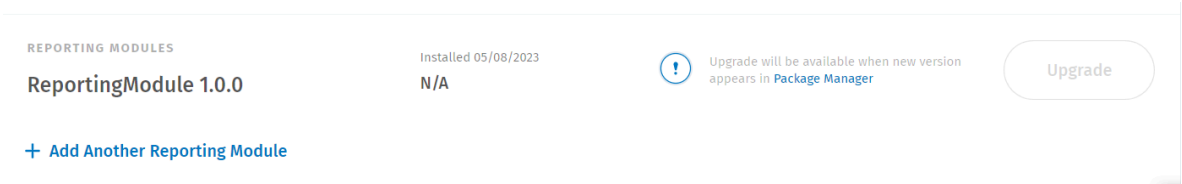
1. On the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.
2. Browse to the workspace whose details you want to view, then click its corresponding card, or click (expand) to the right of the card.

Results

The workspace card is expanded in the content pane, displaying all the details of the selected workspace in view mode.

If new versions are available for the reporting module installed in this workspace, the **Upgrade** button will be available in the bottom-right corner.

Note:
If there are no available new versions, the **Upgrade** button will be disabled.

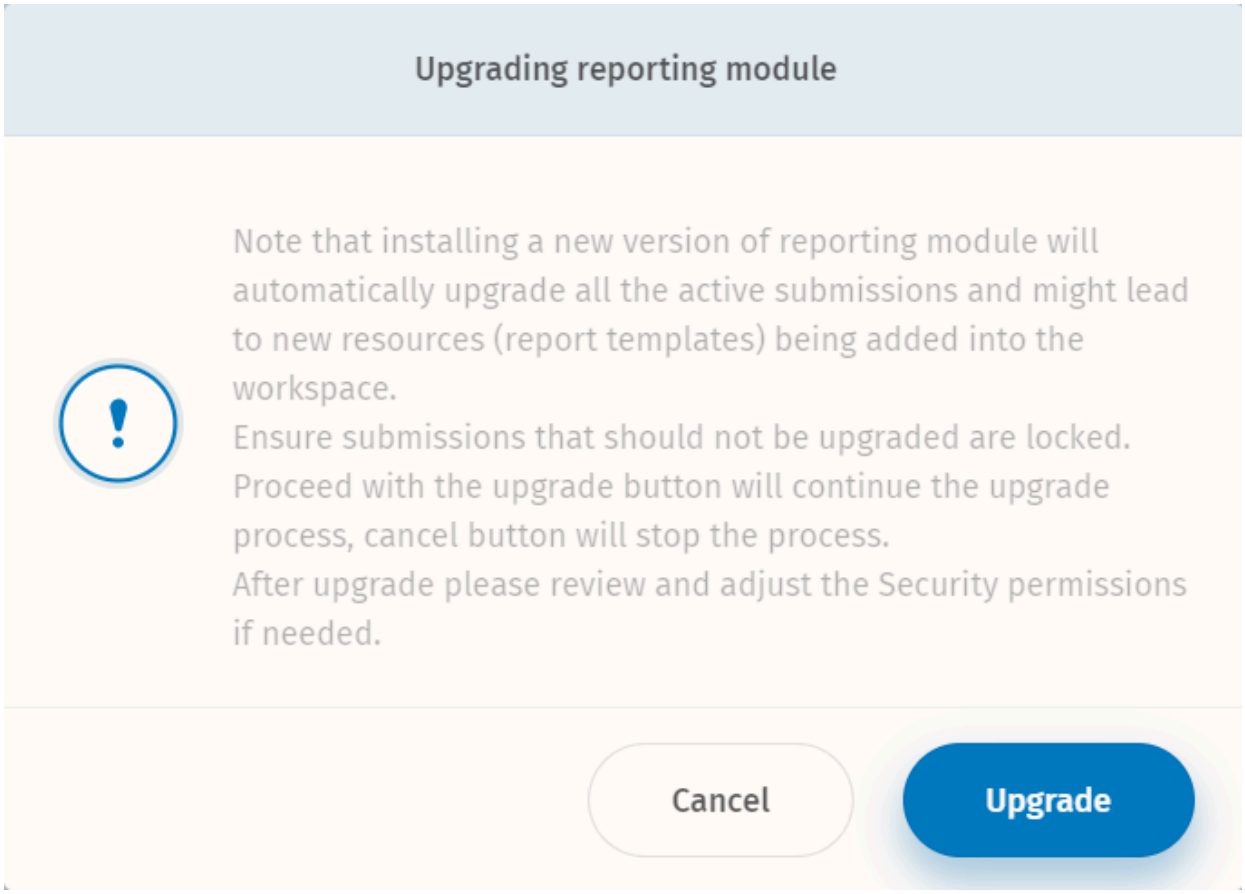


To upgrade the reporting module installed in the selected workspace, perform the following steps.

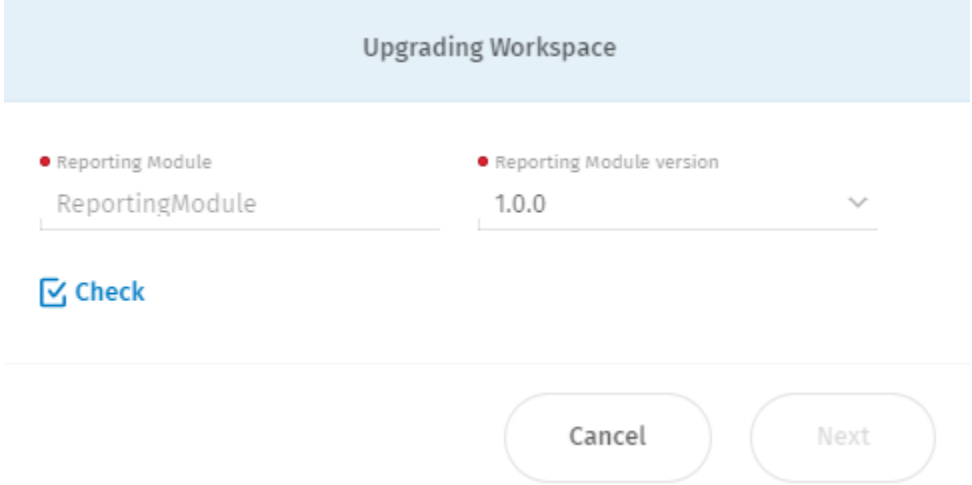
3. Click **Upgrade**.

Results

A dialog will inform you that currently active submissions will be upgraded automatically to the new Reporting Module. To proceed with upgrading the workspace, click **Upgrade**.

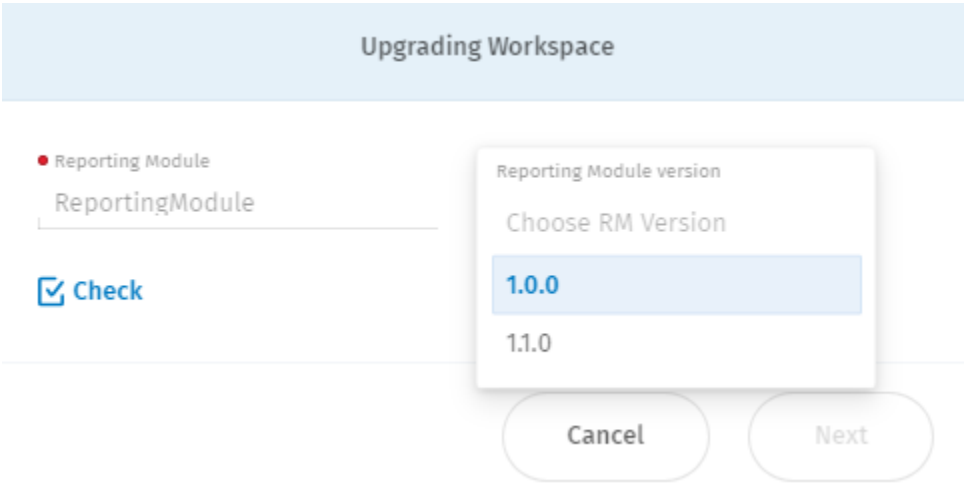


For the selected workspace, the **Upgrading workspace** dialog is displayed.

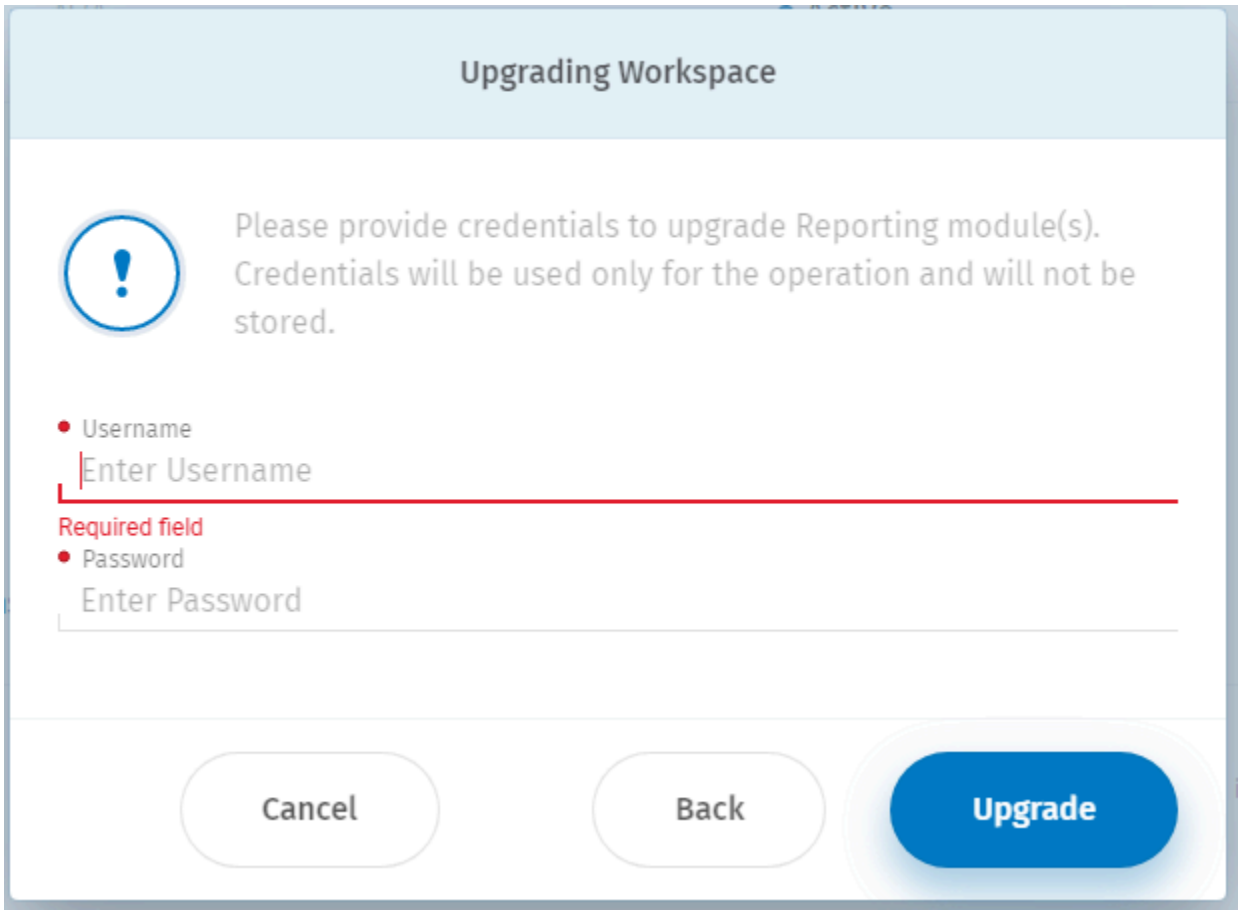


4. In the **Version** list, select the reporting module version you want to upgrade to and click **Next**.

Results



5. In the **Upgrading Workspace** dialog, provide the following information:
   - **Username** - Enter the user name of a user who has permission to connect to the databases.
   - **Password** - Enter the password of a user who has permission to connect to the databases.



6. Click **Upgrade**.

Results

If all the details you provided are correct, the connections are established successfully. The upgrade process starts and you are prompted to the expanded view of the workspace that you are upgrading. A progress bar indicates that the upgrade is in progress.

Results

The upgrade process may take several minutes up to several hours, depending on multiple factors, including the complexity of the reporting module. When the process is finished, you are notified through a message that appears in the **top bar options** > **Notifications**. For details on viewing notifications, see Viewing notifications.

13.11 Optimizing workspaces

Warning:
This functionality will be available for existing workspaces after upgrading to a higher OneSumX Regulatory Engine version.

To optimize your workspace, do the following

Step 1: Diagnose your workspace.

Step 2: Optimize your workspace to install either Oracle or SQL Server:

- Oracle
- SQL Server

13.11.1 Diagnosing workspaces

Prerequisites

The platform has been upgraded to 2024.05.07 or higher.

Context

This functionality allows you to check the consistency of a workspace.

Procedure

1. Run the diagnostic script on the business database (the one from the workspace): *file:diagnostic.sql*.
   a. For Oracle, use the script from the following path *tools/admin-center-schema-diagnostic.oracle.sql*.
   b. For SQL Server, use the script from the following path *tools/admin-center-schema-diagnostic.mssqlserver.sql*.
2. If script returns a record equal to Yes, choose one of the options:
   - Recreate workspace from a scratch.
   - Use the repairing script: *file:repair.sql* from the following path *tools/admin-center-schema-repair.sql*.
3. Run the applicable workspace optimization.

- Optimize the workspace for the Oracle installation.
- Optimize the workspace for the SQL Server installation.

13.11.2 Optimizing workspace for Oracle installation

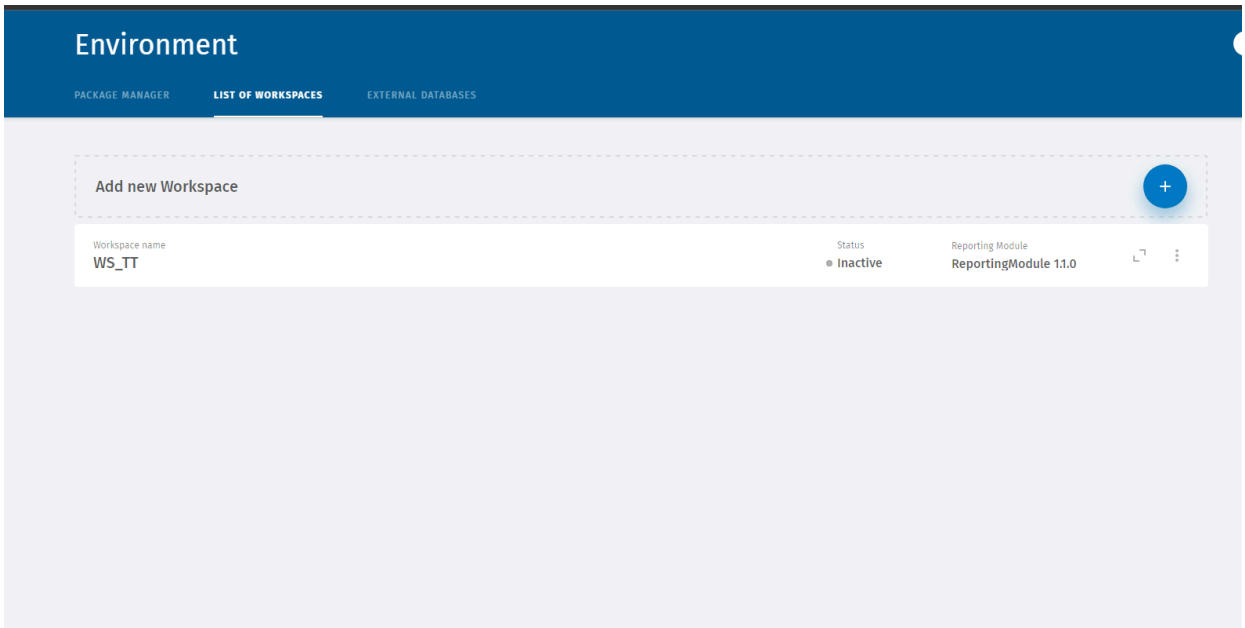
Context

This functionality allows users with the Manage workspaces permission to upgrade an existing workspace in the Admin Center.

Procedure

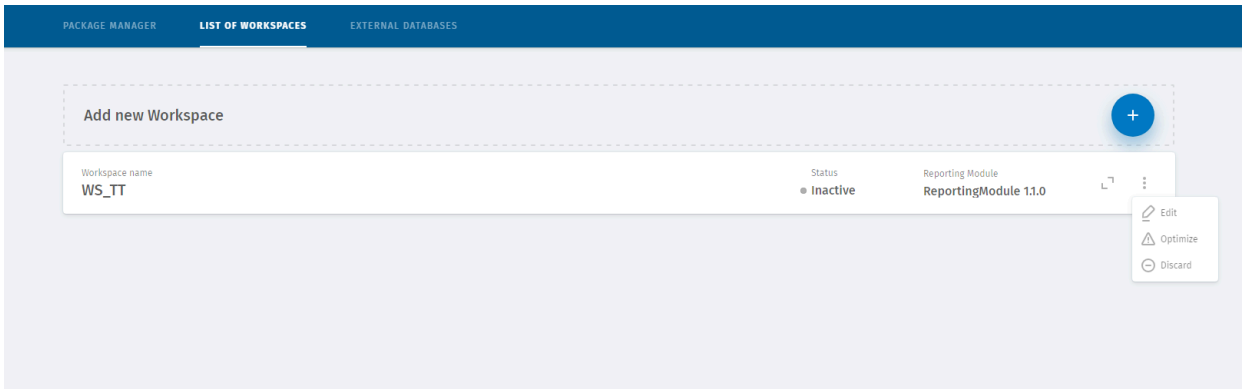
1. In the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.

   #### Results

2. In the content pane, look for the required workspace to run the optimization and, to the right of its corresponding card, click . Then, click **Optimize**.

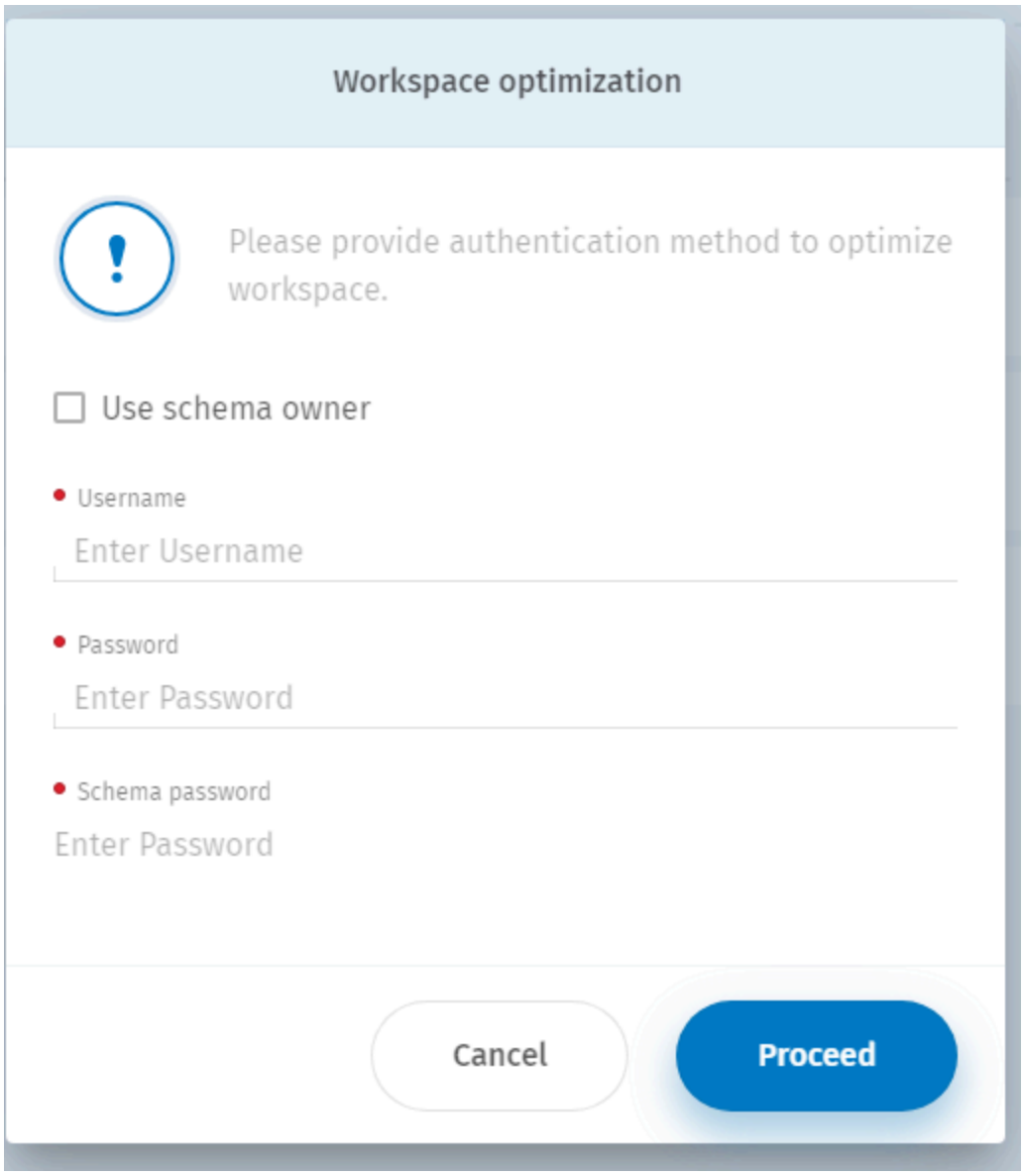
Results



3. In the **Optimizing Workspace** dialog, you get a database backup warning. If you did indeed create that backup, click **Next**. Otherwise, create the backup first.
4. Provide the following information:

   - **Username**: User who has the permission to alter the schema.

   - **Password**: Password of that user.



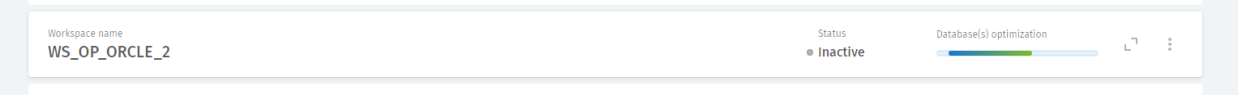
- Grants required for custom alter user (for example, level2user)

```
CREATE USER level2user identified by 123level2_user DEFAULT TABLE SPACE users
QUOTA unlimited ON users;
grant create session to level2user with admin option;
grant alter session to level2user with admin option;
grant create synonym to level2user with admin option;
grant create role to level2user with admin option;
grant select_catalog_role to level2user with admin option;
grant drop user to level2user;
grant create user to  level2user;
grant alter user to level2user;
grant select any table to level2user;
grant create any table to level2user;
grant insert any table to level2user;
grant update any table to level2user;
grant delete any table to level2user;
grant alter any table to level2user;
grant drop any table to level2user;
```

```
grant alter any role to level2user;
grant drop any role to level2user;
grant grant any role to level2user;
grant grant any object privilege to level2user;
grant create any index to level2user;
grant create any sequence to level2user;
grant select any sequence to level2user;
grant drop any sequence to level2user;
grant create any trigger to level2user;
grant create any procedure to level2user;
grant execute any procedure to level2user;
grant drop any procedure to level2user;
grant create any type to level2user;
grant create any view to level2user;
```

Results

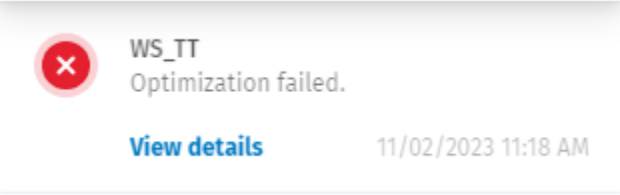
The optimization process starts.



5. After the process is executed, depending on the result, a notification appears.

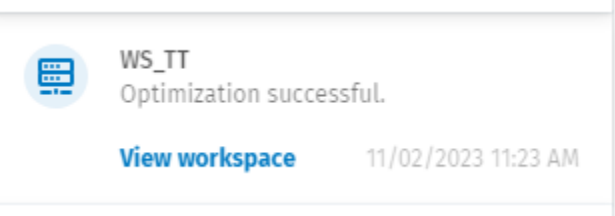
Results

If the optimization fails, the following notification will appear.



If the optimization succeeds, the following notification will appear.

Tip:
If the optimization is successful, the workspace will be returned to its previous status.



13.11.3 Optimizing workspace for SQL Server installation

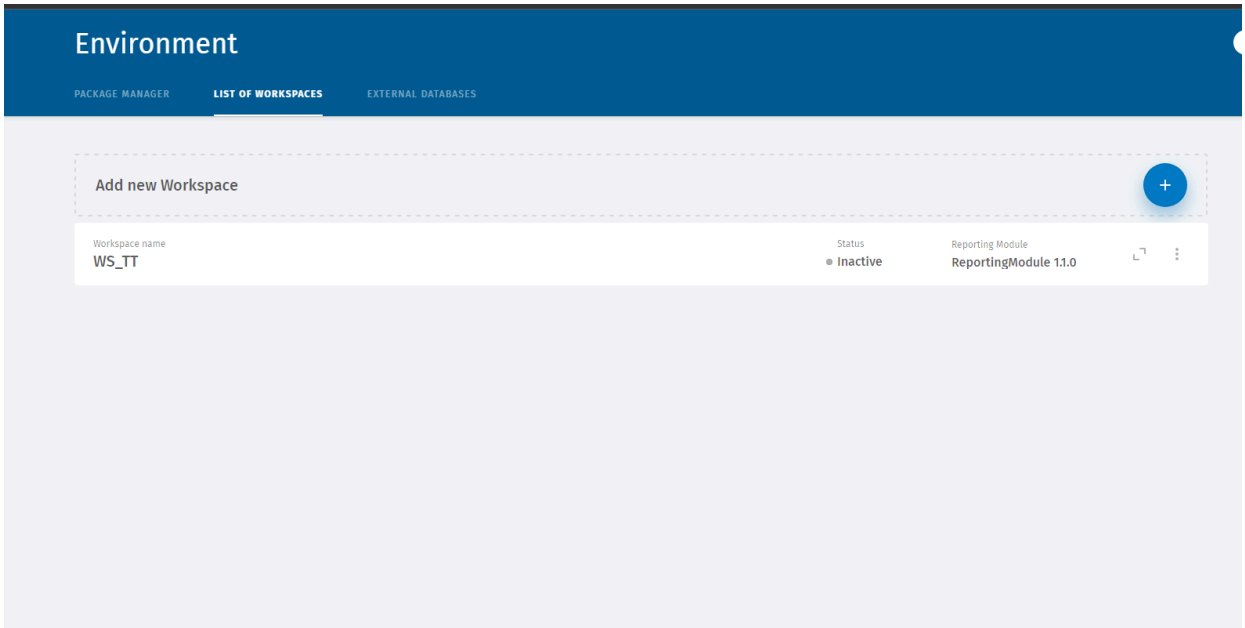
Context

This functionality allows users with the Manage workspaces permission to upgrade an existing workspace in the Admin Center.

Procedure

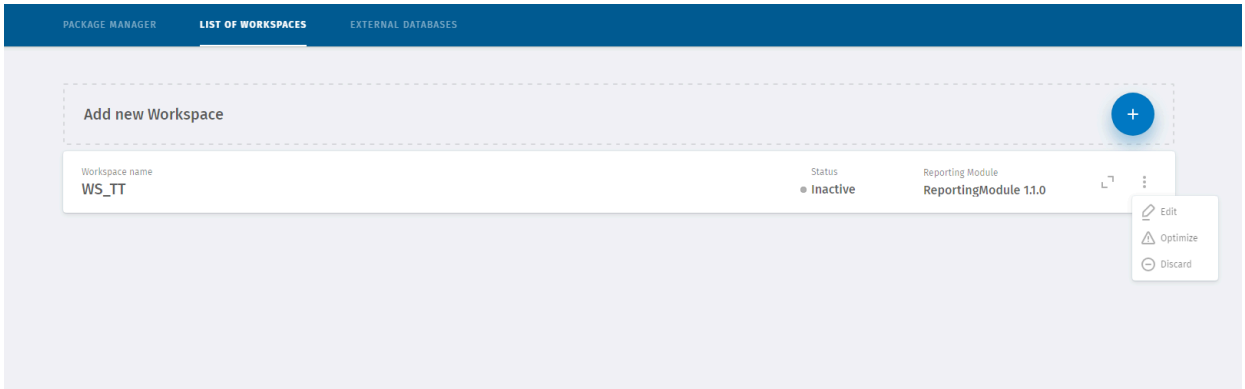
1. In the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.

   #### Results

2. In the content pane, look for the required workspace to run the optimization and, to the right of its corresponding card, click . Then, click **Optimize**.

   #### Results

3. In the **Optimizing Workspace** dialog, you get a database backup warning. If you did indeed create that

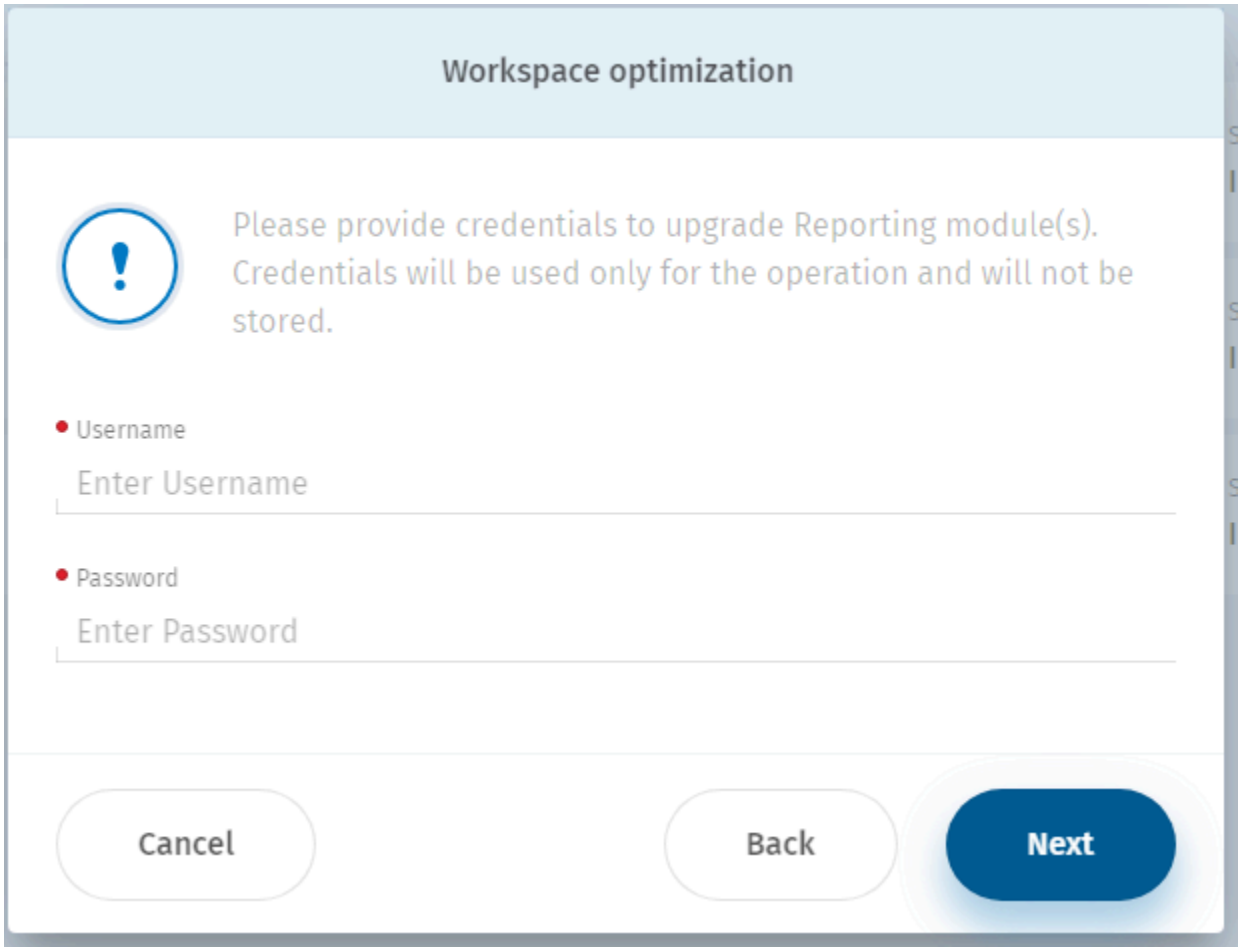
backup, click **Next**. Otherwise, create the backup first.

4. Provide the following information:

   - **Username**: User who has the permission to alter the schema.
   - **Password**: Password of that user.

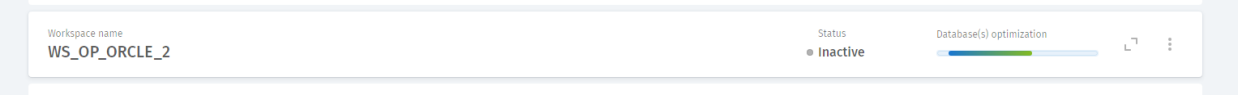
   The default admin user (sa) can be used without any additional grants, or a custom user can be created and have the required grants assigned.

```
use master;
CREATE LOGIN level2user WITH PASSWORD = '123level2_user';
GO;
use NEW_SCHEMA;
CREATE USER level2user FOR LOGIN level2user;
GRANT CREATE TABLE TO level2user;
GRANT ALTER ON SCHEMA::dbo TO level2user;
GRANT REFERENCES TO level2user;
GRANT SELECT TO level2user;
GRANT UPDATE TO level2user;
GRANT INSERT TO level2user;
GRANT DELETE TO level2user;
GRANT CONNECT TO level2user;
GRANT CREATE PROCEDURE TO level2user;
GRANT ALTER TO level2user;
GRANT EXECUTE to level2user;
GO
```



Results

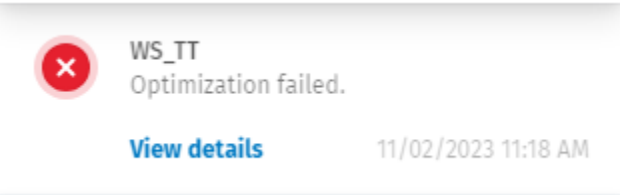
The optimization process starts.



5. After the process is executed, depending on the result, a notification appears.

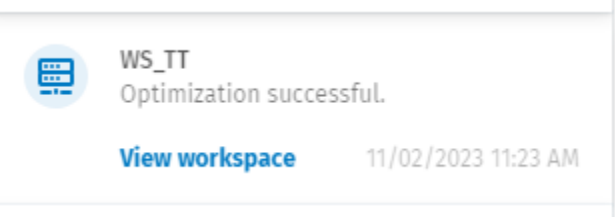
Results

If optimization fails, the following notification appears.



If optimization succeeds, the following notification appears.

Tip:
If the optimization is successful, the workspace will be returned to its previous status.



13.12 Disabling audit comments

You must have the **Change workspace properties** permission enabled to see the option.

Procedure

1. In the main menu, click **Environment**, then click the **LIST OF WORKSPACES** submenu.
2. Select the workspace that requires audit comments disabled, click ⋮ and select **Disable Audit Comments**.

Results

Note:
If you want to enable audit comments, select **Enable Audit Comments** in the same manner for a workspace where this option was disabled.

3. Click **Disable**.

Results

With audit comments disabled, the **Add Comment** dialog and **Comment** column on the **Audit** page will no longer be visible at Data sets and Submission level.

14 Managing resource groups

A resource group is a collection of workspaces (resources) on which users can perform different actions, according to their security roles. A System Administrator grants users access to one or several resource groups.

For example, if a user has the Regulatory Analyst role and is granted access to Workspace 1, they can create submissions or datasets for that specific workspace in Reporting Hub.

You can manage the resource groups as follows, depending on your permissions:

- Creating resource groups
- Granting users access to resource groups
- Removing users from resource groups
- Viewing resource groups and their details
- Adding workspaces to resource groups
- Deleting resource groups

Note:
When you upgrade OneSumX Regulatory Engine from version 4.0.x to 4.1.x, report-, enrichment process-, data set- and submission resources will be created for all the existing workspaces. The mentioned resources will be turned on in all the existing resource groups so that the workspaces that already exist are usable as they were before the upgrade. After the upgrade is completed, you can configure each resource individually.

14.1 Creating resource groups

Prerequisites

- You are logged in to OneSumX Regulatory Engine, to Admin Center.
- You have the **Security Management** permission.
- There is at least one workspace created. For details on creating workspaces, see Creating resource groups.

Procedure

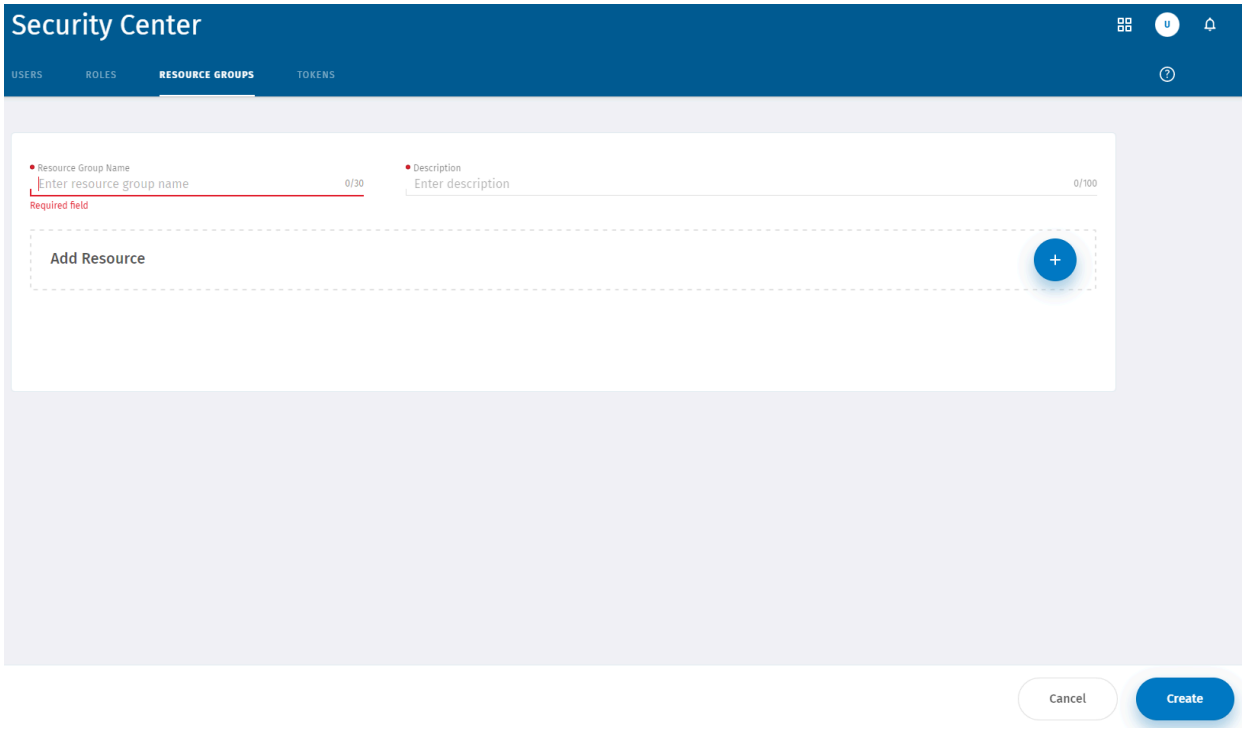
1. On the main menu, click **Security Center**, then click the **RESOURCE GROUPS** submenu.

   All the resource groups available for OneSumX Regulatory Engine are displayed in the form of cards. By default, the list is sorted by resource name, in an ascending order.

2. In the content pane, in the **Add Resource Group** area, click .

   #### Results

   A template form is displayed. A red dot precedes the mandatory fields.



3. Provide details for the new resource group, as follows:

   - **Resource Group Name**- Enter a name for the new resource group (up to 30 characters, no special characters are allowed).
   - **Description**- Enter a description for the new resource group (up to 100 characters).

4. In the **Add Resource** area, click .

Results

The **Add Workspace to Resource Group** dialog is displayed.

5. Select the workspace you want to add to the resource group and click **Add**.

Results

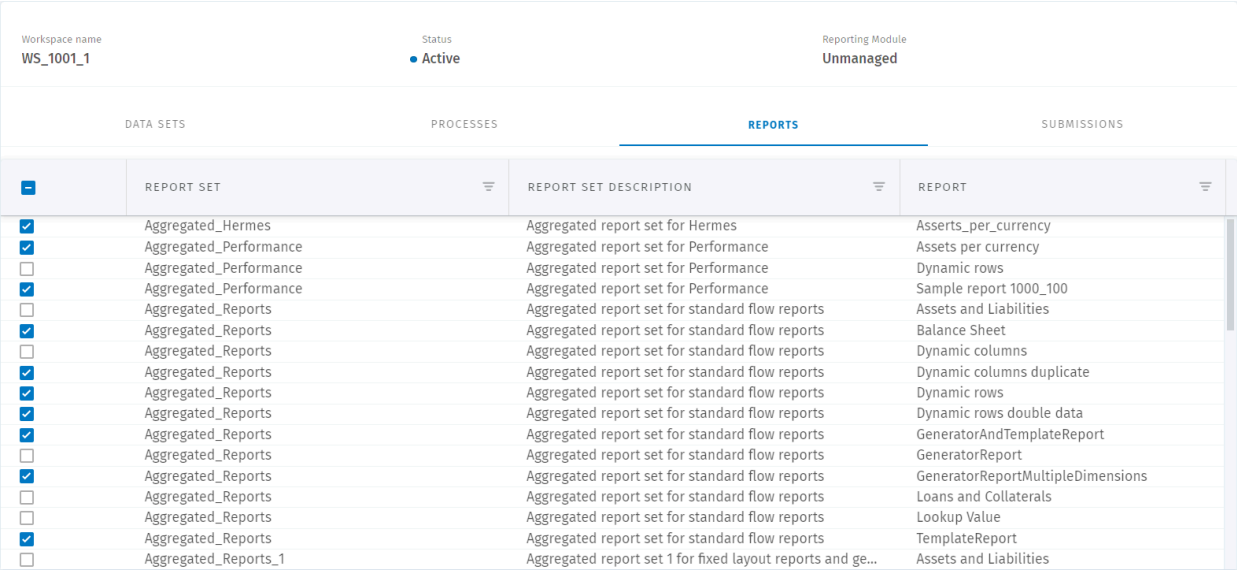
The workspace details are displayed in the content pane:

- The DATA SETS tab - Displays the data sets allocated to the workspace.
- The PROCESSES tab - Displays all the processes associated to the workspace.
- The REPORTS tab - Displays all the reports associated to the workspace.
- The SUBMISSIONS tab - Displays all the submissions associated to the workspace.

6. On the **DATA SETS** page, select the data sets you want to include in the resource group.
7. On the **PROCESSES** page, select the processes you want to include in the resource group.

Note:
In Reporting Hub, the users assigned to this resource group will be able to use only the enrichment processes that you select for this workspace.

8. On the **REPORTS** page, select the reports you want to include in the resource group.



REPORT SET	REPORT SET DESCRIPTION	REPORT
Aggregated_Hermes	Aggregated report set for Hermes	Asserts_per_currency
Aggregated_Performance	Aggregated report set for Performance	Assets per currency
Aggregated_Performance	Aggregated report set for Performance	Dynamic rows
Aggregated_Performance	Aggregated report set for Performance	Sample report 1000_100
Aggregated_Reports	Aggregated report set for standard flow reports	Assets and Liabilities
Aggregated_Reports	Aggregated report set for standard flow reports	Balance Sheet
Aggregated_Reports	Aggregated report set for standard flow reports	Dynamic columns
Aggregated_Reports	Aggregated report set for standard flow reports	Dynamic columns duplicate
Aggregated_Reports	Aggregated report set for standard flow reports	Dynamic rows
Aggregated_Reports	Aggregated report set for standard flow reports	Dynamic rows double data
Aggregated_Reports	Aggregated report set for standard flow reports	GeneratorAndTemplateReport
Aggregated_Reports	Aggregated report set for standard flow reports	GeneratorReport
Aggregated_Reports	Aggregated report set for standard flow reports	GeneratorReportMultipleDimensions
Aggregated_Reports	Aggregated report set for standard flow reports	Loans and Collaterals
Aggregated_Reports	Aggregated report set for standard flow reports	Lookup Value
Aggregated_Reports	Aggregated report set for standard flow reports	TemplateReport
Aggregated_Reports_1	Aggregated report set 1 for fixed layout reports and ge...	Assets and Liabilities

Note:

If at least one report is not selected for the resource group, the users assigned to this resource group will have access to limited operations in Reporting Hub when working with submissions:

- On the **SUBMISSIONS** page (**Reporting Hub** > **Reporting** > **SUBMISSIONS**), the top action panel in the Reports tab is not available.
- The **Validation All** button in the Validation section is not available; only the **Download All** button.
- The **Run and Update Variance** button in the Variance section is not available; only the **Export Variance** button.
- The actual report is filtered out in the Configuration of the active submission and is not available in the list.

9. On the **SUBMISSIONS** page, select the submissions you want to include in the resource group.
10. In the bottom-right corner, click **Add**.

Results

The workspace is added to the resource group list.

11. Optionally, to add more workspaces to the resource group you are creating, repeat steps 4 to 10.
12. Click **Create** to finish the creation of the resource group.

Results

The resource group is created and it is displayed in the resource group list.

What to do next

To allow access to users to the newly created resource group, follow the steps described in section Granting users access to resource groups.

14.1.1 Granting users access to resource groups

Prerequisites

- You are logged in to OneSumX Regulatory Engine, to Admin Center.
- You have the **Security Management** permission.

Procedure

1. On the main menu, click **Security Center**, then click the **RESOURCE GROUPS** submenu.
2. In the content pane, look for the resource group to which you want to grant users access and click its corresponding card.

Results

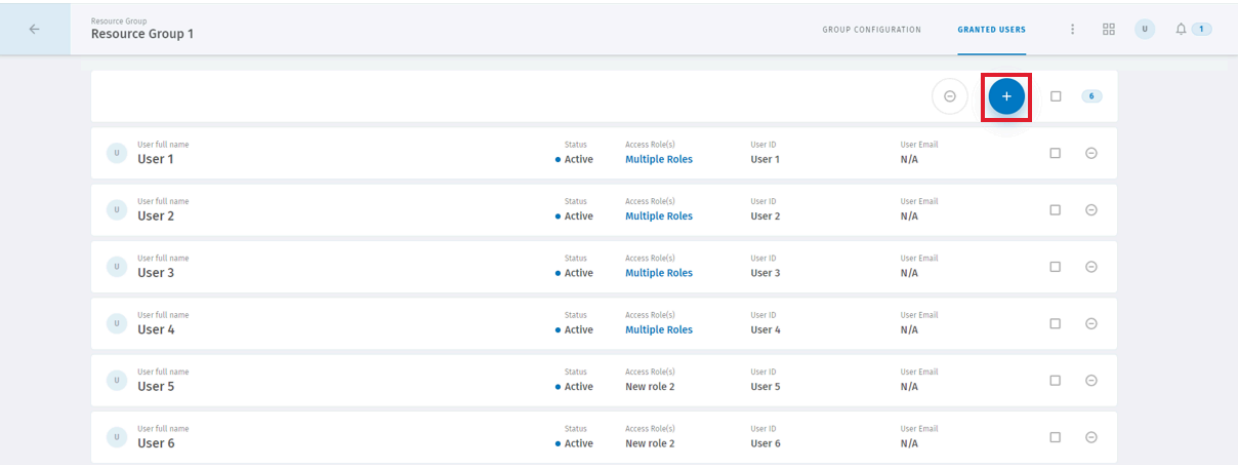
The **GROUP CONFIGURATION** menu is displayed, showing the workspaces included in the selected resource group.

3. On the resource group menu, click **GRANTED USERS**.

Results

The list of all the users who have access to the selected resource group is displayed.

4. On the top toolbar, click .



Results

The **Grant User(s) Access to the Resource Group** dialog is displayed.

Note:
Only the users who have not been granted access to the selected group are displayed in the list.

5. Perform one of the following actions:

   - To grant access for specific users to the resource group, select their corresponding checkboxes.

- To grant access to all the users in the list to the resource group, select the checkbox next to **SELECT ALL USERS**.

6. Click **Grant**.

Results

You have granted the selected users access to the resource group. The list of users who have access to the selected resource group is updated.

14.2 Removing users from resource groups

Prerequisites

- You are logged in to OneSumX Regulatory Engine, to Admin Center.
- You have the **Security Management** permission.

Procedure

1. On the main menu, click **Security Center**, then click the **RESOURCE GROUPS** submenu.
2. In the content pane, browse to the resource group from which you want to remove users' access and click its corresponding card.
3. On the resource group menu, click **GRANTED USERS**.

   ### Results

   The list of all the users who have access to the selected resource group is displayed.

4. Perform one of the following actions:
   - To remove a specific user's access from the resource group, look for the specific user and click the **Remove from Group** button to the right of its corresponding card.
   - To remove several users from the security role, select their corresponding checkboxes to the right of their cards, then, click the **Remove from Group** button in the content pane, on the top bar.
   - To remove all the users from the security role, click in the content pane, on the top bar.
5. Click **Yes, Remove** in the confirmation dialog.

Results

The selected users' access is removed from the resource group. The list of users who have access to the resource group is updated.

14.3 Viewing resource groups and their details

Prerequisites

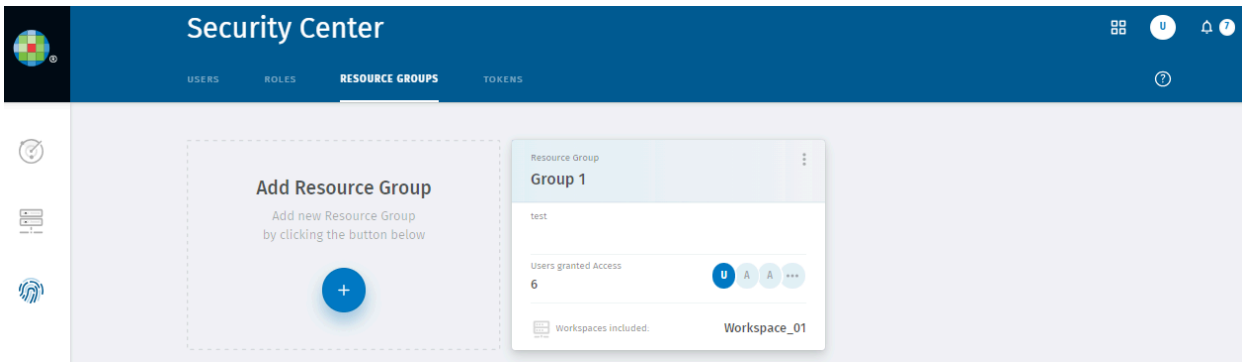
- You are logged in to OneSumX Regulatory Engine, to Admin Center.

Procedure

1. On the main menu, click **Security Center**, then click the **RESOURCE GROUPS** submenu.

   ### Results

   All the resource groups available for OneSumX Regulatory Engine are displayed in the form of cards. By default, the list is sorted by resource name, in an ascending order.

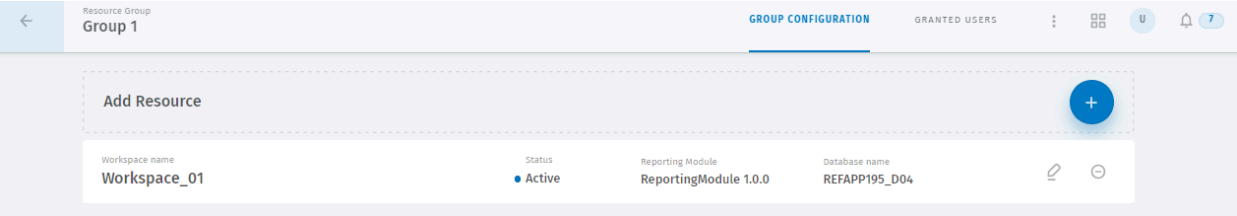
   Each resource group card displays the following information:

   - Resource Group - the card header displays the name of the resource group.
   - The resource group description - a brief description of the group is displayed.
   - Users granted Access - the number of users assigned to the group is displayed, as well as their username or avatar. The information is updated whenever users are assigned or removed from the group. If only one user is assigned to a group, the user's name is displayed, instead of the number of users. If there are more than three users assigned to the group, there is a button that you can click to see the entire list of users.
   - Workspaces included - the number of workspaces included in the group. If there is only one workspace, then the name of the workspace is displayed, instead of the number of workspaces. The information is updated whenever workspaces are added or removed from the group.

2. To view the details of a resource group, click its corresponding card.

   ### Results

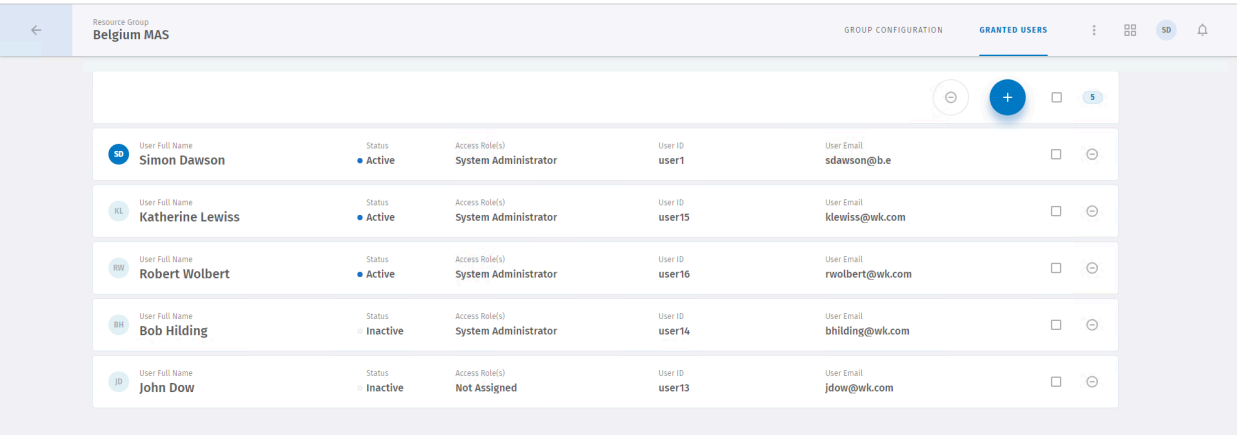
   The **GROUP CONFIGURATION** menu is displayed, showing the list of workspaces included in the group. For details regarding workspaces, see Managing workspaces.



3. To view the users who have access to the selected resource group, click the **GRANTED USERS** menu.

   All the users who have access to the selected resource group are displayed in the content pane, in the form of cards.

   For details on granting users access to resource groups, see Viewing resource groups and their details.



4. To close the resource group detailed view, click the left arrow on the menu bar.

Results

The list of all the resource groups available in OneSumX Regulatory Engine is displayed.

14.4 Adding workspaces to resource groups

Prerequisites

- You are logged in to OneSumX Regulatory Engine, to Admin Center.
- You have the **Security Management** permission.

Procedure

1. On the main menu, click **Security Center**, then click the **RESOURCE GROUPS** submenu.
2. In the content pane, browse to the resource group to which you want to add a workspace and click its corresponding card.

Results

The **GROUP CONFIGURATION** menu is displayed, showing the workspaces included in the selected resource group.

3. In the **Add Resource** area, click .

Results

The **Add Workspace to Resource Group** dialog is displayed. You can view all the workspaces (and their applications) that are currently not assigned to the selected resource group.

4. Select the workspace you want to add to the resource group and click **Add**.

Results

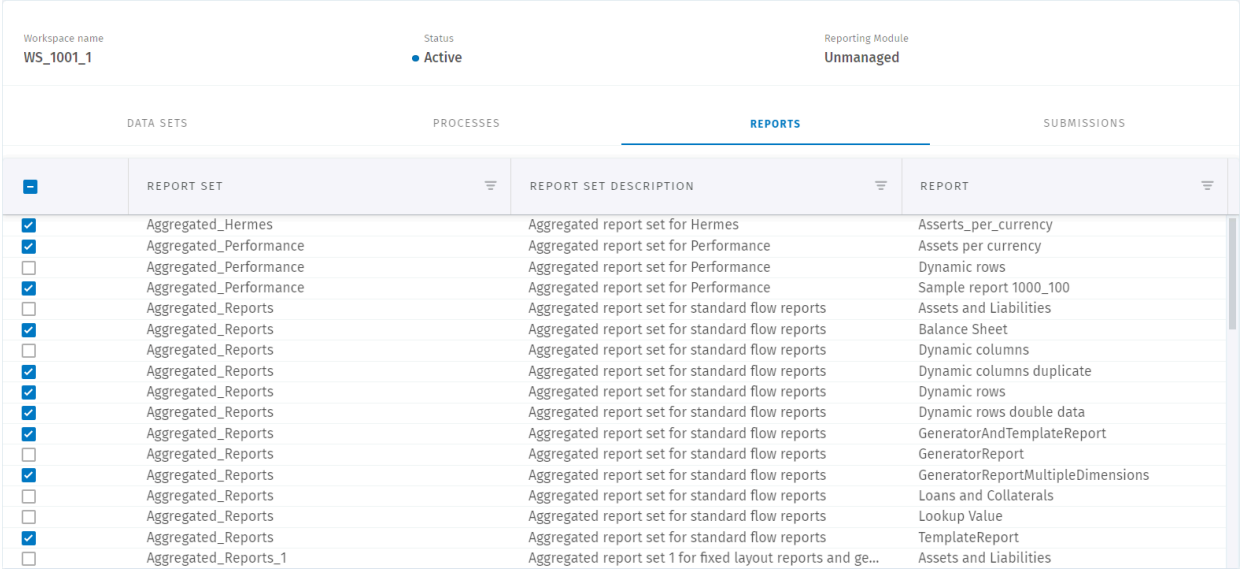
The workspace details are displayed in the content pane:

- The DATA SETS tab - Displays the data sets allocated to the workspace.
- The PROCESSES tab - Displays all the processes associated to the workspace.
- The REPORTS tab - Displays all the reports associated to the workspace.
- The SUBMISSIONS tab - Displays all the submissions associated to the workspace.

5. On the **DATA SETS** page, select the data sets you want to include in the resource group.
6. On the **PROCESSES** page, select the processes you want to include in the resource group.

Note:
In Reporting Hub, the users assigned to this resource group will be able to use only the enrichment processes that you select for this workspace.

7. On the **REPORTS** page, select the reports you want to include in the resource group.



REPORT SET	REPORT SET DESCRIPTION	REPORT
Aggregated_Hermes	Aggregated report set for Hermes	Asserts_per_currency
Aggregated_Performance	Aggregated report set for Performance	Assets per currency
Aggregated_Performance	Aggregated report set for Performance	Dynamic rows
Aggregated_Performance	Aggregated report set for Performance	Sample report 1000_100
Aggregated_Reports	Aggregated report set for standard flow reports	Assets and Liabilities
Aggregated_Reports	Aggregated report set for standard flow reports	Balance Sheet
Aggregated_Reports	Aggregated report set for standard flow reports	Dynamic columns
Aggregated_Reports	Aggregated report set for standard flow reports	Dynamic columns duplicate
Aggregated_Reports	Aggregated report set for standard flow reports	Dynamic rows
Aggregated_Reports	Aggregated report set for standard flow reports	Dynamic rows double data
Aggregated_Reports	Aggregated report set for standard flow reports	GeneratorAndTemplateReport
Aggregated_Reports	Aggregated report set for standard flow reports	GeneratorReport
Aggregated_Reports	Aggregated report set for standard flow reports	GeneratorReportMultipleDimensions
Aggregated_Reports	Aggregated report set for standard flow reports	Loans and Collaterals
Aggregated_Reports	Aggregated report set for standard flow reports	Lookup Value
Aggregated_Reports	Aggregated report set for standard flow reports	TemplateReport
Aggregated_Reports_1	Aggregated report set 1 for fixed layout reports and ge...	Assets and Liabilities

Note:

If at least one report is not selected for the resource group, the users assigned to this resource group will have access to limited operations in Reporting Hub when working with submissions:

- On the **SUBMISSIONS** page (**Reporting Hub** > **Reporting** > **SUBMISSIONS**), the top action panel in the Reports tab is not available.
- The **Validation All** button in the Validation section is not available; only the **Download All** button.
- The **Run and Update Variance** button in the Variance section is not available; only the **Export Variance** button.
- The actual report is filtered out in the Configuration of the active submission and is not available in the list.

8. On the **SUBMISSIONS** page, select the submissions you want to include in the resource group.
9. In the bottom-right corner, click **Add**.

Results

The workspace is added to the resource group list.

10. Optionally, to add more workspaces to the resource group you are creating, repeat steps 4 to 10.
11. In the bottom-right corner, click **Save**.

Results

The workspace is added to the resource group.

Note:

Changes to the resource group permissions are applied with a delay of approximately five minutes. This is due to the internal permission propagation mechanism.

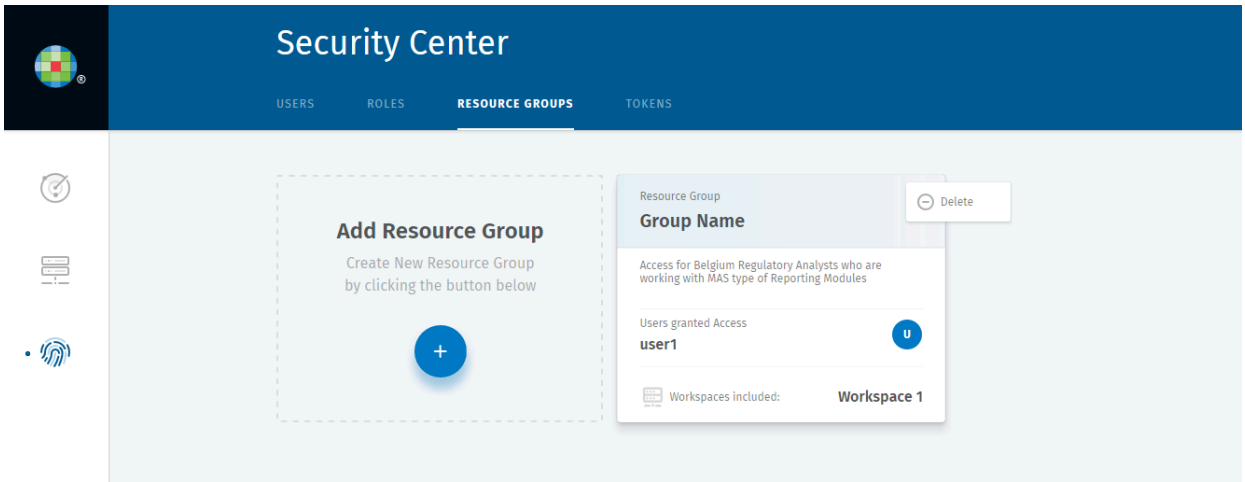
14.5 Deleting resource groups

Prerequisites

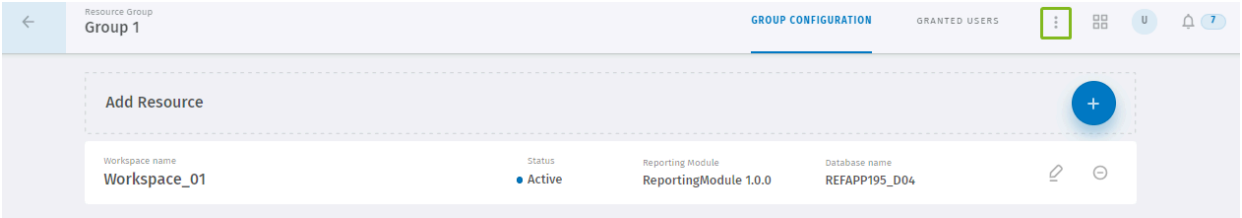
- You are logged in to OneSumX Regulatory Engine, to Admin Center.
- You have the **Security Management** permission.

Procedure

1. On the main menu, click **Security Center**, then click the **RESOURCE GROUPS** submenu.
2. In the content pane, look for the resource group that you want to delete, click its ⋮ menu, then click **Delete**.



Alternatively, look for the role you want to delete, click its corresponding card, click the ⋮ menu on the toolbar, then click **Delete**.



3. Click **Yes, Delete** in the confirmation message.

Results

The resource group is deleted and can't be retrieved anymore.

15 Risk Engine Model

Docker Compose creates the model for Risk Engine automatically. Alternatively, the model can be set up manually as follows:

1. Setting up environment
2. Importing Risk Engine model
3. Updating database
4. Generating JAR

   The following steps are mandatory within the current Risk Engine model.

5. Creating legal entity
6. Creating consolidation hierarchy
7. Updating base currency

Note:
You need to modify and configure the Risk Model as per your needs. Contact Support for more details, see Contact.

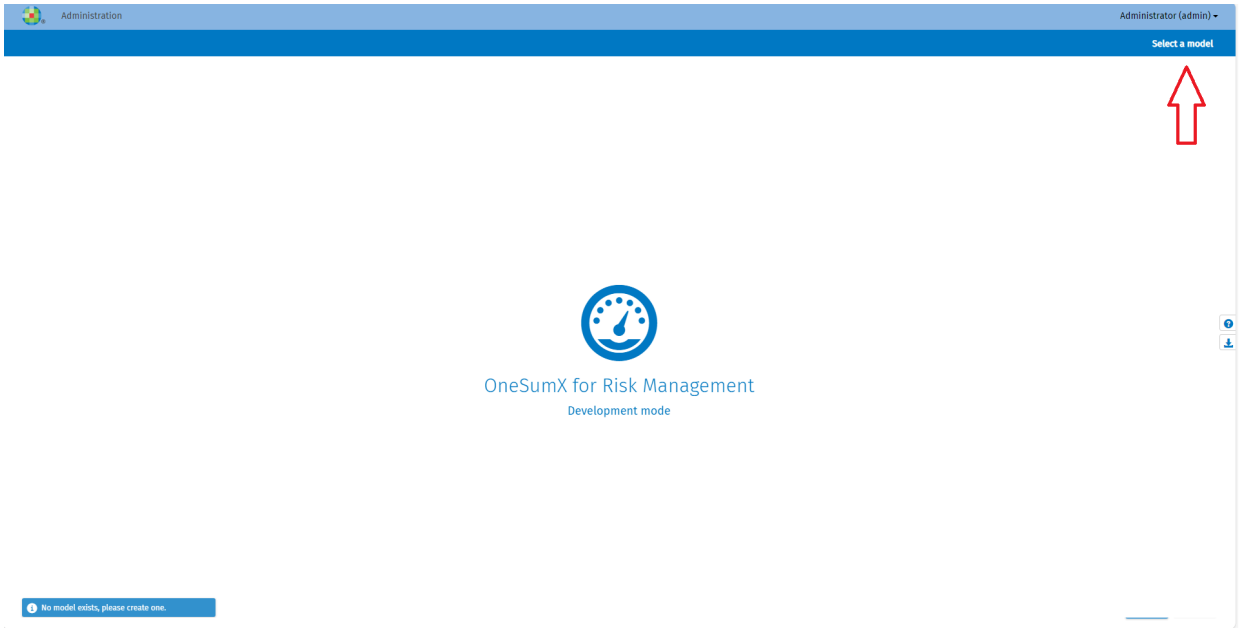
15.1 Setting up environment

Context

This step is only required if you set up the Risk Engine model manually.

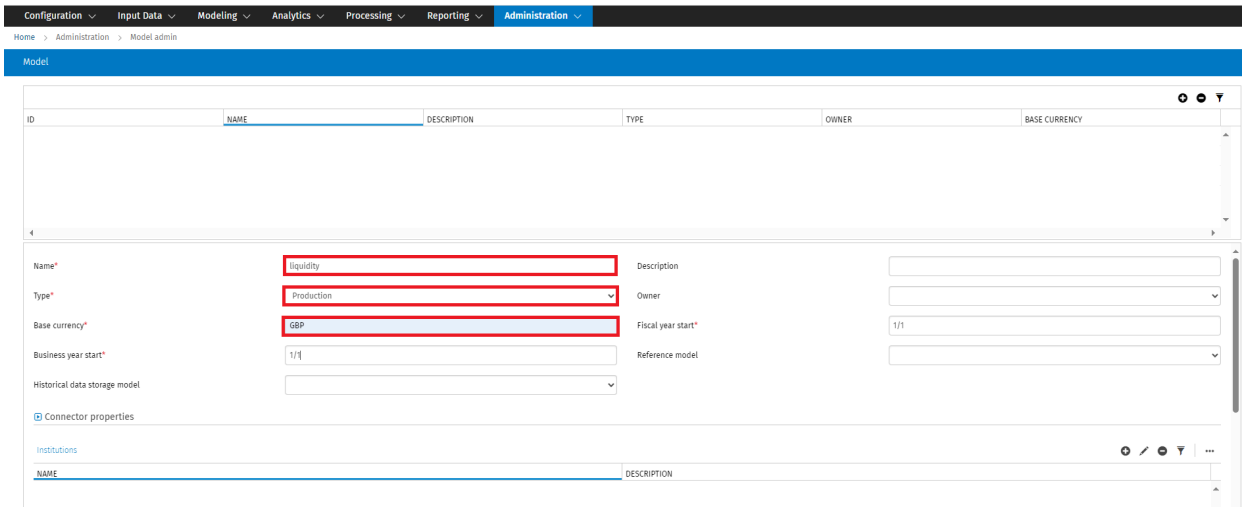
Procedure

1. In your browser, open Risk Engine.
2. Click menu option **Select a model** to configure a model.



3. Create a new model and configure the environment settings that enable the loading of the model configurations and calculations. Provide the applicable values for the following fields:

   - **Name**: liquidity
   - Type: Production
   - Base currency: Currency acronym (for example, HKD, AUD, CAD, EUR)



4. Save the model.

15.2 Importing Risk Engine model

Context

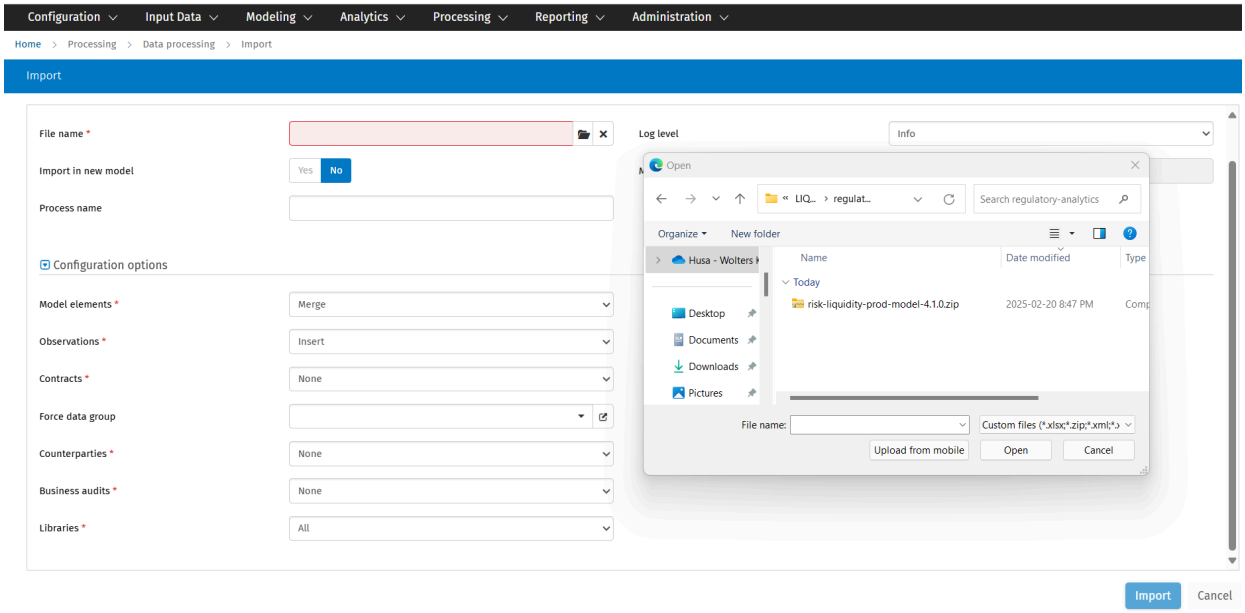
Import the Risk Engine model as follows:

Procedure

1. Select **Processing** > **Data processing** > **Import**.
2. In the **File name** field, click the folder icon ( ). Select the *risk-liquidity-prod-model-4.1.0.zip* file and click **Open**.

Results

The import screen appears.



3. Set Import in new model to **No**.
4. Set the **Configurations options** as follows.

Fields	Value
Model elements	**Merge**
Observations	**Insert**
Contracts	**None**

Fields	Value
Counterparties	**None**
Business audits	**None**
Libraries	**All**

5. Click **Import**.

Results

The `Import successfully` message appears.

15.3 Updating database

Context

Update the database as follows:

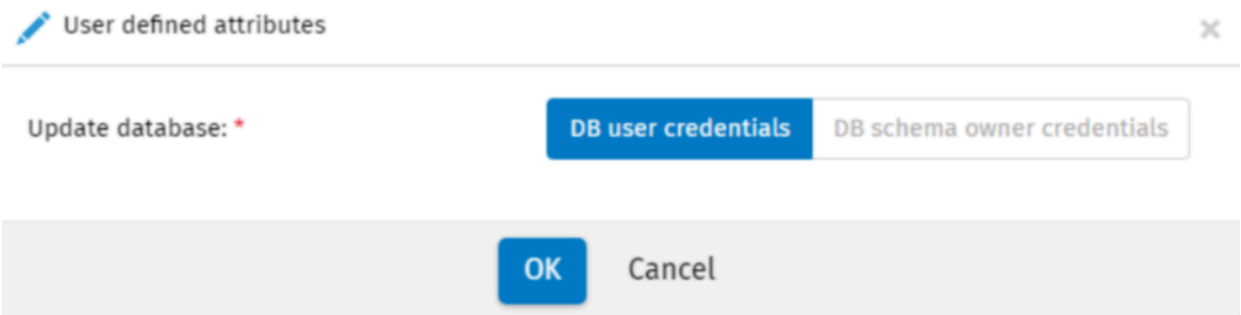
Procedure

1. Navigate to **Configuration** > **User defined expression** > **User defined attributes**.
2. Click on the ellipses in the upper-right corner and select **Update Database**.

Results

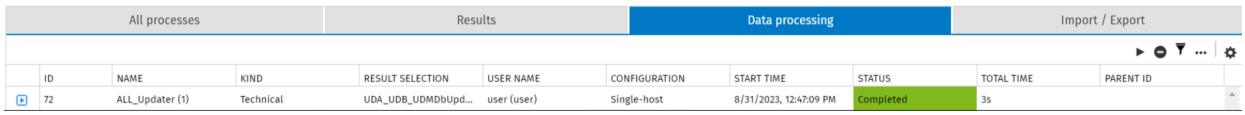
The **User defined attributes** dialog appears.

3. Click **OK**.



Results

Once the process is completed, the **Data Processing** page shows the status as "Completed".



15.4 Generating JAR

Context

Generate the JAR file as follows:

Procedure

1. Navigate to **Configuration** > **User defined expression** > **User defined attributes**.
2. Click on the ellipses in the upper-right corner.
3. Click **Generate jar**.

Results

A dialog appears with the message -Jar file generated.

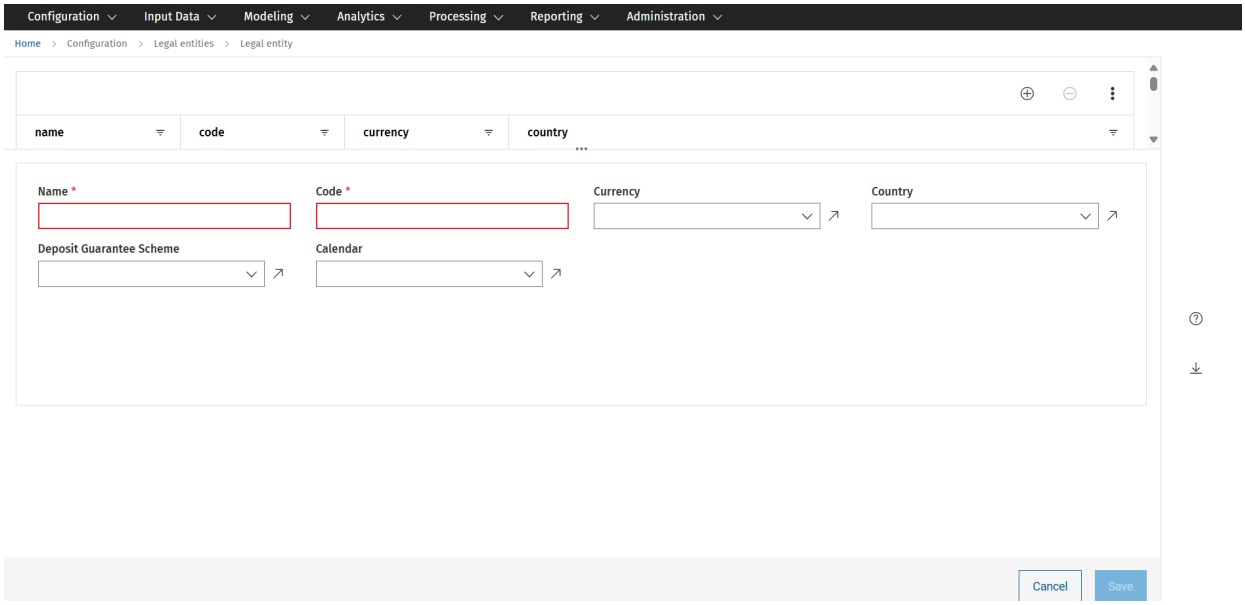
15.5 Creating legal entity

Context

By default the model provides the legal entity as SBE. The institutions must enter their own legal entity. This process needs to be set up again after each upgrade.

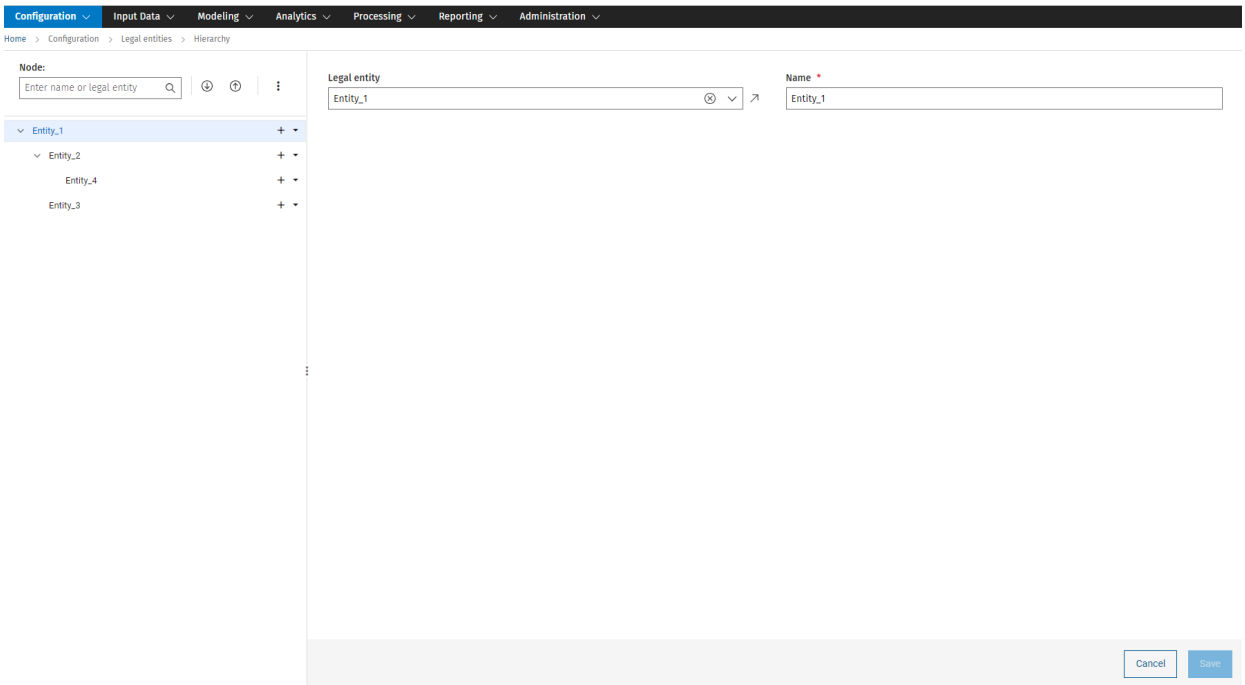
Procedure

1. Navigate to **Configuration** > **Legal entities** > **Legal entity**.
2. Click in the upper-right corner.
3. Provide your legal entity for the following fields:
   - **Name**
   - **Code**



4. Click **Save**.
5. Navigate to **Configuration** > **Legal entities** > **Consolidation hierarchy**.
6. Click in the pane to the left.
7. Provide the consolidation hierarchy with your legal entity for the following fields:

   - **Legal entity**
   - **Name**



8. Click **Save**.

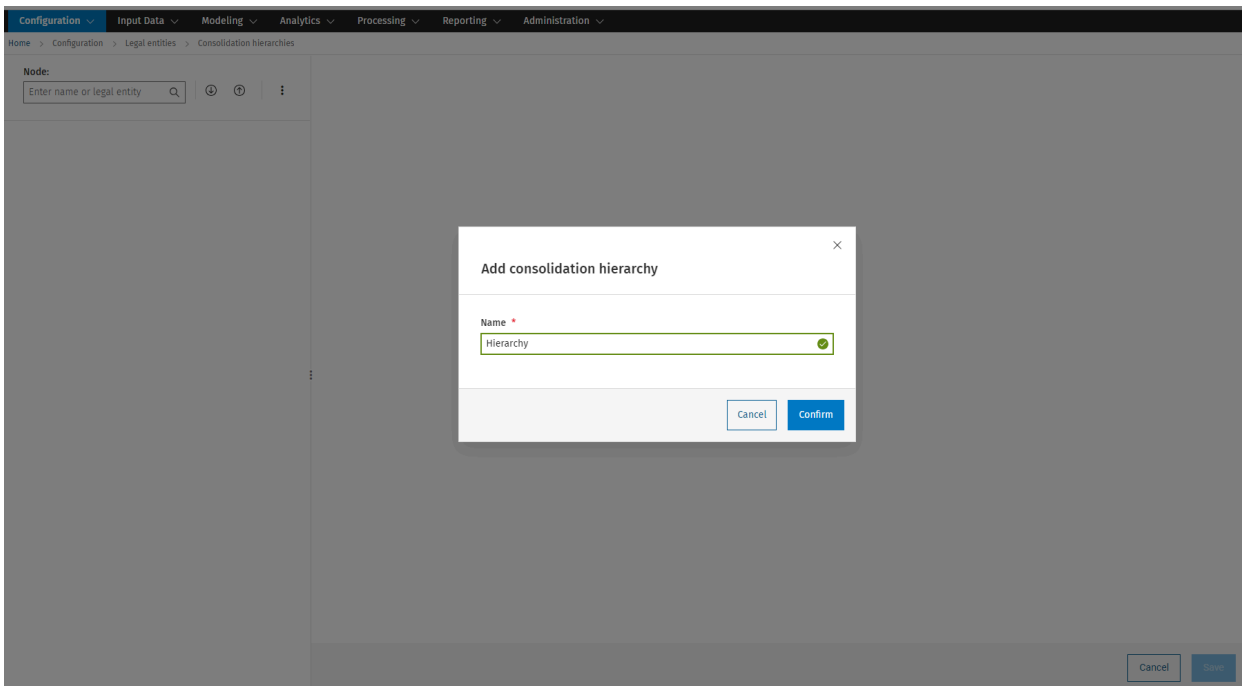
15.6 Creating consolidation hierarchy

Context

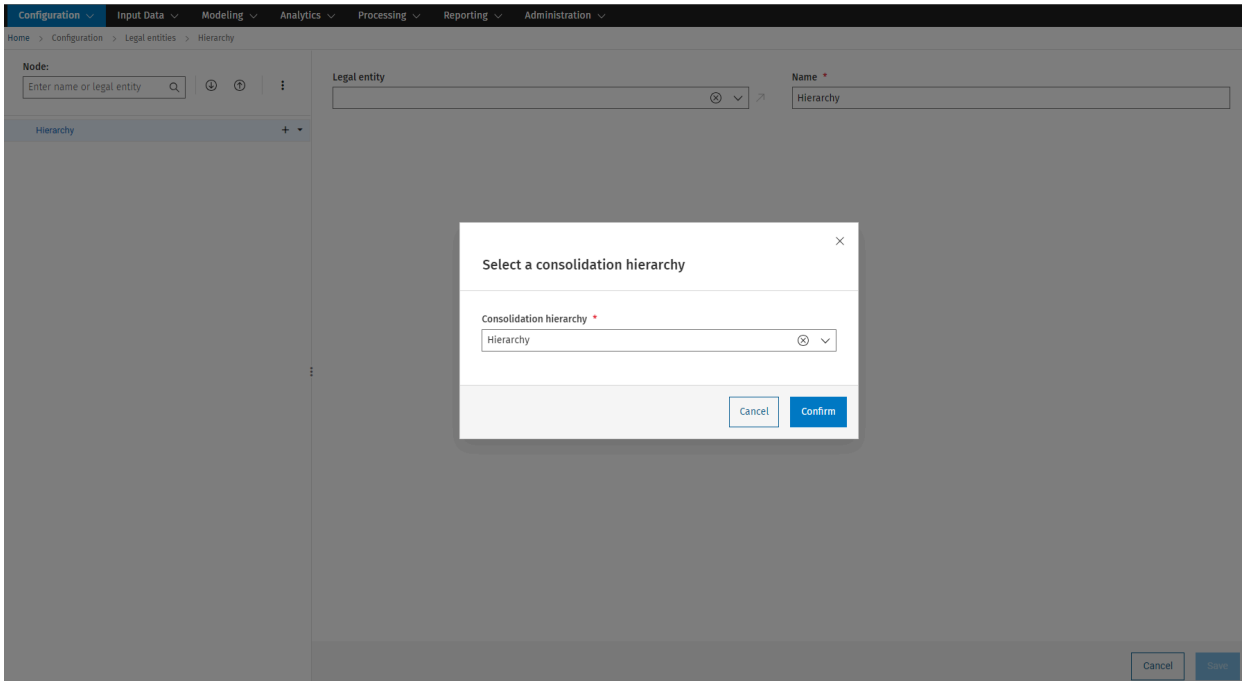
This process to create a new consolidation hierarchy needs to be set up again after each upgrade.

Procedure

1. Navigate to **Configuration** > **Legal entities** > **Consolidation hierarchy** and add the consolidation hierarchy name.

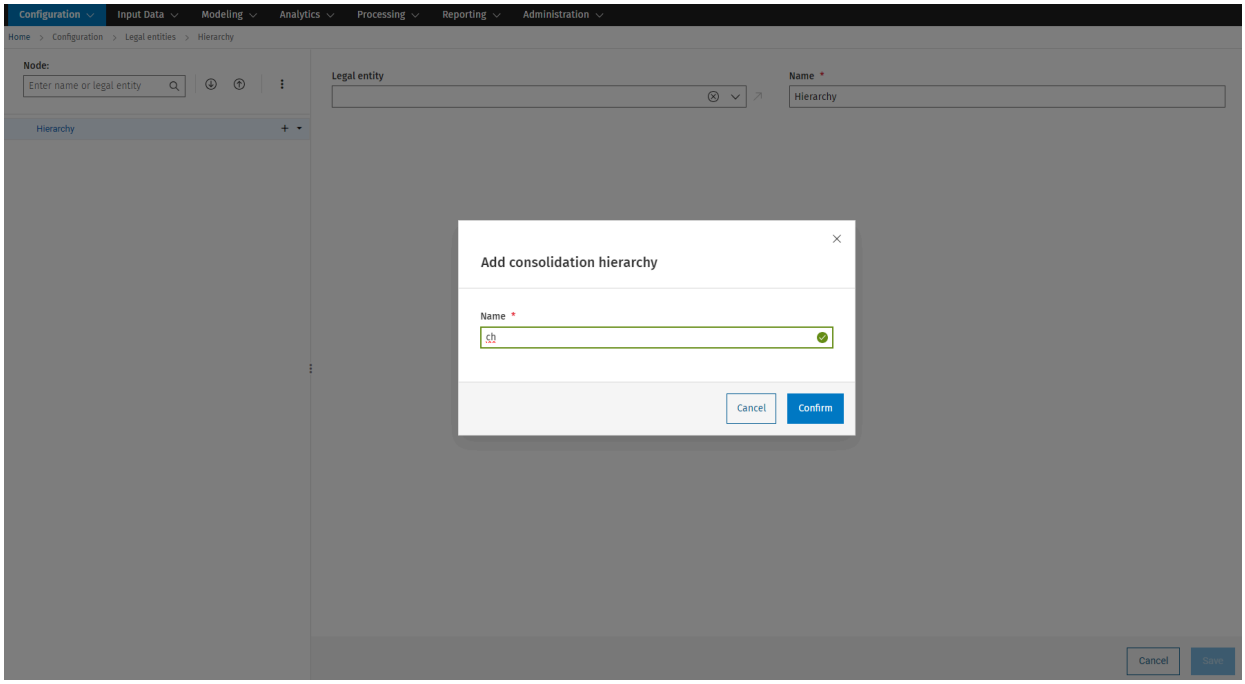


2. Click **Confirm**.
3. Click the button and choose **Select a consolidation hierarchy**.



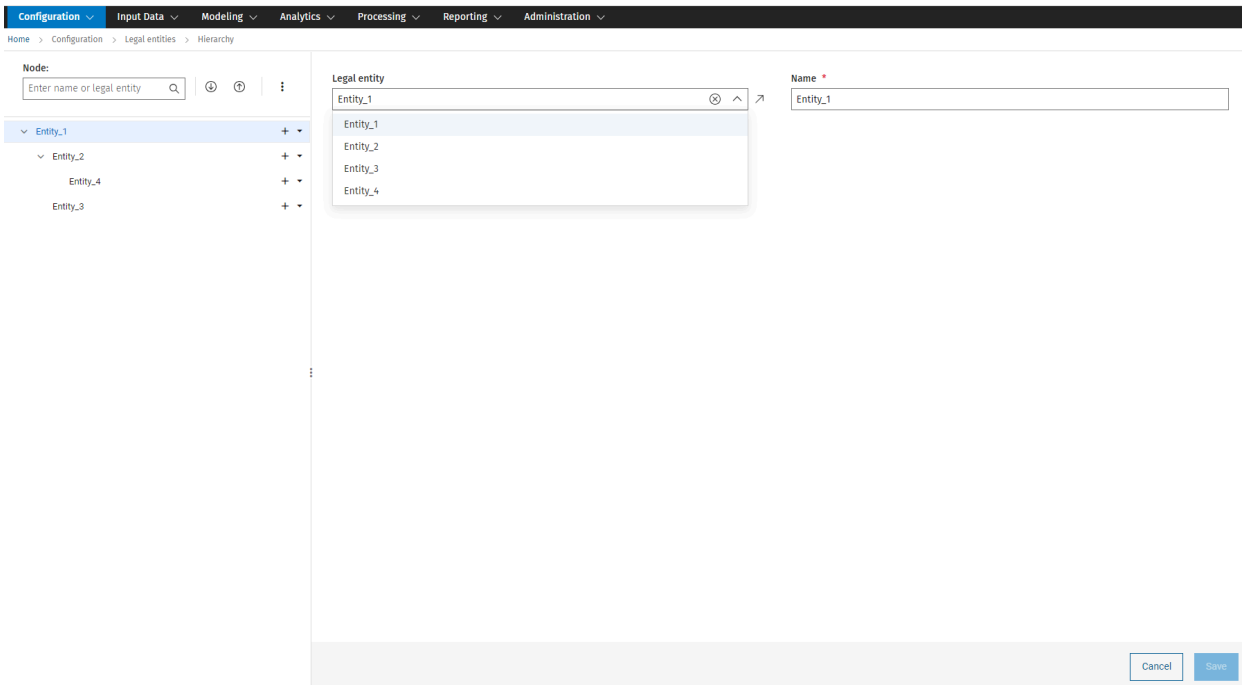
4. Click **Confirm**.

5. Click the button and choose **Add consolidation hierarchy**.
6. Provide a name for the new consolidation hierarchy.



7. Click **Confirm** to create the new consolidation hierarchy.
8. Provide the legal entity for your consolidation hierarchy via the following fields:

   - **Legal entity**
   - **Name**



9. Click **Save**.

15.7 Updating base currency

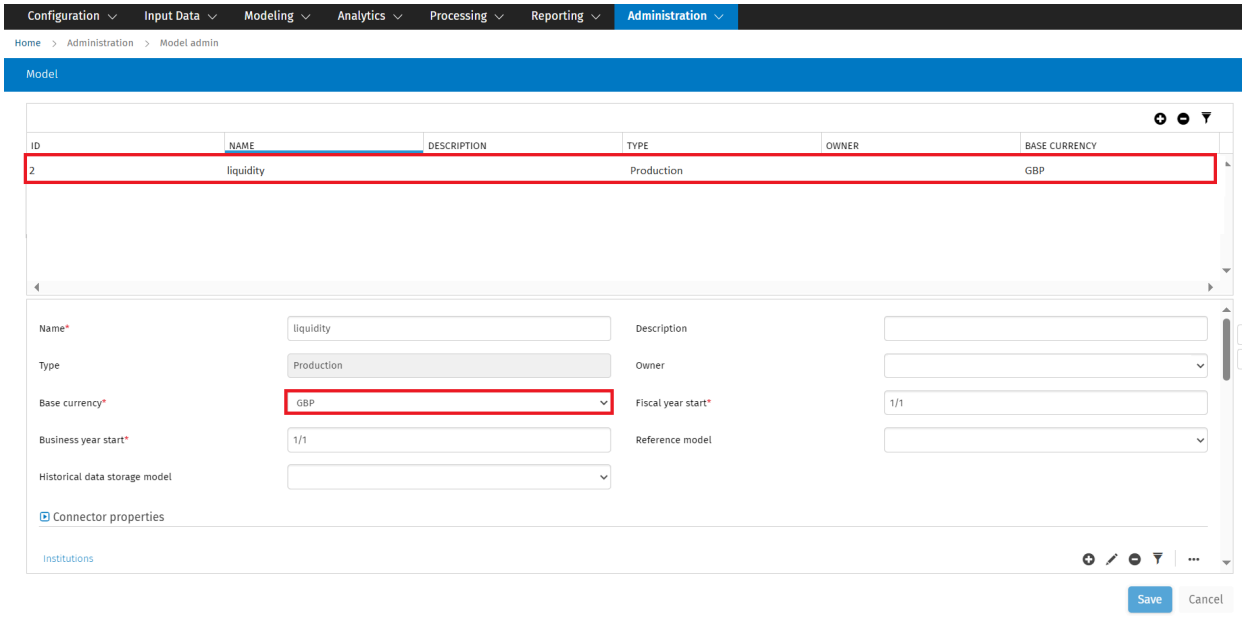
Context

Docker Compose by default provides EUR as base currency. If the model is set up manually, the base currency can be set when setting up the environment.

Change the base currency as follows:

Procedure

1. Navigate to **Administration** > **Model admin**.
2. Select the model and update the base currency.



3. Click **Save**.

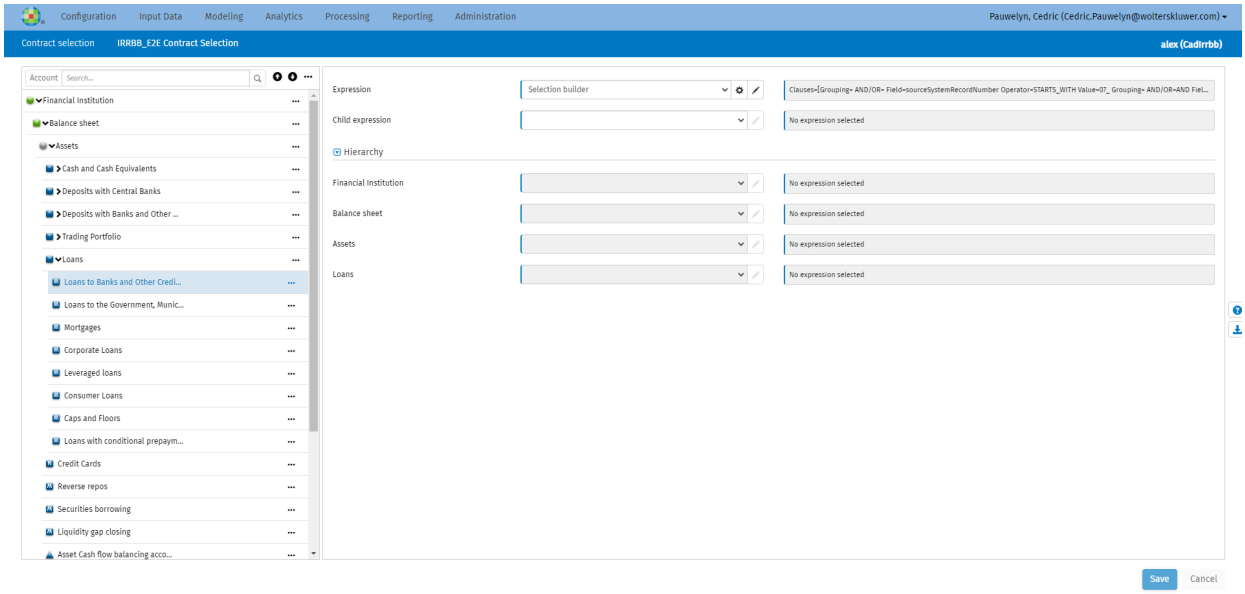
15.8 Configuring contract selection

Context

Exposures in Data Foundation of FDM are not linked to a Chart of Accounts. This is a mandatory attribute for each exposure within the Risk Engine. Therefore, it's important that each exposure is assigned an account when fetched into the Risk Engine. This can be done with the Contract Selection functionality.

Procedure

1. Go to **Configuration** > **Contract Selection** .
2. Each contract must have an account allocated. Write an expression for each Chart of Accounts node in which FDM contracts should flow. This allocation rule can happen in multiple ways, see the OneSumX documention for more detailed information on allocation rules.


Configuration
Input Data
Modeling
Analytics
Processing
Reporting
Administration
Pauwelyn, Cedric (Cedric.Pauwelyn@wolterskluwer.com)
Contract selection
IRRBB_E2E Contract Selection
alex (Cadirrbb)
Account
Search...
Financial Institution
Balance sheet
Assets
Cash and Cash Equivalents
Deposits with Central Banks
Deposits with Banks and Other ...
Trading Portfolio
Loans
Loans to Banks and Other Credi...
Loans to the Government, Munic...
Mortgages
Corporate Loans
Leveraged loans
Consumer Loans
Caps and Floors
Loans with conditional prepaym...
Credit Cards
Reverse repos
Securities borrowing
Liquidity gap closing
Asset Cash flow balancing acco...
Expression
Selection builder
Clauses=[Grouping= AND/OR= Field=sourceSystemRecordNumber Operator=STARTS_WITH Value=07_ Grouping= AND/OR=AND Fiel...
Child expression
No expression selected
Hierarchy
Financial Institution
No expression selected
Balance sheet
No expression selected
Assets
No expression selected
Loans
No expression selected
Save
Cancel

16 Contact

- For questions or issues related to the product or solution, raise a support ticket through our FRR Community.
- For questions about your support ticket or to subscribe additional email addresses to release notifications, email FRR-Support@wolterskluwer.com.
- To request support for implementation from our Professional Services team or to request a training session, contact your account manager or sales representative from Wolters Kluwer.

17 Glossary

ALMM

Additional Liquidity Monitoring Metrics

Allocation

Process of feeding reports with data from Standard Record, in accordance with given allocation rules.

AMM

Additional Monitoring Metrics

CCP

Central Counterparty

COMSWP

Commodity Swap in OneSumX for Risk Management

CRR

Capital Requirements Regulation

DA

Delegated Act

Data Sets

Set fed by user in DF format (Global Data Unit in ReportGenerator)

DF

DataFoundation data model

Dimension

A grouping of descriptive data about an aspect of business such as industry group or geography, that typically has hierarchy of members and is defined by a regulator, for example EBA.

Dimension member

An attribute of data within a dimension hierarchy, for example, the European Union as a member of the geography dimension as is France which is a child member of the European Union.

Direct Allocation

Process of feeding reports from a dedicated standard record, with report, row and column coordinates.

DSL

Domain Specific Language

EBA

European Banking Authority (formerly Committee of European Banking Supervisors)

Enrichment

Calculation and mapping process that delivers data in standard record

ESRB

European Systemic Risk Board

FDM

Financial Data Model – New common data model that will replace in the future all the other data models. It enables in-memory data storage for now.

GDM

Global Data Model – DataFoundation or the Mapping database

IAS

International Accounting Standards

IASB

International Accounting Standards Board

IFRS

International Financial Reporting Standards

IMM

Internal Model Method

Interface

The Interface is the mapping of fields in the back office or source systems to DataFoundation tables.

ISIN

International Securities Identification Numbering system

LCR

Liquidity Coverage Ratio

LDM

Local Data Model – the Param database

LIQREP

Liquidity reporting required by EBA

Lookup tables

System tables holding the list of all possible values that a particular column can contain. Their name starts with the prefix “LKP”. For example, the table LKP_ECON_SECTOR is the lookup table of the column CDE_ECONOMICAL_SECTOR.

Mapping

The mapping of L2 tables from DataFoundation to the LDM and/or RDM.

Metric

Value of data, defined in EBA taxonomy.

NSFR

Net Stable Funding Ratio

ODBC

Open DataBase Connectivity

OneSumX Regulatory Engine (RE)

Enhanced and latest engine.

PE

Processing Engine, included in RE package

RE

New regulatory engine, web-based.

Result Sets

Enriched data set (Data Unit/ Data Set in ReportGenerator)

Standard record

Table with data transformed in Taxonomy language.

Submission

Collection of reports (Work Collection in ReportGenerator)

Taxonomy/Regulator

This is the vocabulary used by the (local or regional) regulator. It is specific to a particular set of reporting requirements.

Transmission

Process of building the XBRL file to be transmitted to the regulator.

Validation

Process of checking the data in reports or result set against EBA validation rules.

Wolters Kluwer vocabulary

This is the vocabulary created by Wolters Kluwer to be used as a common ‘language’ across our applications, aka DataFoundation (DF) or FinancialDataModel (FDM).

Workspace

Section to display opened objects

XBRL

Extensible Business Reporting Language